

TECHNICAL PROPOSAL

FOR

Implementation of the Centralized Integrated Management System
(CIMS) and Business Intelligence Platform (Web & Mobile Based)

For MSME

For

Department of Industries, Bihar

(Tender Notice No.: 1127/UM, <https://eproc2.bihar.gov.in>)

Submitted To



Udyog Mitra

Department of Industries, 1st Floor, Indira Bhawan,

RCS Path, Patna, Bihar-800001,

Telephone No.: - 0612-2547695

E-mail: udyogmitrabihar@yahoo.co.in

Submitted By



Unecops Technologies Limited

C-185, Phase-1, Naraina Industrial Area,

New Delhi-110028

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1 EXECUTIVE SUMMARY

The Government of Bihar, through Udyog Mitra, Department of Industries, has articulated a governance transformation mandate of singular ambition; To design, develop, and deploy a Centralized Integrated Management System (CIMS) and Business Intelligence Platform that will fundamentally redefine how the State identifies, supports, monitors, and empowers its Micro, Small, and Medium Enterprise base. This mandate, anchored in the Raising and Accelerating MSME Performance (RAMP) Scheme and the Department's overarching industrial development objectives, calls for a technology partner of demonstrated institutional capability, deep domain experience, and the process maturity to deliver a platform of this complexity within the governance discipline that a public sector engagement of this significance demands.

Bihar's MSME ecosystem is one of the most dynamic and consequential in the country, encompassing hundreds of thousands of enterprises across diverse sectors, administrative hierarchies spanning State to Panchayat, and a network of schemes, subsidies, and support programmes originating from both State and Central Government. Yet the institutional infrastructure that governs this ecosystem today remains fragmented. Scheme data, enterprise registration records, human resource information, training participation histories, and grievance logs exist in isolation in disparate, non-interoperable systems that limit visibility, impede data-driven decision-making, and create friction for the MSME entrepreneur at precisely the moment when the State should be most accessible to them.

The CIMS platform, as envisioned in this RFP, is the answer to this fragmentation. It is not merely a software project but an institutional transformation initiative that will establish Bihar as a national exemplar of AI-enabled, data-driven MSME governance. We at Uneecops, understand this distinction completely and every dimension of the solution proposed herein has been designed with this transformational intent at its center.

Therefore, We, M/s. Uneecops Technologies Limited are honored to present this Technical Proposal in response to Tender No. 1127/UM, and submit the same with the full confidence of an organization that has spent over one and a half decade building enterprise-grade digital platforms for Government and institutional clients across India, earning, in the process, the highest process maturity certification available in the software industry: CMMi Level 5.

We propose to design, develop, deploy, and support a unified, intelligent, and institutionally scalable CIMS and Business Intelligence Platform comprising of seven natively integrated solution modules, namely the MSME Management Information System (Centralized enterprise data platform with AI-driven analytics and bilingual voice assistance), the Human Resource Management System (full-lifecycle HR governance with biometric integration and statutory payroll compliance), the Training Management Information System (end-to-end training governance with AI-powered content generation), the Grievance Redressal System (multi-channel, escalation-governed complaint management integrated with ODR and TReDS), the Integrated Job Portal with AI-Powered Resume Builder (structured employment facilitation with AI-driven matching) and the Business Intelligence and Analytics Platform (on-premises BI engine fully compliant with all eighteen of the RFPs' Annexure-4 technical specifications, delivering interactive dashboards, geo-spatial visualizations, predictive analytics, and AI-generated gap analysis) and DevOps, Infrastructure, and Security (the cross-cutting technical discipline governing deployment, security posture followed by the 52-month O&M service framework). All of the above, sharing a common data fabric, unified identity and access management architecture, and a single AI and intelligence layer, deployed exclusively at the Bihar State Data Centre, Patna.



Our proposal is anchored in five institutionally significant differentiators namely, CMMi Level 5 process maturity, an AI-first platform architecture embedding the MSME Knowledge Graph, NLP-driven policy ingestion, predictive analytics, and bilingual intelligence as structural capabilities rather than peripheral features; deep e-Governance domain experience drawn from successfully executed IT implementation, BI, HRMS, and PMU engagements for Central and State Government clients; complete data sovereignty with all platform data residing exclusively at the Bihar State Data Centre; and active ISO certifications spanning Quality Management (9001:2015), Information Security (27001:2022), and IT Service Management (20000-1:2018), each operationally embedded within Uneecops delivery framework.

This engagement will be executed and governed by a formally documented Data Governance Framework, defense-in-depth security architecture, and Service Level Agreements covering incident management, security operations, enhancement releases, and continual service improvement. Uneecops Technologies Limited commits to this engagement not merely as a technology vendor but as a dedicated institutional partner, bringing its full organizational depth to the service of Bihar's MSME entrepreneurs, departmental officials, and senior policy makers, in strict compliance with all requirements of Tender No. 1127/UM and its associated Corrigenda.

The overview of our exact understanding of the stakeholders' requirements highlighting the features & functionalities which are to be included in the overall framework is given in this document. The details of the proposed solution and how the proposed solution will get implemented through various phases & the activities to be undertaken are also mentioned. Further, detailed information about total turnaround time involved in this project & other business aspects is also given in this document.

2 UNEECOPS OVERVIEW

Uneecops Technologies Ltd., founded in 1995, is a technology solution and consulting organization, A 2000 Cr. Group Company, we, provide integrated solutions and services to our clients across India. We have PAN India presence with offices in various regions of the country. Ever since its inception, Uneecops has been at the forefront of providing integrated IT service solutions to enterprises as well as government organizations. Serving over 1000 satisfied customers, Uneecops has also successfully established itself in areas such as ERP, HRMS, BI, GIS and web development.

We have our registered office at New Delhi and Development Center in Noida. Uneecops has state-of-the-art infrastructure equipped with high-speed leased line and all forms of communications to ensure constant connectivity. We have a team of 1000+ professionally qualified and highly skilled resources who are allocated to various domains and aligned to different service needs. We provide 24x7 support with transparency of communication, upfront and fast response time. We have a comprehensive web-based support portal which renders highly customized and interactive assistance to our enterprise customers to combat day-to-day challenges.

Having partnered with major OEMs around the world including SAP, NEC, Canon, Tyco, RISO etc., Uneecops provides innovative solutions around flagship products.

Our Assets, Pride and Growth Highlights: -



2000 Cr. revenue and growing, Strong work force of 1000+, 9 centers of excellence. Expanding portfolio of products and services.



1000+ customers globally,

Empanelment with NASSCOM, CII, DGS&D, NICSI; Tech partners – SAP, Qlik Tech, Tableau, Oracle, ESRI, Canon, BenQ, Tyco RISO.



A CMM Level 5 company, Global presence – India, Africa, Singapore, Middle East, New York.

Our Products

HRONE: A robust, scalable, human resources management system to meet the needs of modern businesses and employees



Ctrl A: A breakthrough educational ERP system for educational institutes that enables centralized management of multiple operations.

Unecops Plus +

Innovative software solution designed to fulfil the needs of different industries and enhance their SAP Business-One Experience

Our Services Portfolio



Management Consulting

Complete business consulting solution to help customers improve their business processes and productivity. Process analysis, gap analysis, process design and implementation, audit services to help prevent business loss.



Business Intelligence

End to end implementation and maintenance of QlikView, a flexible Business Intelligence platform that integrates with existing ERP solution and turns available data into knowledge.



Enterprise Resource Planning

End to end SAP implementation and support. Custom SAP add-ons Uneecops PLUS+



Outsourcing

Application, HR and manpower outsourcing solutions, customized to meet the needs to present day businesses.



Web and Mobility

Futuristic web and mobile solutions for rapidly growing organizations. From Drupal, Joomla and Wordpress to Android and iOS, our tech geeks can translate your business needs into smart, scalable solutions.

Uneecops Capabilities in Software Development & Support

Application Development

Through its application services, Uneecops enables businesses to refine their enterprise application strategies by helping them to deploy, manage and modernize their application portfolios, to integrate new technology to existing portfolios to achieve higher speed, agility and cost effectiveness and drive greater business value. We enable organizations in creating and evolving new capabilities to achieve competitiveness and respond rapidly to changing market landscape by leveraging modern technologies and opportunities such as the cloud, information, mobile devices and social media.

Application Modernization

Uneecops' Application modernization service offers the migration of legacy to new applications or platforms, including the integration of new functionality to provide the latest functions to the business. Our Modernization services include Re-Platforming, Re-hosting, Re-coding, Re-architecting, Reengineering, Interoperability, Replacement and Retirement. We also consult on changes to the application architecture to clarify which option should be selected.

Application Integration

Uneecops enterprise application integration helps enterprises to achieve their end business goal with help of technology. Uneecops not only understands the business objective for application/solution development but



also helps enterprises in integrating these and other solutions in the ecosystem so as to meet the business objective. We help clients assess their application integration initiatives by defining application integration projects specific to clients' individual requirements through our tailored application integration approach.

Application Outsourcing

Application outsourcing is the process of assigning projects by a company to an outside entity, which specializes in working with the required applications. This is done to bring out results swiftly, without compromising on the pace and demands of existing business. It relieves the burdens of the internal resources in the company as well.

Application Maintenance

In Application Maintenance we enhance customers' age-old technology into modern solution by making it swift, bug free and risk-free at lower cost. The Customized Package service includes packages that are off-the-shelf for a wide range of industries that are highly configurable and innovative within affordable cost and gives desired results. Uneecops assures transformation in the clients' businesses through cutting-edge technology.

Uneecops Technology Expertise

Uneecops works on both proprietary and Free & Open Source (& Open Standard Software services)-FOSS, which include PHP, Java, .Net, MySQL and Drupal enable organizations to develop critical capabilities to optimally leverage the Application ecosystem while achieving effective cost benefits. Uneecops provides Application strategy and implementation services to optimize and accelerate Open-Source technologies for a wide range of businesses through our extensive technology expertise.

On the mobility front, we work with companies throughout their mobility journey to enable them to maximize the potential of their mobile implementation. Uneecops mobile computing competencies include android, iOS, Windows Mobile platforms. We leverage best practices & solution accelerators for cross-platform mobile application development with its highly skilled architects trained on various device platforms. We develop a mobility strategy with an architecture that provides an end-to-end enterprise mobility solution and evolves as an easy integration to enable access on multiple OS platforms.

2.1 CMMI LEVEL 5 CERTIFICATE



3 STRATEGIC INTERPRETATION OF THE MANDATE

3.1 BRIEF BACKGROUND

Bihar is home to one of the most expansive and structurally diverse MSME ecosystems in the Indian subcontinent, a sector that accounts for a significant share of the State's non-farm employment, industrial output, and entrepreneurial activity across its thirty-eight districts. From the brass artisans of Bhagalpur and the silk weavers of Bhagalpur and Nathnagar, to the food processing enterprises of Muzaffarpur and the light engineering clusters of Patna and Hajipur, Bihar's MSME landscape is characterized by its remarkable sectoral breadth, its deep rootedness in local craft and manufacturing traditions, and its extraordinary potential for value-added growth if provided with the right institutional support, market access, and governance infrastructure.

Yet for all its potential, Bihar's MSME sector has historically operated at a structural disadvantage. Geographic dispersion across largely rural and semi-urban geographies, limited access to formal credit and institutional finance, constrained market linkages beyond local and regional boundaries, fragmented awareness of available Government schemes and subsidies, and the absence of a unified, data-driven governance platform have collectively limited the pace and scale at which Bihar's enterprises have been able to grow, formalize, and compete.

The State Government, through the Department of Industries and Udyog Mitra, have in recent years demonstrated a clear and decisive intent to address these structural constraints, investing significantly in digital governance infrastructure, scheme rationalization, and enterprise support ecosystems. The CIMS and Business Intelligence Platform, as envisaged in this RFP, represents the most ambitious and consequential expression of that intent to date.

Unecops Technologies Limited brings to this engagement not merely a technical response to a procurement requirement, but a genuine and grounded understanding of the Bihar MSME context, its institutional architecture, its policy landscape, its geographic and demographic realities, and the aspirations of the lakhs of entrepreneurs whose livelihoods and futures this platform will directly serve.

3.2 OUR UNDERSTANDING OF THE RAMP SCHEME

The Raising and Accelerating MSME Performance (RAMP) Scheme, launched by the Ministry of Micro, Small and Medium Enterprises, Government of India, with World Bank support, represents a watershed moment in India's approach to MSME governance, transitioning from a purely subsidy-driven support model to a comprehensive, outcomes-oriented institutional strengthening programme that simultaneously addresses market access, credit access, Centre-State coordination, delayed payment resolution, and the greening of the MSME sector.

For Bihar specifically, RAMP carries a significance that extends well beyond the financial allocation it provides. The scheme's emphasis on strengthening State-level MSME governance institutions and building the analytical capacity of State governments, improving the quality and accessibility of scheme delivery, and creating data infrastructure that enables evidence-based policy, speaks directly to Bihar's most pressing institutional needs. A State that has historically faced challenges in translating Central Government scheme

intent into ground-level enterprise outcomes now has, through RAMP, both the mandate and the institutional support to build the kind of digital governance infrastructure that will bridge that gap permanently.

Unecops reading of the RAMP framework in the Bihar context identifies four structural imperatives that the CIMS platform must address to make the scheme's objectives operationally real.

- **Visibility:** The Department must be able to see, in real time, the performance of every scheme across every district, block, and cluster, without depending on manual reporting cycles that introduce delay and distortion.
- **Traceability:** Every rupee of scheme benefit must be traceable from sanction through disbursement to outcome, with beneficiary-level accountability embedded in the platform's data architecture.
- **Convergence:** Bihar's MSME enterprises must experience a single, unified interface to all State and Central Government support programmes, rather than navigating a fragmented landscape of disconnected portals and offices.
- **Intelligence:** The Department must transition from reactive scheme administration to proactive, AI-enabled programme management that identifies gaps, flags underperformance, and surfaces actionable insights before problems become irreversible.

The CIMS platform as proposed by Unecops is designed to deliver precisely these four imperatives, making RAMP's institutional strengthening objectives operationally tangible for the Government of Bihar.

3.3 OBJECTIVES

Unecops understanding of the objectives of this engagement extends across three distinct but interdependent dimensions; the immediate platform delivery objective, the medium-term governance transformation objective, and the long-term institutional legacy objective, each of which has informed the design and architecture of the proposed solution.

- **Platform Delivery Objective: -**
 - Design, Development and deployment of a fully functional, technically compliant, seven-module CIMS and Business Intelligence Platform operational at the Bihar State Data Centre
 - Accessible through web and mobile in English and Hindi
 - Integrated with eleven GoI and State digital platforms
 - Formally handed over to Udyog Mitra at Milestone-4 (The RFP defined Week 35/Month-8) in a production-ready, security certified, user-trained state.
- **Governance Transformation Objective: -**
 - Transition from a data-scarce, manually intensive, scheme-delivery-focused administrative posture to a data-rich, AI-enabled, outcomes-oriented governance model
 - Embedding data-driven decision-making habits at the district and block level through intuitive dashboards and mobile BI tools.
 - Creating the information infrastructure through which the State can confidently measure, report, and improve its MSME programme outcomes to both the State Government and the Ministry of MSME at the Centre.

- **Institutional Legacy Objective: -**

- Create a digital governance asset that is not merely functional but institutionally owned, technically understood, and operationally self-sustaining.
- This will be delivered through comprehensive knowledge transfer, documentation, administrator training, and a 52-month O&M engagement that progressively builds the Department's own operational capability.

3.4 STAKEHOLDER MAPPING

The CIMS platform will serve a stakeholder ecosystem of exceptional breadth and diversity spanning Government institutions at multiple levels, MSME enterprises across all sectors and geographies, individual citizens seeking economic opportunity, and national digital infrastructure platforms. Our solution design has been informed by a careful analysis of each stakeholder's constituency's distinct needs, literacy context, and interaction patterns with the platform.

- **Udyog Mitra & the Department of Industries, Bihar:** Represents the primary institutional client and the platform's governing authority. As the entity responsible for the RAMP Scheme's implementation in Bihar, for the administration of State MSME policy, and for the oversight of the Department's four directorates i.e. Industries, MSME, Handloom and Sericulture, and Food Processing, what Udyog Mitra requires from the CIMS platform is a comprehensive, real-time operational intelligence environment that enables senior officials to monitor programme performance, manage resources, and make evidence-based policy decisions with confidence and speed.
- **District & Block Level Officials:** This constitutes the operational backbone of the Department's scheme delivery apparatus; the administrators who register enterprises, process applications, conduct field visits, resolve grievances, and translate State policy into ground-level outcomes.
 - For this constituency, the CIMS platform must be intuitive, reliable, accessible on modest hardware and variable network connectivity, and practically useful in the day-to-day rhythm of district administration.
 - Their adoption of the platform will ultimately determine whether the CIMS transforms delivery scheme into practice or remains a headquarters-level governance tool.
- **MSME Enterprises and Entrepreneurs:** The ultimate beneficiaries of the CIMS platform with lakhs of Micro, Small, and medium enterprises across Bihar's thirty-eight districts whose access to schemes, credit, markets, training, and grievance resolution the platform is designed to improve. This constituency encompasses a vast spectrum of digital literacy: from urban, formally registered enterprises with sophisticated technology awareness, to rural micro-enterprises for whom the bilingual mobile application and AI voice chatbot may represent their first meaningful digital interface with the State Government. The platform's accessibility, bilingual design, and intuitive self-service architecture are a direct response to this diversity.
- **Departmental Employees:** Internal Users of HRMS and TMIS modules- employees whose attendance, payroll, leave, performance, and training participation will be managed through the platform. For this constituency, the platform must be fair, accurate, transparent, and compliant with applicable

Government of Bihar service rules building trust in the digital HR infrastructure as an institutional asset rather than a surveillance instrument.

- **Job Seekers & Broader Citizen Ecosystem:** This represents the wider public constituency served by the Job Portal, the public information portal, and the grievance redressal interface; citizens who may not be formal MSME enterprises, but who interact with the Department's services as potential employees, as complainants, or as information seekers. For this constituency, the platform's value is measured entirely in terms of accessibility, responsiveness, and outcome whether their grievance was resolved, their job application was acknowledged, their question was answered.
- **Ministry of MSME and Government of India Platforms:** Representing the national institutional stakeholders whose digital infrastructure like Udyam, TReDS, ODR, ONDC, and Ministry portals, the CIMS platform will integrate with and contribute data to. For this constituency, Bihar's CIMS must be a reliable and standards-compliant integration partner that enriches the national MSME data ecosystem rather than operating in isolation from it.
- **The Analytic Cell of the Department of Industries:** This deserves specific recognition as a stakeholder of strategic importance- the internal unit that will be the primary institutional user of the Business Intelligence and Analytics Platform, responsible for translating the CIMS's data and intelligence outputs into policy analysis, programme evaluation, and evidence-based recommendations for the Department's leadership. The success of this constituency's engagement with the platform will determine whether the CIMS's intelligence layer fulfils its transformational potential or remains an underutilized technical capability.

4 APPROACH & METHODOLOGY, UNDERSTANDING OF BUSINESS REQUIREMENTS

4.1 CONTEXT & STRATEGIC IMPERATIVE

The Department of Industries, Government of Bihar, through Udyog Mitra, is undertaking one of the most ambitious digital governance transformations in the MSME sector in the country. The Raising and Accelerating MSME Performance (RAMP) Scheme, anchored in the overarching development mandate of the Department, seeks to create a unified, intelligent, and data-driven governance ecosystem that will fundamentally alter how the State interacts with, supports, monitors, and empowers its Micro, Small, and Medium Enterprise base.

Unecops Technologies Limited recognises that the core business challenges this RFP addresses is not merely one of software development it is one of institutional transformation through technology. The Department currently operates across multiple directorates like Industries, MSME, Handloom and Sericulture, and Food Processing, each of which generates significant volumes of data, processes citizen and enterprise requests, and administer schemes and subsidies. In the absence of a centralized, integrated platform, this data remains fragmented, scheme visibility remains limited, decision-making remains reactive rather than predictive, and the MSME entrepreneur's interface with the State remains friction-laden and opaque.

The centralized Integrated Management System (CIMS) and Business Intelligence Platform, as envisioned by Udyog Mitra, will resolve this institutional fragmentation by creating a single, authoritative, AI-enabled digital platform that serves multiple stakeholder constituencies simultaneously the MSME entrepreneur seeking scheme access and market linkage, the departmental official requiring real-time operational visibility, the

senior administrator demanding analytical insights for policy decisions, and the State's human resource management apparatus requiring a structured and compliant HRMS. Uneecops Technologies fully appreciates the breadth, complexity, and governance significance of this mandate.

4.2 UNDERSTANDING OF STAKEHOLDER ECOSYSTEM

Uneecops's approach is grounded in a thorough understanding of the multi-stakeholder ecosystem within which the CIMS platform will operate. The platform will serve the following distinct user constituencies, each with differentiated needs:

- **MSME Enterprises and Entrepreneurs:** Require seamless self-service access to scheme information, registration and onboarding, grievance filing, job posting, market linkage through the integrated e-commerce and ONDC interface, and training and capacity building opportunities through the TMIS. The platform must serve these users in a bilingual environment, with intuitive navigation suited to varying levels of digital literacy.
- **Departmental Officials at State, District, Block, and Panchayat Levels:** Require real-time operational dashboards, scheme monitoring tools, compliance tracking, employee management through the HRMS, and workflow-enabled grievance resolution with defined turnaround timelines and escalation matrices.
- **Senior Government Officials and Policy Makers:** Require a sophisticated Business Intelligence and Analytics layer that delivers predictive insights, AI-generated gap analyses, scheme effectiveness metrics, and Centre-State programme alignment reporting, all accessible through an integrated MIS dashboard with drill-down visibility across the State's administrative hierarchy.
- **Ministry and GoI Integration Points:** The platform must interact seamlessly with the Udyam Registration Portal, MSME Ministry portals, TReDS, ODR platforms, ONDC, and other GoI digital infrastructure through well-governed API integration, ensuring that Bihar's MSME ecosystem is positioned within (and not isolated from) the national digital public infrastructure landscape.
- **Internal HR and Administrative Functions:** The department's own workforce, spread across its multiple directorates, requires a robust, compliant HRMS that manages the complete employee lifecycle from recruitment and onboarding through attendance, payroll, performance management, leave, and pension with biometric integration and mobile accessibility.

Uneecops Technologies Limited will design, develop and deploy the CIMS platform with this complete stakeholder matrix at the centre of every architectural and functional decision.

4.3 KEY BUSINESS PROBLEMS BEING ADDRESSED

Uneecops's understanding of the business requirement can be distilled into five structural problems that the CIMS platform will resolve:

- **Data Fragmentation and Absence of a Single Source of Truth:** Scheme data, MSME registration data, training records, grievance logs, and HR data currently reside in disparate, non-interoperable

systems. The CIMS will establish a unified data architecture with a centralised, real-time database that serves as the authoritative source of truth for all departmental operations.

- **Lack of Real-Time Scheme Visibility and Beneficiary Traceability:** The Department has limited visibility into who has accessed which scheme, at what stage their application sits, and what outcomes have been delivered. The CIMS will provide a complete beneficiary lifecycle view, from registration through scheme application, approval, disbursement, and outcome measurement.
- **Absence of AI-Driven Intelligence for Policy and Programme Management:** Decision-making is currently constrained by the absence of predictive analytics, gap analysis, and automated insight generation. The AI-powered MSME Knowledge and Intelligence layer within CIMS will transform this automatically ingesting policy documents, generating structured training content, identifying underperforming geographies and sectors, and delivering actionable intelligence through the integrated MIS dashboard.
- **Limited Market Access and Economic Linkage for MSMEs:** Bihar's MSME entrepreneurs face significant barriers to market access, e-commerce participation, trade fair visibility, and export-import linkage. The CIMS platform will directly address this through its integrated e-commerce interface, ONDC linkage, demand and supply module, and directory of exporters and importers.
- **Absence of Integrated Human Resource Governance:** The Department manages a significant workforce across multiple directorates without a unified HRMS. This results in attendance management gaps, payroll inconsistencies, and limited performance visibility. The HRMS module within CIMS will fully digitise and govern the HR lifecycle for all departmental employees.

4.4 APPROACH & METHODOLOGY

4.4.1 DELIVERY PHILOSOPHY

Uneecops Technologies Limited will approach the design, development, and deployment of the CIMS and Business Intelligence Platform through a delivery philosophy that is built on three foundational principles: -

- **Citizen & Enterprise Centricity:** Every design decision will prioritize the ease of access, navigability, and utility of the platform for its primary beneficiaries which are the MSME entrepreneur and the departmental official. User Experience (UX) and User Interface (UI) design will be guided by the diverse digital literacy landscape of Bihar's MSME ecosystem, ensuring that the platform is accessible, intuitive, and bilingual.
- **Governance by Design:** Given that this is a public sector digital infrastructure project, Uneecops will embed governance, compliance, audit, and accountability mechanisms at every layer of the platform, from access control through SSO, RBAC, RLS, and OLS, to comprehensive audit logging, data classification, retention policy enforcement, and application security compliance with CERT-In guidelines, IT Act 2000 and 2008, ISACA standards, and ISO 27001.

- **Scalability & Institutional Longevity:** The platform will be architected for long-term institutional use designed to absorb new schemes, new directorates, and new GoI integration points as they emerge over the 60-month engagement period, without requiring fundamental re-design. The architecture will ensure that the Department is not locked into a static system but owns a living, extensible digital infrastructure.

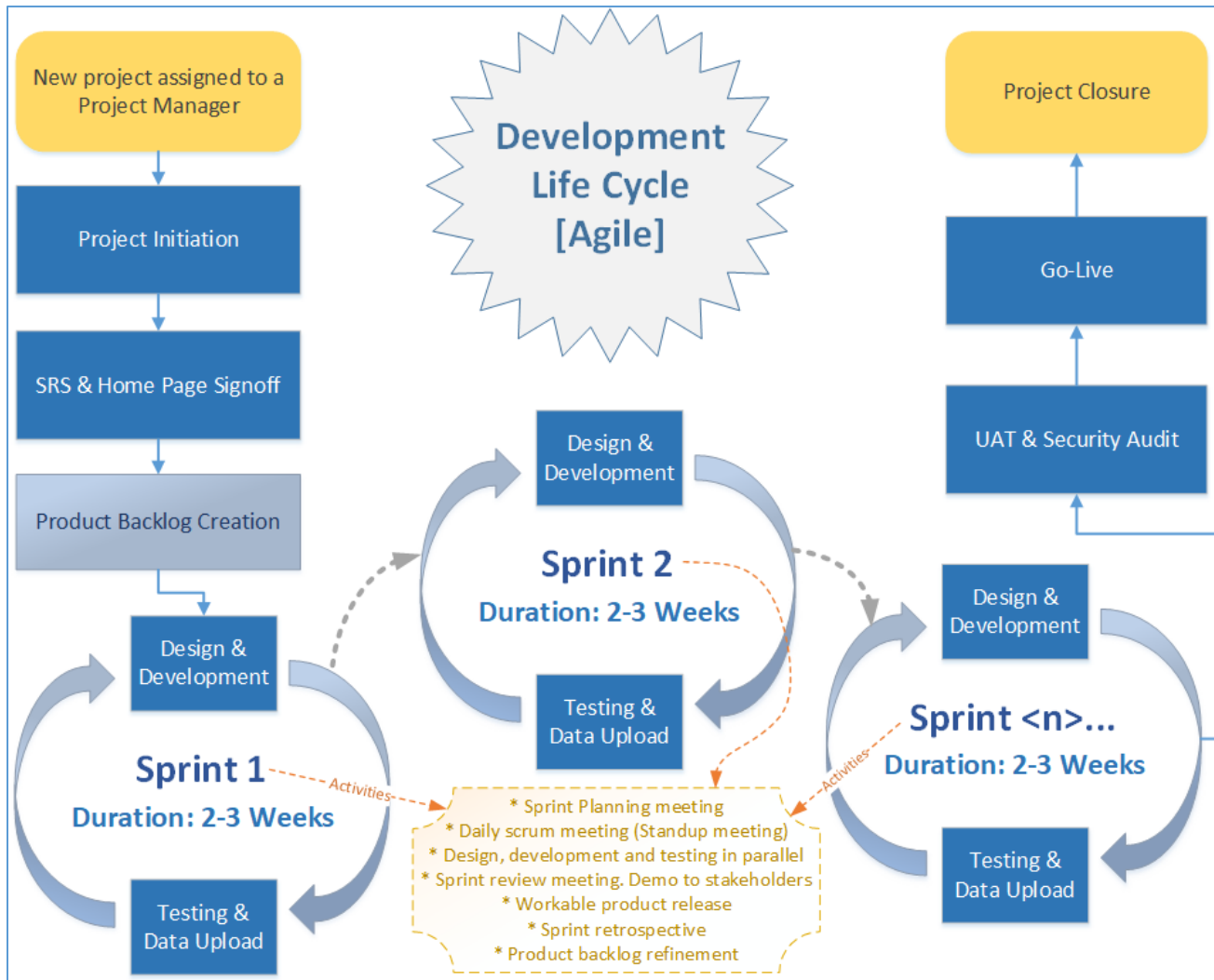
4.4.2 DELIVERY METHODOLOGY- AGILE WITH CMMI LEVEL 5 PROCESS GOVERNANCE

Uneecops Technologies will execute the CIMS project through an Agile delivery framework governed by CMMi Level 5 process discipline, a combination that delivers the responsiveness and iterative quality of Agile while maintaining the process rigour, risk management, and predictability that a public sector engagement of this magnitude demands.

- **Agile Delivery Framework:** The development lifecycle will be organized into two-week sprint cycles across four parallel delivery tracks i.e. MSME MIS and Integration, HRMS and TMIS, Grievance and Job Portal, and BI/Analytics and AI Platform. This parallel-track approach ensures that all seven modules progress concurrently, with sprint reviews at fortnightly intervals providing the Department with regular, demonstrable progress against committed deliverables.
- **CMMI Level 5 Process Governance:** As a CMMi Level 5 certified organization, Uneecops will apply its institutionalized process framework across all phases of delivery viz. “requirements management”, “design verification”, “code review”, “defect prevention”, “root cause analysis”, and “continuous process improvement”. This certification is not merely a credential; it represents a delivery culture where quality is built into the process rather than inspected at the end.
- **Sprint Review & Stakeholder Visibility:** At the conclusion of every sprint, a formal sprint review will be conducted with designated stakeholders from Udyog Mitra. The working software will be demonstrated, feedback will be incorporated into the subsequent sprint backlog, and progress will be tracked against the master delivery plan. This ensures that the Department always maintains full visibility and control over the platform’s evolution.

4.4.2.1 AGILE DELIVERY FRAMEWORK

We are Proposing Here Customized Agile Software Development Methodology



4.4.2.1.1 DEFINITION OF SCRUM

Scrum is a software product development strategy that organizes software developers as a team to reach a common goal — creating a ready-for-market product. It is a widely used subset of Agile software development.

4.4.2.1.2 SCRUM FRAMEWORK

Roles

- Product Owner
- Scrum Master
- Team

Scrum Ceremonies

- Sprint Planning
- Daily Stand-up Meeting
- Sprint Review
- Sprint Retrospective

Artefacts

- Product Backlog
- Sprint Backlog
- Burn down Charts

4.4.2.1.3 ROLES: PRODUCT OWNER

- Creating, prioritizing and managing product backlog along-with acceptance criteria. Product backlog should contain all of the work to be performed. Product backlog prioritization and sequencing should be based on business value.
- Provide clear guidance to team about the features to be delivered next.
- Participation in development events including planning, refinement, review, retrospective of the sprint and daily scrum meetings. Collaboration with the team to develop and deliver the work.
- PO inspects the product progress at the end of every Sprint and has complete authority to accept or reject work done.
- Within weekly refinement sessions, PO works with the development team to define, elaborate, estimate, order or delete product backlog items.
- Share project progress status update and metrics with stakeholders.
- External stakeholder should not communicate directly with the Scrum Team. Instead the Scrum Product Owner should collect and discuss required functionalities with the different Stakeholders (e.g. customer). These requirements are then combined and filtered before giving it to the team in the form of prioritized Scrum Product Backlog Items.

4.4.2.1.4 ROLES: SCRUM MASTER

- Remove barriers and impediments that are preventing the team from completing work.
- Facilitate Scrum ceremonies (sprint planning meeting, daily stand-up, sprint review meeting and sprint retrospective meeting) and enforce processes.
- Protecting the team from outside interruptions and distractions.
- Mediating the general conflict of goals between development team (high technical quality) and product owner (more features).
- Encouraging the use of Agile Engineering Practices within the development team.
- Ensures workshops, sessions and ceremonies for the upcoming Sprint are scheduled.
- Capture and track impediments, risks and dependencies until resolution.
- Updation of backlog and/or tool with tasks and commitments.

4.4.2.1.5 ROLES: SCRUM TEAM

- Delivers working product on-time, on scope and on quality
- Shares development responsibilities, assists, trains and mentors Team members to meet “Sprint” challenges.
- Strives to value the individual and increase team recognition over self-recognition.
- Always is prepared to challenge self and the Scrum Team in (DoD) “Definition of Done.”
- Scrum team as a whole is accountable for the delivery.
- The Scrum Team is self-organizing i.e. its members determine how to proceed with the items on Sprint backlog.
- Ensures delivery of potentially shippable product at the end of each sprint.

4.4.2.1.6 SCRUM SOFTWARE DEVELOPMENT STARTS WITH PRODUCT BACKLOG

- Scrum software development starts with a wish list of features — a.k.a. product backlog.
- The Product Backlog will have complete user stories as per the requirement freezed from customer.
- Then with the help of Dev team these user stories should be divided in small achievable tasks.
- Tasks are created in such manner that estimation should not be more than 8 hours per task.
- Scrum relies on an Agile software development concept called Sprints.
- Sprints are periods of time when software development is actually done.

- A Sprint usually lasts from one week to one month to complete items from the backlog.
- The goal of each Sprint is to create a sellable product (demo able product for client).
- Each Sprint ends with a Sprint review.
- Then the team chooses another piece of backlog to develop — which starts a new Sprint.
- Sprints continue until the project deadline, or the project budget is spent.

4.4.2.1.7 DEFINITION OF READY

- User Story defined.
- User Story dependencies identified.
- User Story sized by Delivery Team (Tasks created under user stories).
- Scrum Team accepts User Experience artefacts (wireframes and design).
- Performance criteria identified, where appropriate.
- Person who will accept the User Story is identified.
- Team has a good idea what it will mean to Demo the User Story.

4.4.2.1.8 DEFINITION OF DONE

There might be different DoD at various levels:

- DoD for a user story.
- DoD for a sprint.
- DoD for a release (e.g. writing release notes)

Story Definition of Done

- All story should have QA Pass status.
- The story should have working code supported by unit test.
- The story should have well defined acceptance criteria.
- The code should be reviewed (code review).
- Code must be completely checked in to the source control system and the build should pass with respective test cases.
- The product owner must accept the story.

Sprint Definition of Done

- Product owner should have defined a sprint goal.
- All stories completed for the sprint must be accepted by the product owner
- All the test cases should be passed for the stories in the sprint.
- All code must have gone through a code review process.
- If there is a database involved, the database scripts must have been automated and tested.

Release Definition of Done

- Product is deployed to the test box and makes it to staging.
- There are deployment documents for the release.
- Training manuals are available for users.
- All stories for the release are completed and accepted.
- The release does not have any level one bugs.

4.4.2.1.9 SCRUM CEREMONIES

- Sprint Planning
- Sprint Review
- Sprint Retrospective
- Daily Scrum meeting

Sprint Planning

- ✓ Product owner, scrum master and scrum team select items from the product backlog.
- ✓ Sprint Backlog is created.
- ✓ This is a time box meeting.

Sprint Review

- ✓ Team presents what is accomplished during the sprint. A demo will be conducted.

Sprint Retrospective

- ✓ The whole team will discuss what they will like **Start doing, Stop doing and Continue doing.**

Daily Scrum meeting

- ✓ Time boxed for 15 minutes

✓ Agenda

- What did you do yesterday?
- What will you do today?
- Is anything in your way?

Sprint Goal

- ✓ A short statement of what the work will be focused on during the sprint.

Example:

- ✓ Sprint 7 – Implement basic cart functionality including add, remove and update.
- ✓ Sprint 8- The checklist process –pay for an order, pick shipping, order gift wrapping.

4.4.2.1.10 BENEFITS OF SCRUM

- Revenue-Using Scrum, new features are developed incrementally in Sprints. At the end of each Sprint a potentially releasable Increment of Software/Product is available. This enables the product to potentially be released much earlier in the Development cycle enabling benefits to be realised earlier than otherwise may have been possible where we waited for the entire Product to be “complete” before a release.
- Quality-Maintaining quality is a key principle of development with Scrum. Testing occurs every Sprint, enabling regular inspection of the working Product as it develops. This allows the Scrum team early visibility of any quality issues and allows them to adjust if necessary.
- Transparency-Scrum development encourages active Product Owner and Stakeholder involvement throughout the Products development. Transparency is therefore much higher for key stakeholders, both of the project’s progress and of the product itself, which in turn helps to ensure that expectations are effectively managed.
- Risk Mitigation- Transparency in Scrum development helps to ensure that any necessary decisions can be taken at a suitable time, while it can still make a difference to the outcome.

4.4.2.2 PHASE-WISE DELIVERY APPROACH

Unecops will structure the CIMS engagement into three clearly defined phases, strictly aligned with the milestone framework prescribed in the RFP:-

Phase 1: Inception, Design, and Architecture (Weeks 1 to 8)	This phase will encompass detailed requirements validation through structured stakeholder workshops with Udyog Mitra and departmental representatives, finalization of the solution architecture, technology stack confirmation, data
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	architecture design, integration blueprint development for all GoI systems, and the establishment of the development and testing environment at the Bihar State Data Centre. All design artefacts including the Solution Architecture Document, Data Flow Diagrams, Interface Control Documents, and UI/UX prototypes will be formally reviewed and approved by the Department before development commences.
Phase 2: Iterative Development, Integration, and Testing (Weeks 9 through 24-> Milestone 3)	The platform will be built iteratively across the four parallel delivery tracks through sprint-based development cycles. Each module will progress through development, unit testing, system integration testing, and user acceptance testing in a structured sequence. The BI and Analytics layer will be developed in parallel with the transactional modules, with progressive data integration from the MSME MIS and HRMS modules feeding the analytics engine. By the completion of Milestone 3 at 24 weeks, all core modules will be in an advanced stage of development with integration testing substantially complete.
Phase 3: User Acceptance Testing, Pilot Deployment, and Go-Live (Weeks 25 through 35 -> Milestone 4)	This phase will encompass final UAT with departmental end users across all modules, resolution of all UAT observations, application security audit and vulnerability remediation, load testing for a minimum of 100 concurrent users, bilingual interface validation, training of departmental users and administrators, data migration and onboarding support, and formal go-live on the Bihar State Data Centre infrastructure. Milestone 4, at approximately 35 weeks from the Contract Effective Date, will mark the formal handover of the fully operational CIMS platform to Udyog Mitra.
Phase 4: Operations and Maintenance	Following go-live, Uneecops will provide comprehensive Operations and Maintenance

(Month 8 through Month 60)	support for a period of 52 months, concluding at Month 60 of the engagement. This phase will encompass application support and bug resolution, performance monitoring and tuning, database management and backup, incorporation of scheme and policy changes as directed by the Department, security patch management, periodic re-audits as required, enhancement releases as per the Department's evolving requirements, and continuous user support and training. The O&M framework will be governed by defined Service Level Agreements covering response times, resolution timelines, uptime commitments, and escalation protocols, all of them structured to ensure that the platform remains operationally robust and institutionally current throughout the full 60-month engagement.
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4.4.2.3 QUALITY ASSURANCE & TESTING STRATEGY

Uneecops's quality assurance approach for the CIMS platform will be multi-layered and continuous rather than terminal. Basically, an automated testing suite covering unit tests, regression tests, and API integration tests will be established from Sprint 1 and maintained throughout the development lifecycle, ensuring that new features do not introduce defects into previously tested functionality.: -

Shift-Left Testing	Testing activities will commence at the requirements and design stage, not at the end of development. Requirement reviews, design walkthroughs, and code inspections will be conducted as standard practice across all sprints.
Unit Testing	In this test, applications shall be divided into smaller units, and testing will be carried out on those units independently.
Smoke Testing	In this type of testing, the main functionality of the application will be tested. For e.g.: • Data flow in the application • UI design and functionalities

Integration Testing	After unit testing, the units or modules are to be integrated which gives rise to integration testing. The purpose of integration testing is to verify the functionality, performance, and reliability between the modules that are integrated.
Performance & Load Testing	The platform will be subjected to rigorous performance and load testing simulating a minimum of 100 concurrent users at peak load, prior to go-live, in accordance with the security audit requirements specified in the RFP. Basically, the purpose of performance testing is to simulate the volume of a live system and make sure that the solution responds within expectations. • Online response time for screen navigation • Online response time for processing data against databases • The time to run batch processes • Downtime for batch processing or periodic maintenance • File sizes and storage capacity
Regression Testing	Regression test verifies that the application which is already tested using other testing methods still performs correctly after it was changed or interfaced with other software. Changes may include enhancements, patches, configuration changes, etc. • This test is a very brief run-through of the functionality of a computer program, system, calculation, or other analysis, to assure that part of the system or methodology works roughly as expected. This is often prior to a more exhaustive round of testing.
Security Testing	A dedicated application security testing cycle covering SQL injection, XSS, CSRF, directory traversal, authentication vulnerabilities, buffer overflows, and all other categories enumerated in the RFP will be conducted prior to go-live and the results submitted to the Department's appointed Certified Auditor for formal compliance certification. All identified vulnerabilities will be remediated within the 15-day period mandated by the RFP.
User Acceptance Testing	It is usually the last phase of the testing process. During UAT, actual users test the application to make sure the application can handle required tasks in real-world scenarios, according to specifications. For this assignment, structured UAT cycles will be conducted with representative end users from across the Department's directorates at the district, subdivision, block, and state headquarters levels to ensure that the platform meets the operational realities of its intended users and not merely the formal requirements specification.

4.4.2.4 POST IMPLEMENTATION PLAN AND METHODOLOGY

Unecops will be responsible for providing managed application management services for the entire project duration post Go-Live of the newly developed application. During this period, we will be responsible for carrying out any maintenance activities for overall system stabilization, database maintenance, and application maintenance.

4.4.2.5 APPLICATION MAINTENANCE SUPPORT

The post implementation support model for enhancement migrations can be achieved through robust release management. Incorporation of operational changes into the IT production environments is always an area of concern. The approach being pursued is to formalize and adopt processes that enable improved acceptance of change into the production environment that includes:

Approach	Benefits
Production acceptance	Error reduction through the controlled release of software to the live environment
Production control	Proper control and safeguarding of software assets
Quality assurance	An ability to absorb high rates of change to the live systems
Release management	Smoother transitions of releases from the development activities (projects) to the Customer's business environment

4.4.2.6 RISK MANAGEMENT APPROACH

4.4.2.6.1 OVERVIEW

Unecops recognises that a project of this complexity spanning seven functional modules, multiple GoI integration points, on-premises deployment at the Bihar State Data Centre, an AI and BI layer, and a 60-month engagement horizon, carries inherent delivery risks. Unecops will establish a formal Risk Management framework from project inception, encompassing: -

- Continuous risk identification and classification across technical, integration, stakeholder, and operational dimensions
- Quantified risk assessment with defined probability and impact ratings
- Mitigation strategies and contingency plans for all high and medium-rated risks
- Fortnightly risk review as a standing agenda item in sprint reviews
- Formal escalation protocols for risks that materialise beyond defined thresholds

Key risk areas that Uneecops has proactively identified include GoI API availability and stability for integration with Udyam, TReDS, and ONDC; data migration complexity arising from legacy MSME registration records; user adoption across varying levels of digital literacy; and the evolving policy landscape requiring the platform to absorb new schemes and directives within the O&M period.

4.4.2.6.2 RISK MANAGEMENT

Uneecops will follow a comprehensive risk management strategy. In any project, risks are the event or uncertainty that may impede our ability to achieve project objectives effectively. As a common risk management practice “Internal Control” will be the means of reducing a risk rather than living with hence, the risk management will be primarily addressed by internal control. We will be identifying Inherent risks, residual risks, strategic risks, and operational risks and log them in the logbook.

4.4.2.6.3 RISK PLANNING

In this process we will discuss various approaches and agree on the most appropriate approach. We will plan and execute risk management activities for the project. An overall risk management plan for the contract will be created in collaboration with all the key stakeholders. A centralized project manager and PMO will collaborate to determine the risk tolerance that will be acceptable to the project. Tolerance will be based on boundaries such as time, scope, cost, safety and quality. The tolerance levels will guide the project and determine if a task, stage or the entire project is at risk of failure due to breach of perceived tolerance on a particular risk. Risks are best managed with joint responsibility of both the project manager and the PMO. The PMO will identify any external risks and make decisions based on the project manager’s recommendations.

4.4.2.6.4 RISK IDENTIFICATION

- **Risk Identification:** Risk identification will be one of our key activities. Our aim will be to analyze which risks affect the service and determine their attributes. This step will require all the relevant business and technical stakeholders to come together and identify risks in their areas and help formulate their attributes. As we know, risk identification is a highly scalable activity, hence a small group of stakeholders will be involved at the commencement of a problem analysis and subsequently will involve other stakeholders. Since this system implementation and subsequent support activities constitute a large project, a series of structured workshops will be addressing many potential risk categories.

- Risks will be stored in a risk log. The log will maintain the details of the risk with a description, analysis, planning and tracking & control as well as the status. The risk log will be continually updated with new risks and changes to existing risks.

Risk Category	Risk Description
High	Data migration to be high risks because of incorrect entity relationship, data duplicity and redundancy, large number of unused tables, and the same data stored in another system. Migration automation seems to be challenging. Risk mitigation action: Need data in text file format.
High	Existing document migration will be a challenging task because associated metadata/index data are incorrect. Same document type has been referred to by a different name and other metadata. Risk mitigation action: Need re-indexing of documents after the files are imported to a network folder. The task will take much more time and effort than planned.
Medium	Department staff will be only 20% available for the project. Hence the requirement specification may take much longer time than planned. Risk mitigation action: Request client's project manager to assign at least one SME full-time for next two weeks.

4.4.2.6.5 CHALLENGES AND RISKS LIKEY TO BE ENCOUNTERED ALONG WITH THEIR MITIGATION PLAN

#	Challenges	Mitigation Approach
1	Clarity in understanding requirements	▪ Engage with the Client to seek clarification ▪ Define deliverables and their acceptance criteria ▪ Define interface requirements and acceptance criteria ▪ Develop a common project charter, goals and objectives ▪ Share the project vision
2	Delay owing to Infrastructure not available	▪ Jointly develop project plans ▪ Measure and track progress ▪ Conduct regular periodic reviews
3	Interdependency of various operation work streams may lead to	During the early stages of the project, the key interface points will be defined and where possible, standard integration methods will

	cascading delays affecting other components as well	be used to ensure that there is little impact of one channel affecting the development of another channel.
4	Interface requirements with other complex systems, or other systems not under UNEECOPS control may not be understood	UNEECOPS will work with the other vendors and the authorities in ensuring a cohesive design, development, and testing approach wherein the standards will be defined, and any changes communicated through formal reporting structures.
5	Milestone Slippages	Proper interaction channels will be defined, and reporting methods agreed to ensure that any delays are detected during the early stages and handled. Other mitigation approaches will be: Use of proven estimation methodologies Multistage Estimation Validation of estimation by experts Critical path identification Joint periodic reviews with customer
6	Inadequate testing due to schedule pressure	- Testing commences early in the development process for early capture of defects - Incremental testing approach Test plan and acceptance criterion decided upfront
7	Security vulnerabilities introduced in the application	- Train coders and give them a checklist to prevent vulnerabilities being introduced in the code. - Review of code for security vulnerabilities in software Security vulnerability testing of developed code
8	Delay in review/ decision cycles	Regular monitoring of the progress through the different levels of strategic / tactical management groups will help in ensuring that such delays are kept to minimum

9	Multi-location development/ operational environment may result in insufficient communication/ delays in reacting to situations.	UNEECOPS proposes a PMO that will be closely engaged with the different locations and will have the capability to handle situations remotely where possible.
10	Risk of Key persons leaving or transferred from the client or the development team.	UNEECOPS takes pride in established processes and document controls, which minimizes the impact of key persons leaving a project. In addition, we will work with Client to ensure that appropriate measures and controls are in place to minimize our dependency on key personnel across organizations.
11	Continuous Data Capture & upload	Extended Application support to capture the data aggregation, removal of data duplication and correction at each workflow pitstop level
12	Effective usage of the software	Periodical status meeting would be conducted with Education department Officers to identify the issues and their resolution. An indicative communication plan has been proposed under section "Communication & Reporting".
13	Change Request & Acceptance	Sensitization of Leadership among stakeholders at each level
14	Software Awareness among the Different Stakeholders	Need to conduct Workshop & Video Conferencing
15	Delays in getting approval from department or relevant stakeholders for signoff should be provided within agreed timeline delays may impact the schedule SRS, UAT, Integrations	All the milestones will be captured and highlighted in the status report & Followed up

Some Portal Application Specific Risks

1	Connectivity in Remote Districts	Offline-capable mobile app with sync, data compression, SMS fallback, edge caching at district level
3	Bilingual Complexity (English-Hindi)	Unicode fonts, professional translation, parallel editing CMS, linguistic QA process
3	MSME Digital Literacy Variations	Multi-tier training, Hindi video tutorials, , district champions (T-O-T model)
4	Festival Calendar Impact	Training calendar aligned with Patna festivals, flexible rescheduling, self-paced modules
5	High Concurrent User Load	Load testing at 5x expected users, auto-scaling infrastructure, CDN for static content, Redis caching
6	Coordination challenges with department, line departments, MSMEs, and external agencies	• Dedicated stakeholder management plan • Weekly coordination meetings with structured agenda • Single Point of Contact (SPOC) for each stakeholder group • Escalation matrix for unresolved coordination issues
7	Data Migration Issues: Legacy data quality issues causing migration failures or data loss	• Comprehensive data profiling and quality assessment • Data cleansing and validation scripts before migration • Phased migration approach with rollback capability • Parallel run period for data verification • Full backup retention for 90 days post-migration

4.4.2.6.6 QUALITATIVE & QUANTITATIVE RISK ANALYSIS

- We will conduct qualitative risk analysis for prioritizing risks for subsequent analysis or action based upon the probability of their occurrence and their impact on the service provision. This will be based on our past experience, historical data of the managed service and/or the customer perception of the risk. Quantitative analysis is a process where we will be numerically analyzing the effect on overall project objectives of identified risks. The objective is to determine the quantitative impact measured against the elements of cost, schedule, scope, quality, benefit and resources. This input will be crucial to understanding how to respond to a risk.

- Once the risks identified and clearly defined each risk item will be then assessed (scored) in terms of its likelihood of occurrence (probability) and its potential impact on the project if it occurs (importance/ consequences) in accordance with the matrix and scoring definitions.

4.4.2.6.7 RISK REPORTING & PLANNING

This process will help us develop various options and actions to enhance opportunities and to reduce threats to the objectives of the service. In order to develop a response strategy several questions will be discussed. Some of the following questions will be considered:

- Can we live with this risk?
- Can we do anything to mitigate or avoid the risk within a reasonable budget and timeframe? If yes, what would be the measurable goals so we can tell we are done mitigating the risk?
- Would it be just as effective to deal with the risk if and when it becomes an issue?

After the various response options are identified and evaluated, a selection process will occur to determine the response to be put in place for that particular risk. The selection will be based on the balance between the cost of the action and the cost/ impact of the risk occurring and impacting the project. A standard response strategy will be chosen from the following:

- Avoid: Eliminate a specific threat by eliminating its cause, usually through a change in project strategy.
- Mitigate: Develop and execute a specific action plan to reduce the expected impact or probability of occurrence of a risk event.
- Accept with Contingency Plan: Active acceptance, which will include developing a contingency plan which will be executed should the risk event occur.
- Accept without Contingency Plan: Passive acceptance, ignore the risk until the risk event occurs and then action plan.
- Transfer: Deflect the risk by transferring it to another party. Transferring a risk will not eliminate the risk but rather convert it to a different risk, which should be easier to manage.
- Monitor: If not currently possible, cost effective, or desirable to take action, watch this risk and select one of the other strategies should a change in the situation warrant it.

Following the selection process, planning and resourcing is required to implement the response actions that are selected. This involves identifying the type of resources required, creating a plan of action and gain approval from management. The relevant stage and project plans are updated to reflect the changes.

4.4.2.6.8 RISK MONITORING & CONTROL

This will be the process of tracking identified risks, monitoring residual risks, identifying new risks, executing risk response plans, and evaluating their effectiveness throughout the service delivery life cycle. We will effectively identify, plan for mitigation, take action as planned, control/mitigate all risks associated with the

project. A team of experts (risk management professionals) will oversee the risk management activities and ensure the coverage of identification and mitigation plan.

4.4.2.7 CAPACITY BUILDING PLAN

Unecops understands that technology deployment in a public sector context succeeds or fails on the strength of its adoption not merely its technical correctness. Accordingly, a structured Capacity Building programme will be embedded into the project delivery plan, covering:

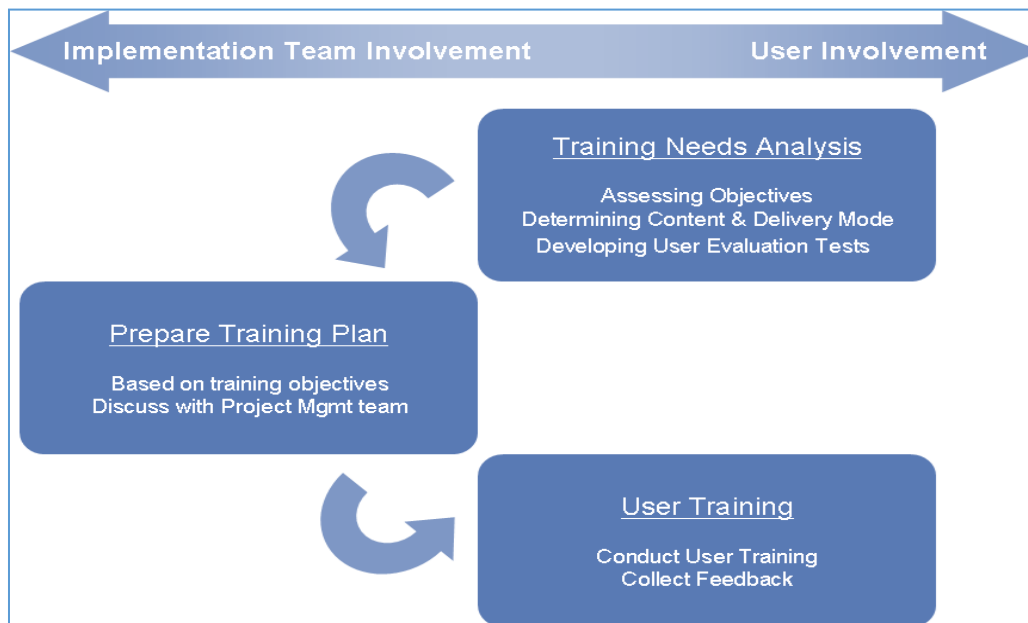
- **Stakeholder Engagement:** Regular structured interactions with Udyog Mitra leadership, departmental heads, and district-level officials throughout the development lifecycle, ensuring alignment, managing expectations, and building institutional ownership of the platform.
- **End-User Training:** Role-based training programmes for all categories of users namely administrators, data entry operators, grievance officials, HR managers, and senior analytical users delivered in bilingual format (Hindi and English) through both classroom and digital channels.
- **Administrator and Super-Administrator Training:** Dedicated technical training for the Department's designated system administrators, covering platform configuration, user management, report creation, data governance procedures, and first-level troubleshooting.
- **Knowledge Transfer:** Comprehensive documentation including System Administration Manuals, User Guides, API Documentation, and Data Dictionary will be delivered in both digital and print formats, ensuring that the Department possesses full institutional knowledge of the platform independent of Unecops's ongoing support.

4.4.2.8 TRAINING MANAGEMENT METHODOLOGY

Keeping in view the diversity of the project with respect to its requirements, there are significant differences in capacities of client to undertake such initiatives. The two important components of capacity building are: -

- Administrator & Operational Level Training
- End User training

We will follow a three-step training model for both the training as shown in the figure: -



Training Effectiveness Evaluation-

- Uneecops shall evaluate the effectiveness of all end users' trainings using feedback form.
- Uneecops shall be responsible for analyzing the feedback and arrange for conducting refresher training, wherever needed. Issue modified or new document -Uneecops's project manager will:

Issue new or modified documents- -Uneecops's project manager will:

- Issue new or modified documents, proposal amendment etc.
- Ensure recipients acknowledge receipt of the modified or new documents
- Where practical, ensure superseded documents are withdrawn from use
- Advise quality consultant to execute the change

Note: Once the change is approved, it will form a part of the overall project scope.

Roles & Responsibilities Matrix -

- **Uneecops**
 - Uneecops will appoint a training supervisor/training lead
 - Uneecops will develop courseware
 - Uneecops will conduct the training at the training venue of the client
 - Uneecops will prepare the training calendar, course material and training schedule
 - Uneecops will communicate to the client in writing about the training venue, training calendar and the batches

- **Udyog Mitra, Bihar**

- The client should identify qualified trainees with relevant IT experience and training competency within each department who will be directly trained by Uneecops and will be responsible for interfacing with Uneecops for all the training initiatives.
- The client will give written go ahead for the training schedule.
- The client project coordinators (Trainees) work closely with Uneecops to ensure that the identified users attend the training on schedule and training sessions take place without disruption to the training calendar.
- The client will provide reviews and signoffs within a mutually agreed timeframe.

Tier 1: Train-the-Trainer (3T-TTT) Program

- **Audience:** 10-15 departmental trainers from the Udyog Mitra department and line departments
- **Duration:** 3 days intensive training
- **Content:**
 - Complete system functionality (all modules)
 - Training delivery techniques
 - Troubleshooting common user issues
 - Assessment and certification methods
- **Deliverables:**
 - TTT certification upon successful completion
 - Trainer toolkit (presentations, exercises, quick reference guides)
 - Monthly refresher webinars for trainers
- **Outcome:** Internal champions capable of conducting independent training sessions

Tier 2: Administrator Training

- **Audience:** System administrators from Udyog Mitra, Bihar (5-8 personnel)
- **Duration:** 2 days hands-on training
- **Content:**
 - Backend functionalities and configurations
 - User and role management
 - Master data management
 - Report generation and MIS dashboards
 - Troubleshooting and issue resolution
 - Security and access control
- **Deliverables:**
 - Administrator manual (English + Hindi)
 - Video tutorials for each admin function (15+ modules)
- **Outcome:** Self-sufficient administrators for day-to-day portal management

Tier 3: MSME End-User Training

- **Audience:** MSME owners and representatives

- **Duration:** Practical sessions in batches of 10-15 (identified by department)
- **Content:**
 - MSME MIS
 - Human Resource Management System (HRMS)
 - Training Management Information System
 - Grievance Redressal System
 - Integrated Job Portal with AI-Powered Resume Builder
 - Business Intelligence and Analytics Platform
 - Dashboard and reports interpretation
- **Deliverables:**
 - User manual (English + Hindi) - printed + digital
 - Video walkthrough tutorials in English & Hindi
 - Quick reference cards (laminated)
- **Outcome:** Confident MSME users driving portal adoption and utilization

Tier 4: Government Officials Training

- **Audience:** Officials from line departments, banks, and nodal agencies (200+ officials)
- **Duration:** 2 days classroom training
- **Content:**
 - MIS dashboard usage and interpretation
 - Scheme management and approval workflows
 - Application review and processing
 - Report generation for decision-making
 - Monitoring and evaluation tools
- **Deliverables:**
 - Official user guide with role-specific modules
 - Dashboard interpretation handbook
 - Quarterly refresher sessions
- **Outcome:** Efficient scheme processing and data-driven decision-making

Patna-Specific Training Adaptations

Bilingual Training Materials:

- All documentation in English + Hindi (Unicode)
- Video tutorials with Hindi voiceover and subtitles
- Training presentations in dual language

Local Context Integration:

- Case studies featuring successful Bihar MSMEs
- Examples from local industries

Cultural Sensitivity:

- Training schedules aligned with regional festival calendar
- Avoidance of major local festivals

- Training timings accommodating local working hours

Training Delivery Modes

Classroom Training:

- Face-to-face sessions at Udyog Mitra, Bihar Headquarters in Patna only
- Interactive demonstrations with live system
- Q&A sessions with trainers

Virtual Training:

- Microsoft Teams/Zoom sessions for remote participants
- Screen sharing for step-by-step guidance
- Recorded sessions for later reference

Hybrid Model:

- Blend of in-person and online for maximum reach
- Synchronous online participants alongside classroom
- Same content delivered across both modes

On-Demand Learning:

- Video tutorial library accessible 24×7

Training Effectiveness Measurement

Pre-Post Assessment:

- Pre-training assessment (baseline knowledge)
- Post-training assessment (knowledge gain)
- Improvement percentage calculation
- Target: Minimum 70% improvement per trainee

Attendance Tracking:

- Excel based tracking
- Target: >85% attendance in scheduled sessions

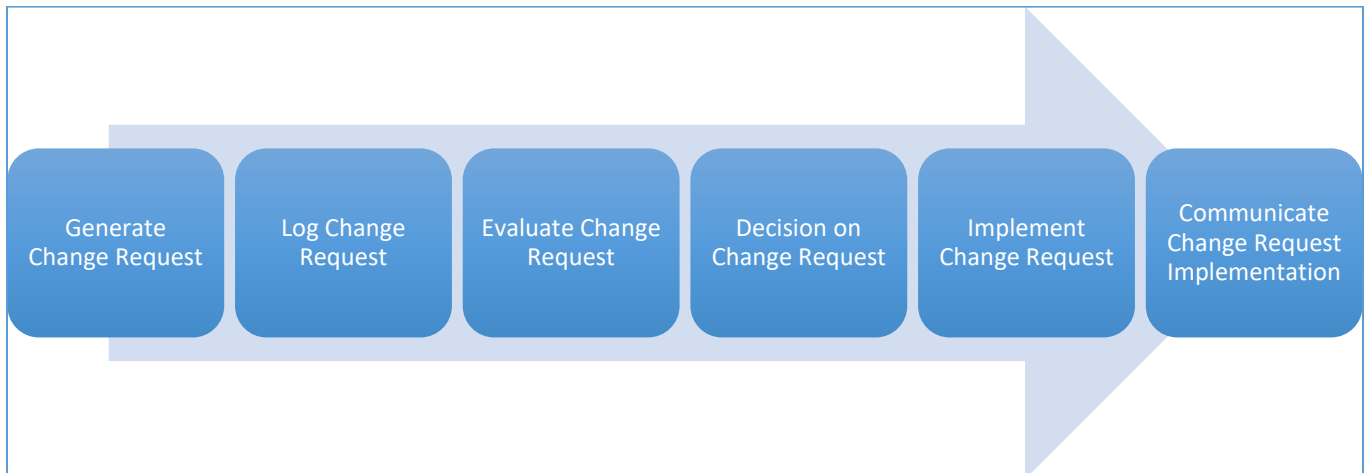
4.4.2.9 CHANGE MANAGEMENT MECHANISM (METHODOLOGY)

4.4.2.9.1 PURPOSE

- The Change Management process establishes an orderly and effective procedure for tracking the submission, review, evaluation, approval and implementation of all changes in a project.

4.4.2.9.2 CHANGE MANAGEMENT PROCESS

- For each requested change, following process will be followed.



S.No.	Step	Description
1.	Generation	Team member receives a change request (CR) details from customer or internal team. The change request form with the available description of required change will be filled by requestor and sent to PM.
2.	Logging	Project Manager logs the CR details in change register/PMS tool
3.	Evaluation	The CR would be analyzed and evaluated by Change Control Board (CCB) in the following aspects: • Items or functionality of website/application which will be affected by the proposed change (i.e. impact analysis form should be filled) • Efforts involved (i.e. estimation sheet to be created for CR work) • Impact on delivery schedule (i.e. project time line change) • Priority of change
4.	Decision	• Approval or Rejection of CR by CCB. • Based on the impact analysis and negotiation with the stakeholders, CCB decides whether to approve or reject the change request. • In case CR gets APPROVED, CCB will notify the PM to incorporate the change(s). • In case CR gets REJECTED, CCB will notify the PM to communicate with relevant stakeholders and close the CR.

5.	Implementation	- On CR approval, project team will incorporate the requested change in the website/application. Customer acceptance will be taken on implemented change. - Project plan, SRS document, Requirements Traceability Matrix and User Manual (if required) would also be updated for the change. - Change register/PMS tool should be updated with implementation status of respective CR.
6.	Communication	Relevant stakeholders will be notified by PM, about the implemented change(s)

4.4.2.9.3 ROLES & RESPONSIBILITY MATRIX

Responsibility	Action	Description
Team Lead	Receive Change Request	Team Lead receive change request (CR) from user or internal team members in change request form.
Change Control Board (CCB)	Analyze Change Request	Change Control board analyses the impact of change requests for: - Items or functionality affected by the proposed change - Efforts involved - Effect on Delivery Schedule - Priority of change
Change Control Board	Communicate Changes	Change Control Board communicate and negotiate changes with the relevant stakeholders. Also, CCB identifies the risks related to implementation of change request
Change Control Board	Accept or Reject Change Request	Based on the impact analysis and negotiation with the stakeholders, CCB decides whether to approve or reject the change request.

Responsibility	Action	Description
		In case the change request is rejected, CCB send the notification to the stakeholder and closes the change request In case Change Request is approved, CCB notify the team member to execute the change(s).
Configuration Controller	Hand Over Configuration Items(s)	Hand over the item(s) to the team as per Change Request Form for implementation
Team	Implement Change Request	Development and testing team: • Implements necessary changes to the relevant product/product documentation • Perform reviews and tests to ensure that changes have not caused any unintended effects on the already tested and released baseline • Update Traceability Matrix for the changes • Seeking signoff from the customer/end user
Configuration Controller	Track Status of Change Request	CC updates configuration status report for new released item (s) and update the baselines. CC Update PMWB for status of change requests and track the change request till closure

4.4.2.10 CMMI LEVEL-5 CERTIFIED PROJECT MANAGEMENT PRACTICES

Unecops technologies will adopt an integrated project management framework consisting of a set of four core processes and four key enablers. The core processes of this framework are:

- Strategy management
- Quality management
- Solution delivery
- Service provision

And the key enablers are:

- Relationship management

- Resource management
- Cost management
- Performance management

Our project management approach will cover all activities related to project execution as outlined below for successful execution of the project:-

- Appointment of a project manager
- Supplying technical documentation on the application architecture
- Risk management and mitigation
- Adherence to quality guidelines and necessary control
- People management
- Time schedule adherence
- Project commissioning and support
- Proper documentation at all milestones and at all levels
- Communication management
- Change management
- Project reporting

And above all, always ensure satisfaction and quality levels. We believe that project management is one of the most critical skills required for successful execution of projects. The project managers have been trained formally in our quality management system and project management system. Uneecops formal quality framework mandates the project managers and project leads should have the following skills: -

Integration management	Risk management
Scope management	Cost management
Time management	Team management
Schedule management	Communications management

Besides this, certain global activities on the project pertain to project organization and management, risk management, change management, configuration management and quality assurance and other technical and operational processes will be done in all phases of the project. This will be our overall responsibility. The project management strategy is based on clear structuring of the project work into phases and activities consistent with the methodology leading to appropriate support and deliverables as scheduled. There are some activities common to each of the phases and will be performed for each phase. These are listed below: -

- The project management and development methodology shared with the client will always have to be available to the project teams.
- The organization's institutionalized and managed processes and policy assets will be made available to the project team.

- We will customize the processes for the project and will incorporate any specific process as required by the client. Details of this will have to be provided by the client during the project initiation phase.
- The process deviations will be recorded in the project's defined software process document. The project manager along with the client's project head will approve this document. The approved document is made available to the project team for adherence.

4.4.2.10.1 PROJECT PLANNING & PHASING

Project planning is a crucial element of project management, and this will be taken up as the very first activity in this project. Project planning involves the following activities for onsite, offsite and offshore needs of the project:

- Sizing
- Effort Estimation
- Staffing
- Scheduling

Client and Uneecops will jointly discuss the project roadmap and priorities the items to be resolved. This will be the basis of drawing out the project plan. Both teams will jointly own the project plan and the respective project managers will monitor the actual progress of their teams as per this plan. Uneecops extensively uses project management tools which will be standardized as per the requirements in the quality management system. The tool will have a dashboard view of the progress of the project.

All project stages (initiation, planning, monitoring, and control etc.) will be planned. Plans will be revisited every month or whenever a major change occurs. The software development plan will include:

- Major activities for each phase of the project.
- List of activities and sub-activities for each module, listing the resources applied for the activities.
- Schedule of activities for monitoring and control.
- Quality control checkpoints

All plans will be baseline and versioned for subsequent monitoring and control. A new baseline will be generated at every change to the project's critical path or a major change in plan. Detailed Work Breakdown Structures (WBS) will be developed and the same will be assigned to the module/team leaders and team members for completion.

4.4.2.10.2 PROJECT ORGANIZATION

Project organization is the way a project has to be delivered. A very effective organization is crucial to the successful delivery of projects on time, to budget and to specification and for an effective project organization, we will be:

- Identifying all key roles and in particular defining their responsibilities
- Defining crystal clear terms of reference and accountabilities for all key roles and bodies; e.g., sponsor; steering committee etc.

- Defining clear supplier/partner/customer interfaces, at all levels, and their specific responsibilities
- Defining 'ways of working' of the team, describing how the team will work with key partners, supplier and the customer(s).

4.4.2.10.3 PROJECT STEERING COMMITTEE

Unecops proposes to set up a project steering committee comprising of the project manager from us and the project management team identified by the customer, which will be known as Project Steering Committee (PSC). The PSC will meet once in 15 days (or more frequently when deemed necessary) to monitor the progress of the project and take necessary actions. All these meetings will be minute, action points assigned and tracked to closure for the successful execution of the project.

4.4.2.10.4 PROJECT EXECUTION GROUP

This will include the project leaders from Unecops and the representatives from the respective departments identified by the client. They will be responsible for the successful execution of the project and the day-to-day running of the project. The primary responsibility of this group is to ensure that development, implementation, resolution of problems and escalation of any issues will be done to the execution committee. This group would communicate amongst itself to facilitate clarifications and to keep track of the progress of the project during development phase. The communication could be in the form of meetings, teleconferences or over e-mail. All doubts and clarifications would be logged and maintained on a day-to-day basis in the issues' register. This would be shared with the execution committee on a regular basis.

4.4.2.10.5 STATUS REPORTING

Our team will prepare/submit a fortnightly report to the project coordinator of the client. The report will cover the items of exception that require action. Our project manager will monitor the progress of the activities against the plan and prepare a consolidated monthly report to project manager of the client. This report will cover the project status and deviations against plan. The status report will contain: -

- Activities completed since last report
- Activities planned until the next report
- Progress against plan
- QA status
- Deliverables to the client since last report
- Deliverables due to the client in the next reporting period
- Outstanding issues

There will be regular status meetings within us and the client teams at various levels to discuss the status and issues. Our project manager will review the status of the project on a regular basis and compare it with the plan. Any deviations will be highlighted, and appropriate corrective action will be initiated.

4.4.2.10.6 CONFIGURATION MANAGEMENT STRATEGY

One of the biggest advantages of using ITIL best practice that organizations can leverage is the standard step of documenting all the configuration items that are being used. Configuration Management Data Base (CMDB) can be maintained and changes to CMDB and any incidents due to such changes can be tracked. CMDB indicates the latest configurable items and change management can help in doing impact analysis and decision management with respect to configuration items.

Configuration management takes care of the following:

- Planning – Deciding what level of configuration is required and how it will be achieved.
- Identification – Identifying and uniquely naming all assets.
- Controlling – Establishing and operating the mechanisms for identifying products when complete configuration will be frozen so that further changes are made with approval.
- Reporting – Recording and reporting all current and historical information concerned with each work product.
- Reviewing – Reviewing or auditing to ensure there is conformity between documented baselines and actual business environment.

4.4.2.10.7 PROJECT COMMUNICATION PLAN

As a process, fortnightly communication with the client is mandatory. This could be in the form of a conference call/visit/meeting or e-mail. The communication processes and documents to be used for communication, that is, status reports, minutes of meeting, and other formats would be frozen after mutual discussion. In case of specific requirements from the client, relevant processes would be d Progress review meetings shall be scheduled by the project managers (Unecops and the client) and shall be attended by the appropriate team members. A record of the minutes of each meeting shall be prepared and signed off.

4.4.2.10.8 PROJECT STATUS REVIEW & REPORTS

Progress review meetings shall be scheduled by the project managers (Unecops and the client) and shall be attended by the appropriate team members. A record of the minutes of each meeting shall be prepared and signed off. Suggested Progress Review Meeting Agenda:

Sl. No.	Title	Description
1	Meeting minutes	Review minutes from previous meeting with comments
2	Open action items	Review all outstanding action items
3	Progress review	Review with participating Project Managers regarding most recent project schedule. Update Schedule and develop action plan

Sl. No.	Title	Description
4	Technical discussion	Discuss technical issues
5	Action items	Assign responsibilities for new action items
6	Administrative matters	Discuss administrative matters
7	Next meeting	Schedule place and time for next meeting

All deviations/changes to the scope and other facets are documented. Changes, which have a financial, technical or delivery impact, are diagnosed further and impact analysis is done. Client approval is obtained before carrying out the changes. There are structured formats available. This ensures that the client is not surprised at the 11th hour and interaction is smooth. Every endeavor is made to ensure that the client is kept informed of status and difficulties, if any, in the project.

4.4.2.11 BACKUP METHODOLOGY

Tentative backup methodology:

- Backup of data at datacenter – Local Copy
- Installation and configuration of agent on servers
- Backup schedule will be finalized in consultation with the client.

4.4.2.12 SECURITY MODEL

The proposed solution will be integrated within its security features to identify the vulnerabilities of system at various levels and also protect the system from tampering or modifications whether intentional or unintentional by authorized/unauthorized users. The application will provide security through access control, audit trails, authentication, and authorization at applications as well as database and network level. All applications would go through the security/access control layer which determines what applications and resources that user can access, and the type of access rights read/write/approve etc. Security framework would provide a framework for efficiently securing all business functionality.

- **Authentication Dual Authentication:** Application will support strong user authentication. Password based authentication will ensure the passwords are secure, sufficiently complex and are changed on a regular basis. Authentication module would be pluggable and configurable for different types of authentication mechanisms.
- **Authorization:** Authorization is the function of specifying access rights to resources. Access control uses authentication to verify the identity of user. Solution will provide access control at application as well as

database level. When a user tries to access a resource, the access control process of the application will check whether that user has been authorized to use the resource.

- **Role based Authorization:** In this mode of authorization roles of the logged-in user are matched against the roles of accessed resource. Different actors involved in any application will be segregated according to their ROLE and access will be provided based on the ROLE assigned to a particular actor. Actors having the same ROLE will be categorized in a GROUP and a group can have multiple actors but only with the same ROLE. This will provide a logical grouping of different actors on the basis of their role.
- **Session Management:** The proposed solution will maintain the system state securely. The system would maintain state in database, memory, file, and registry or within client. Stateless protocols like HTTP provide no means for the server to maintain user state. The server will deploy a session a server-side mechanism to provide continuous session-like experience to users. When a user sends his first request, the server will generate a unique session identifier (SID) and send it to the client. The client will then send the SID back along with each subsequent request. The SID will function as an identification token for users, and servers will use the SID to maintain session data.

4.4.2.13 EXIT MANAGEMENT ACTIVITIES

- Knowledge Management is an integral part of our mandated scope of work, and we shall ensure that the application is handed over in a totally operational condition to the satisfaction of the department.
- Uneecops will hand-over the latest version of the source code to the departmental authorities
- An exit management plan shall be prepared in consultation with the department to ensure continuity of services and provided 1 month prior to the termination of the contract.
- Uneecops will undertake knowledge transfer to the department or its nominated agency/vendor including planning documents, bill of materials, technical specifications of all equipment, user manuals, guides, network architecture, change requests if any reports, documents, and other relevant items
- Uneecops shall handover complete source code along with all previous versions of the code and the credentials of production environment.
- Uneecops shall provide necessary handholding and transition support to department staff or its nominated agency
 - This will include conducting detailed walkthroughs and demonstrations
 - Handing over all documentation (e.g., BRD, FRS, SRS, HLD, LLD, etc.) including up to date policies, procedures, asset registers, configuration documents etc. and all such documentation is handed over to the department.
 - Addressing the queries/clarifications of the new agency with respect to the working / performance levels of the infrastructure
 - Conducting Knowledge Transfer sessions consisting of structured knowledge transfer covering system architecture, source code walkthrough, deployment and rollback procedures, backup and DR processes.
 - Allow restricted yet reasonable access to some of our team members from implementation &/or O&M phase to understand the methods of delivery of the services employed by the bidder and to obtain appropriate knowledge transfer

- These sessions will focus on specific IT concerns related to database tasks and activities and configuration management to the future vendor representative /department officials.
- It is expected that the department shall facilitate ensuring that the incoming vendor representatives & department officials co-operate and work alongside UTL to gain familiarity with the database, tools, processes & organizational assets to ensure smooth transitioning.
- It is expected that the PM of incoming vendor, UTL PM and designated department nodal person meet-up 10 days prior to transition completion to determine if any further training or knowledge transfer is required.
- It is expected that the department will provide regular signoffs on transition completion progress report to UTL. The established transition checklist will be utilized for this purpose.

5 SOLUTION PROPOSITION, TECHNICAL AND BUSINESS REFERENCE ARCHITECTURE FRAMEWORK

5.1 SOLUTION PROPOSITION- VISION STATEMENT

Unecops Technologies Limited proposes to design, develop, and deploy a unified, intelligent, and institutionally scalable **Centralized Integrated Management System (CIMS) and Business Intelligence Platform** for Udyog Mitra, Department of Industries, Government of Bihar a platform that will serve as the definitive digital governance backbone for Bihar's MSME ecosystem for the full duration of the 60-month engagement and well beyond.

The proposed CIMS is not a collection of independently developed software modules assembled under a common interface. It is conceived and will be architected as a **natively integrated, data-coherent platform** in which every module from MSME MIS to HRMS, from Grievance Redressal to the AI-Powered Knowledge and Intelligence layer shares a common data fabric, a unified access and identity framework, and a single analytical engine. This architectural philosophy ensures that the platform delivers not merely transactional efficiency but genuine institutional intelligence the ability for the Department to see, understand, and act on its MSME ecosystem in real time.

5.2 SOLUTION OVERVIEW

The CIMS platform as proposed by Unecops Technologies will comprise seven functionally distinct but architecturally integrated solution modules, underpinned by a shared infrastructure layer, a cross-cutting security and access management framework, and a Business Intelligence and Analytics engine that aggregates and analyses data from all modules to deliver a unified intelligence view to decision-makers at every level of the Department's administrative hierarchy. The seven modules are as follows: -

5.2.1 MODULE-1: MSME MIS

- The centralized enterprise data platform for Bihar's MSME ecosystem, covering registration and onboarding, scheme management, DIC profiling, market linkage, e-commerce integration, GIS mapping, compliance monitoring, financial tracking, and AI-driven MSME performance analysis.

5.2.2 MODULE-2: HUMAN RESOURCE MANAGEMENT SYSTEM (HRMS)

- A full-lifecycle HR governance platform for the employees of the Department of Industries across all its directorates, covering recruitment, onboarding, attendance (with biometric integration), leave management, payroll, performance appraisal, tour management, asset management, and retirement processing.

5.2.3 MODULE-3: TRAINING MANAGEMENT INFORMATION SYSTEM (TMIS)

- An end-to-end training governance platform covering trainer management, course creation, learner registration and enrolment, attendance tracking, assessment, certification, training analytics, and AI-driven content generation linked directly to the MSME Knowledge and Intelligence layer.

5.2.4 MODULE-4: GRIEVANCE REDRESSAL SYSTEM

- A multi-channel, escalation-driven grievance management platform covering grievance intake (Email, SMS, Chat, Call), routing, escalation matrix management, turnaround time tracking, resolution recording, feedback capture, and integration with ODR platforms and TReDS.

5.2.5 INTEGRATED JOB PORTAL WITH AI-POWERED RESUME BUILDER

- A structured employment facilitation platform connecting Bihar's job seekers with MSME sector employment opportunities, featuring AI-based resume building, MSME employer onboarding, job listing and application management, and departmental job visibility.

5.2.6 BUSINESS INTELLIGENCE AND ANALYTICS PLATFORM

- A comprehensive, on-premises Business Intelligence and Analytics engine delivering interactive dashboards, ad-hoc query capability, KPI scorecards, geospatial visualizations, predictive analytics, automated insight generation, and AI-generated gap analysis reports aggregating data from all transactional modules and external GoI data sources.

5.2.7 DEVOPS, INFRASTRUCTURE & SECURITY

- The cross-cutting technical discipline governing the platform's development pipeline, deployment infrastructure at the Bihar State Data Centre, application security posture, monitoring and alerting, performance management, and the 52-month O&M service framework.

5.3 BUSINESS REFERENCE ARCHITECTURE

5.3.1 OVERVIEW

The Business Reference Architecture defines how the seven solution modules relate to one another at the level of business process, data exchange, and stakeholder service independent of the underlying technology. It establishes the logical structure of the platform as experienced by its users and as governed by the Department. The architecture is organized around three concentric service rings: -

5.3.1.1 RING-1: CITIZEN AND ENTERPRISE SERVICE LAYER

This is the outermost ring the interface through which Bihar's MSME entrepreneurs, job seekers, and citizens interact with the platform. It encompasses the MSME self-registration and onboarding journey, scheme discovery and application, the Job Portal, the Grievance filing interface, the Training and Capacity Building portal, and the e-commerce and market linkage interface. All citizen and enterprise touchpoints will be accessible through both the web portal and the mobile application (iOS and Android) and will be available in Hindi and English. This ring is designed for the broadest possible audience and prioritizes accessibility, simplicity, and bilingual usability.

5.3.1.2 RING-2: DEPARTMENTAL OPERATIONS LAYER

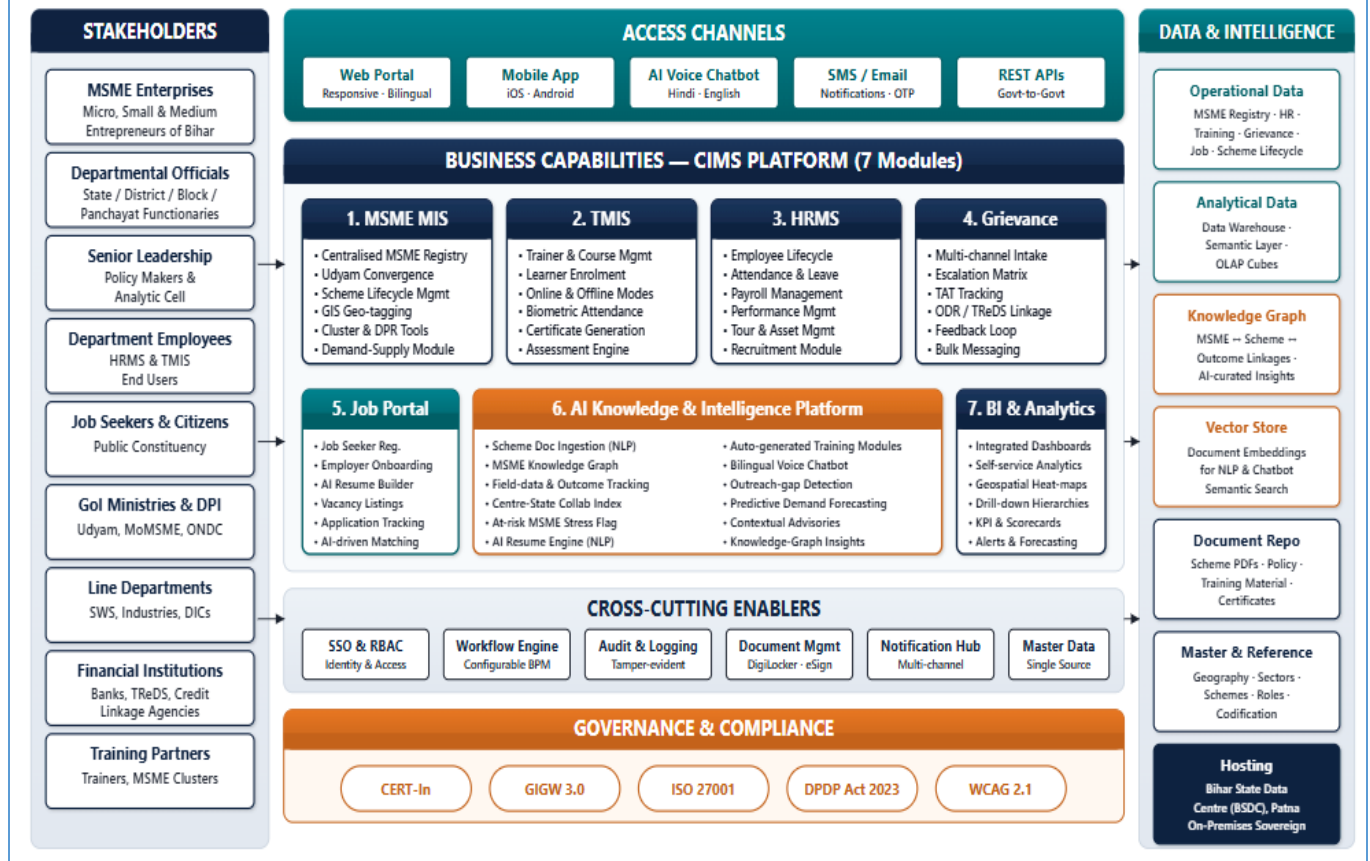
This is the middle ring; the operational workspace of departmental officials across all levels of the Department's administrative hierarchy. It encompasses the MSME MIS operational dashboard, scheme management and beneficiary tracking, HRMS administration, grievance processing and escalation, TMIS administration, GIS-based field monitoring, compliance tracking, and district and block-level MIS. This ring is designed for trained departmental users and provides the workflow, approval, escalation, and reporting tools required for day-to-day governance operations.

5.3.1.3 RING-3: INTELLIGENCE AND POLICY LAYER

This is the innermost ring which houses the analytical and decision-support environment for senior officials, policy makers, and the Analytic Cell established at the Department's headquarters. It encompasses the Business Intelligence and Analytics platform, the AI-Powered MSME Knowledge Graph, predictive analytics and demand forecasting, scheme effectiveness measurement, Centre-State programme alignment reporting, and AI-generated gap analysis. This ring transforms the raw operational data generated in Rings 1 and 2 into structured, actionable intelligence that informs policy decisions, scheme design, and programme management at the highest levels of the Department.

Business Reference Architecture

Bihar CIMS & BI Platform — Stakeholders, Channels, Capabilities, Outcomes



Business Reference Architecture

Bihar CIMS & BI Platform — Stakeholders, Channels, Capabilities, Outcomes

STAKEHOLDERS

MSME Enterprises

Micro, Small & Medium
Entrepreneurs of Bihar

Departmental Officials

State / District / Block /
Panchayat Functionaries

Senior Leadership

Policy Makers &
Analytic Cell

Department Employees

HRMS & TMIS
End Users

Job Seekers & Citizens

Public Constituency

GoI Ministries & DPI

Udyam, MoMSME, ONDC

Line Departments

SWS, Industries, DICs

Financial Institutions

Banks, TReDS, Credit
Linkage Agencies

Training Partners

Trainers, MSME Clusters

ACCESS CHANNELS

Web Portal

Responsive · Bilingual

Mobile App

iOS · Android

AI Voice Chatbot

Hindi · English

SMS / Email

Notifications · OTP

REST APIs

Govt-to-Govt

BUSINESS CAPABILITIES — CIMS PLATFORM (7 Modules)

1. MSME MIS

- Centralised MSME Registry
- Udyam Convergence
- Scheme Lifecycle Mgmt
- GIS Geo-tagging
- Cluster & DPR Tools
- Demand-Supply Module

2. TMIS

- Trainer & Course Mgmt
- Learner Enrolment
- Online & Offline Modes
- Biometric Attendance
- Certificate Generation
- Assessment Engine

3. HRMS

- Employee Lifecycle
- Attendance & Leave
- Payroll Management
- Performance Mgmt
- Tour & Asset Mgmt
- Recruitment Module

4. Grievance

- Multi-channel Intake
- Escalation Matrix
- TAT Tracking
- ODR / TReDS Linkage
- Feedback Loop
- Bulk Messaging

5. Job Portal

- Job Seeker Reg.
- Employer Onboarding
- AI Resume Builder
- Vacancy Listings
- Application Tracking
- AI-driven Matching

6. AI Knowledge & Intelligence Platform

- Scheme Doc Ingestion (NLP)
- MSME Knowledge Graph
- Field-data & Outcome Tracking
- Centre-State Collab Index
- At-risk MSME Stress Flag
- AI Resume Engine (NLP)
- Auto-generated Training Modules
- Bilingual Voice Chatbot
- Outreach-gap Detection
- Predictive Demand Forecasting
- Contextual Advisories
- Knowledge-Graph Insights

7. BI & Analytics

- Integrated Dashboards
- Self-service Analytics
- Geospatial Heat-maps
- Drill-down Hierarchies
- KPI & Scorecards
- Alerts & Forecasting

CROSS-CUTTING ENABLERS

SSO & RBAC

Identity & Access

Workflow Engine

Configurable BPM

Audit & Logging

Tamper-evident

Document Mgmt

DigiLocker · eSign

Notification Hub

Multi-channel

Master Data

Single Source

GOVERNANCE & COMPLIANCE

CERT-In

GIGW 3.0

ISO 27001

DPDP Act 2023

WCAG 2.1

DATA & INTELLIGENCE

Operational Data

MSME Registry · HR ·
Training · Grievance ·
Job · Scheme Lifecycle

Analytical Data

Data Warehouse ·
Semantic Layer ·
OLAP Cubes

Knowledge Graph

MSME ↔ Scheme ↔
Outcome Linkages ·
AI-curated Insights

Vector Store

Document Embeddings
for NLP & Chatbot
Semantic Search

Document Repo

Scheme PDFs · Policy ·
Training Material ·
Certificates

Master & Reference

Geography · Sectors ·
Schemes · Roles ·
Codification

Hosting

Bihar State Data
Centre (BSDC), Patna
On-Premises Sovereign

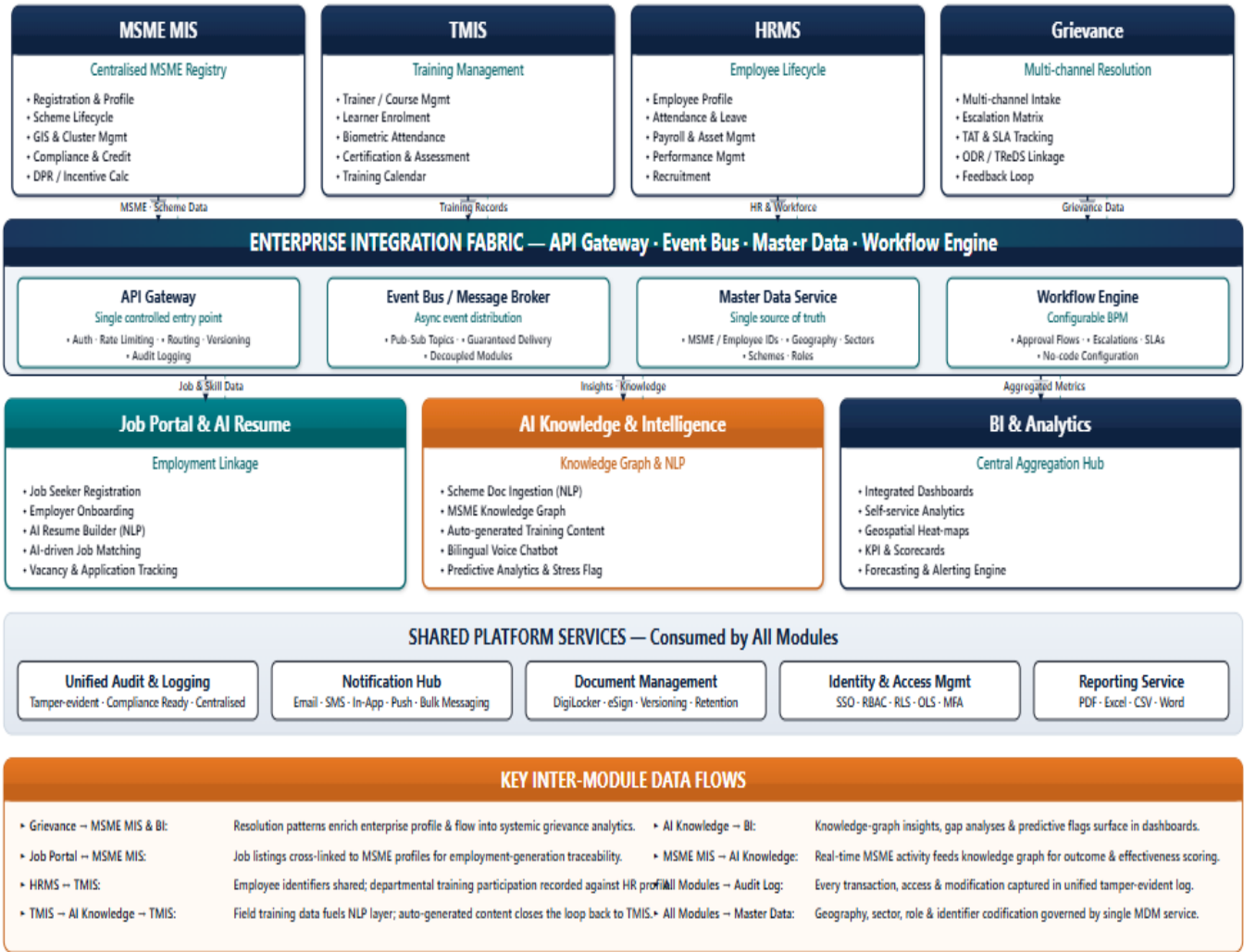
5.3.2 MODULE INTEROPERABILITY & DATA FLOW

The seven modules of the CIMS platform will interoperate through a well-governed internal data exchange framework. The key data flows as follows: -

MSME MIS-> BI/Analytics	All MSME registration data, scheme application data, compliance records, financial indicators, and GIS data will flow continuously into the BI and Analytics engine, where it will be aggregated, cleansed, and made available for dashboard visualizations, ad-hoc analysis, and AI-driven insight generation.
HRMS-> BI/Analytics	Employee attendance, payroll, performance, and leave data from the HRMS will feed the departmental HR analytics dashboards within the BI platform, enabling senior officials to monitor workforce utilisation, performance trends, and HR compliance across all directorates.
TMIS->AI Knowledge Platform	Training records, learner progress data, assessment outcomes, and trainer performance data from the TMIS will feed the AI Knowledge and Intelligence layer. Simultaneously, the AI layer will push structured, policy-derived training content back into the TMIS creating a self-reinforcing loop between programme intelligence and capacity building.
Grievance System-> MSME MIS & BI/Analytics	Grievance data including nature of grievance, resolution timelines, and recurring issue patterns will be linked to the MSME MIS to enrich the enterprise profile and will flow into the BI platform for systemic grievance analysis and escalation reporting.
Job Portal → MSME MIS	Job listings posted by MSME enterprises will be cross-linked to their MSME registration profiles, ensuring traceability between enterprise activity and employment generation which is a key metric for scheme effectiveness measurement.
All Modules → Unified Audit Log	Every transaction, access event, approval, and data modification across all modules will be captured in a unified, tamper-evident audit log, supporting the Department's governance, accountability, and compliance requirements.

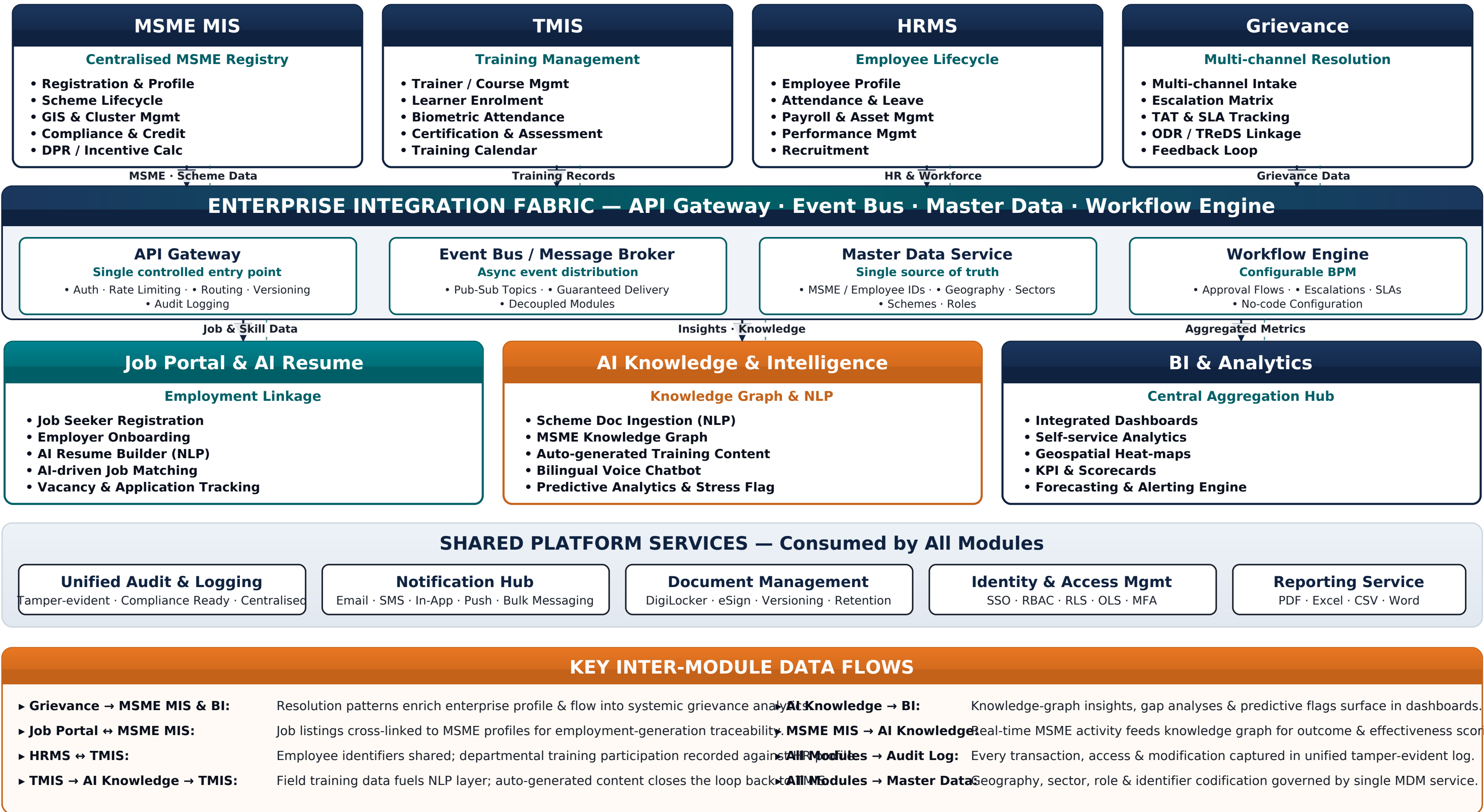
Module Interoperability & Data Flow

Bihar CIMS & BI Platform — Inter-Module Integration Backbone & Cross-Cutting Data Flows



Module Interoperability & Data Flow

Bihar CIMS & BI Platform — Inter-Module Integration Backbone & Cross-Cutting Data Flows

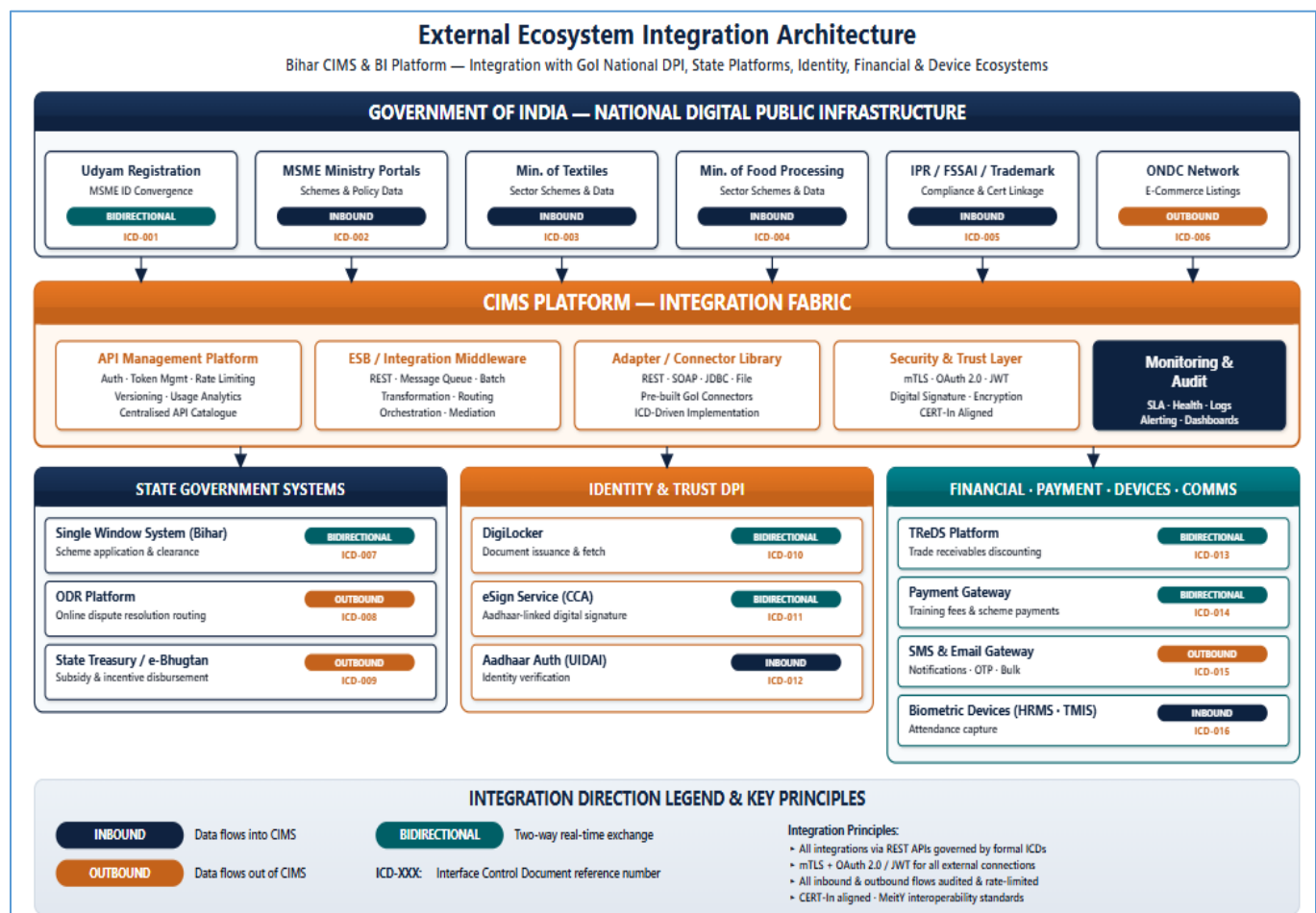


5.4 EXTERNAL ECOSYSTEM INTEGRATION ARCHITECTURE

The CIMS platform will be positioned as a node within (not isolated from) the national digital public infrastructure landscape. Unecops proposes the following external integration points, all to be implemented through secure, standards-compliant REST API interfaces governed by formal Interface Control Documents:

External System	Integration Purpose	Direction
Udyam Registration Portal (GoI)	MSME unique identification convergence onboarding validation and data enrichment	Bi-directional
MSME Ministry Portals (GoI)	Real-time scheme data retrieval and policy updates	Inbound
Ministry of Textiles Portal	Sector-specific scheme and policy data	Inbound
Ministry of Food Processing Industries Portal	Sector-specific scheme and policy data	Inbound
ONDC Network	E-commerce product and service listing for Bihar MSMEs	Outbound
TReDS Platform	Trade receivables discounting integration for MSME financial access	Bi-Directional
ODR Platform	Grievance and dispute resolution routing	Outbound
Single Window System (Bihar)	Integrated scheme application and clearance tracking	Bi-Directional

Biometric Devices (HRMS and TMIS)	Attendance marking for departmental employees and training participants	Inbound
Payment Gateway	Training fee and scheme-related payment processing	Bidirectional
IPR / FSSAI / Trademark Portals (GoI)	Information linkage for MSME compliance and certification	Inbound



External Ecosystem Integration Architecture

Bihar CIMS & BI Platform — Integration with GoI National DPI, State Platforms, Identity, Financial & Device Ecosystems

GOVERNMENT OF INDIA — NATIONAL DIGITAL PUBLIC INFRASTRUCTURE

Udyam Registration
MSME ID Convergence

BIDIRECTIONAL

ICD-001

MSME Ministry Portals
Schemes & Policy Data

INBOUND

ICD-002

Min. of Textiles
Sector Schemes & Data

INBOUND

ICD-003

Min. of Food Processing
Sector Schemes & Data

INBOUND

ICD-004

IPR / FSSAI / Trademark
Compliance & Cert Linkage

INBOUND

ICD-005

ONDC Network
E-Commerce Listings

OUTBOUND

ICD-006

CIMS PLATFORM — INTEGRATION FABRIC

API Management Platform

Auth · Token Mgmt · Rate Limiting
Versioning · Usage Analytics
Centralised API Catalogue

ESB / Integration Middleware

REST · Message Queue · Batch
Transformation · Routing
Orchestration · Mediation

Adapter / Connector Library

REST · SOAP · JDBC · File
Pre-built GoI Connectors
ICD-Driven Implementation

Security & Trust Layer

mTLS · OAuth 2.0 · JWT
Digital Signature · Encryption
CERT-In Aligned

Monitoring & Audit

SLA · Health · Logs
Alerting · Dashboards

STATE GOVERNMENT SYSTEMS

Single Window System (Bihar)
Scheme application & clearance

BIDIRECTIONAL

ICD-007

ODR Platform
Online dispute resolution routing

OUTBOUND

ICD-008

State Treasury / e-Bhugtan
Subsidy & incentive disbursement

OUTBOUND

ICD-009

IDENTITY & TRUST DPI

DigiLocker
Document issuance & fetch

BIDIRECTIONAL

ICD-010

eSign Service (CCA)
Aadhaar-linked digital signature

BIDIRECTIONAL

ICD-011

Aadhaar Auth (UIDAI)
Identity verification

INBOUND

ICD-012

FINANCIAL · PAYMENT · DEVICES · COMMS

TReDS Platform
Trade receivables discounting

BIDIRECTIONAL

ICD-013

Payment Gateway
Training fees & scheme payments

BIDIRECTIONAL

ICD-014

SMS & Email Gateway
Notifications · OTP · Bulk

OUTBOUND

ICD-015

Biometric Devices (HRMS · TMIS)
Attendance capture

INBOUND

ICD-016

INTEGRATION DIRECTION LEGEND & KEY PRINCIPLES

INBOUND

Data flows into CIMS

BIDIRECTIONAL

Two-way real-time exchange

OUTBOUND

Data flows out of CIMS

ICD-XXX: Interface Control Document reference number

Integration Principles:

- All integrations via REST APIs governed by formal ICDs
- mTLS + OAuth 2.0 / JWT for all external connections
- All inbound & outbound flows audited & rate-limited
- CERT-In aligned · MeitY interoperability standards

5.5 TECHNICAL REFERENCE ARCHITECTURE

5.5.1 ARCHITECTURAL PRINCIPLES

The Technical Reference Architecture of the CIMS platform will be governed by the following non-negotiable architectural principles:

On-Premises First	All data, application logic, and analytics processing will reside and execute within the Bihar State Data Centre (BSDC), Patna, as mandated by the RFP. No MSME data, employee data, or departmental operational data will be processed or stored outside the State's sovereign data infrastructure. A cloud-based High Availability (HA) failover configuration will be provisioned as a secondary layer, but the Bihar SDC will always remain the primary and authoritative deployment environment.
API-First Design	All inter-module communication and external integrations will be conducted through well-defined, versioned, and documented REST APIs. This ensures that the platform is modular, extensible, and capable of accommodating new integration partners, whether GoI systems or emerging State-level digital infrastructure without requiring fundamental architectural change.
Security By Design	Security controls will be embedded at every architectural layer from the network perimeter through the application tier to the database. The platform will not rely on perimeter security alone but will implement defense-in-depth across all layers, in compliance with CERT-In guidelines, IT Act 2000 and 2008, ISO 27001:2022, and ISACA standards.
Scalability & High - Availability	The architecture will be designed to scale horizontally adding capacity in response to growing user loads and data volumes without service interruption. High Availability will be engineered into the platform's deployment topology, ensuring continuity of service for both citizen-facing and departmental-facing functions.
Separation of Concerns	Transactional data management, analytics processing, AI/ML workloads, and document management will be maintained on distinct, purpose-optimized infrastructure layers, ensuring that the performance and availability of one layer does not constrain the operation of another.

5.5.2 TECHNOLOGY ARCHITECTURE LAYERS

The CIMS platform's technical architecture will be organized across six distinct layers: -

Layer 1: Presentation Layer (Frontend)	• The platform will deliver its user interface through two primary channels, a responsive web portal and native mobile applications for iOS and Android. Both channels will share a common library and design system, ensuring consistent user experience across devices and form factors. • The web portal will be architected such that it is accessible across the broadest possible range of devices including low-specification hardware commonly used in district and block-level government offices without requiring dedicated client software installation. • The mobile applications will be developed natively for iOS and Android, providing full access to all citizen-facing and field-level departmental functions with offline capability where network connectivity is intermittent. • The UI/UX design framework will be informed by the Government of India's National Design System standards, ensuring that the platform's interface is familiar and accessible to users who interact with other GoI digital services. • All citizen and enterprise-facing interfaces will be fully bilingual (English and Hindi) with the language selection persistent across sessions.
Layer 2: Application Layer (Backend Services)	• The application layer will be built on a microservices architecture, with each of the seven functional modules implemented as a discrete, independently deployable service cluster. • This architectural choice directly supports the parallel-track Agile development methodology, as individual module teams can develop, test, and deploy their service clusters without dependency-induced delays from other modules. • Each microservice will expose its functionality through a well-defined REST API, governed by an API Gateway that manages authentication, rate limiting, request routing, and audit logging for all inter-service and external API calls. • The API Gateway will serve as the single controlled entry point for all API traffic both internal (inter-module) and external (GoI integrations and third-party systems). • The key application-layer capabilities will include: - ○ Workflow Engine: A configurable, rules-based workflow engine governing scheme application approvals, grievance escalations, HR leave and payroll approvals, and training certification allowing the Department to configure workflows without requiring code changes. ○ Notification Engine: A multi-channel notification service capable of delivering real-time alerts, status updates, and scheduled communications via Email, SMS, and in-application

	notifications serving all modules from a single, centralized notification infrastructure. ○ Document Management Service: A secure, structured document repository for scheme application documents, MSME registration certificates, training materials, HR records, and grievance attachments with version control, access logging, and retention policy enforcement. ○ Reporting Service: A centralized report generation service supporting on-demand and scheduled report production across all modules, with export capabilities in PDF, Excel, CSV, and Word formats.
Layer 3: Data Layer	The data architecture of the CIMS platform will be organized into three tiers, each purpose-optimized for its function: Transactional Data Store (OLTP): A relational database management system will serve as the primary store for all transactional data like MSME registrations, scheme applications, HR records, grievance logs, training records, and job portal data. The data model will be designed with a master data management discipline ensuring that MSME enterprise identifiers, employee identifiers, and departmental reference data are maintained as governed master entities with a single, authoritative definition across all modules. Analytical Data Store (OLAP): A columnar, in-memory analytical database will serve as the data foundation for the BI and Analytics platform. Data will flow from the OLTP layer to the OLAP layer through an automated ETL (Extract, Transform, Load) pipeline, operating on both scheduled full-load and real-time incremental modes. The OLAP layer will be optimized for the high-performance, multi-dimensional query workloads generated by dashboard rendering, ad-hoc analysis, and AI-driven insight generation. AI/ML Data Store and Knowledge Graph: A dedicated data store for the AI and Machine Learning workloads encompassing the MSME Knowledge Graph, the document ingestion pipeline for scheme and policy documents, the NLP model store for the bilingual voice assistance chatbot and AI resume builder, and the training data repositories for predictive analytics and demand forecasting models. The complete data architecture will be governed by a formally documented Data Governance Framework covering data ownership, data quality standards, data classification, retention policies, and audit trail requirements which are detailed separately under this proposal.

Layer 4: Integration Layer	The Integration Layer will govern all data exchange between the CIMS platform and external systems both within the Bihar Government's digital ecosystem and with GoI national platforms. It will comprise: Enterprise Service Bus (ESB) / Integration Middleware: A lightweight integration middleware will manage the transformation, routing, and orchestration of data flows between the CIMS platform and external systems. This middleware will support multiple integration patterns of synchronous REST API calls for real-time data exchange, asynchronous message queuing for high-volume bulk data transfers, and scheduled batch processing for periodic data synchronization. API Management Platform: All externally exposed APIs including the GoI integration endpoints and the ONDC, TReDS, and ODR interfaces, will be managed through a formal API Management platform that provides security enforcement, access token management, rate limiting, usage analytics, and versioning control. Data Ingestion Pipeline (AI Layer): A dedicated automated data ingestion pipeline will continuously monitor GoI scheme portals, Ministry notification feeds, and departmental policy repositories, ingesting newly published documents and guidelines into the AI Knowledge Platform for automated parsing, structuring, and conversion into training and advisory content for the TMIS.
Layer 5: Security Layer	The Security Layer is not a discrete component within the architecture but a cross-cutting discipline that permeates every other layer. The CIMS platform's security architecture will comprise: Identity and Access Management (IAM): A centralized IAM service implementing Single Sign-On (SSO) across all modules, Role-Based Access Control (RBAC) governing which users can access which functions, Row-Level Security (RLS) ensuring that users see only the data appropriate to their administrative jurisdiction, and Object-Level Security (OLS) governing access to specific data objects and reports within the BI platform. Web Application Firewall (WAF): A WAF will be positioned at the perimeter of the application layer, providing real-time protection against OWASP Top 10 vulnerabilities including SQL injection, Cross-Site Scripting (XSS), HTTP injection, Cross-Site Request Forgery (CSRF), directory traversal, and file inclusion attacks. Data Encryption: All data at rest within the Bihar SDC will be encrypted using industry-standard symmetric encryption. All data in transit

	between the client and the server, between microservices, and between the CIMS platform and external systems will be encrypted using TLS 1.3 or higher. No unencrypted data transmission will be permitted within any layer of the platform. Security Information and Event Management (SIEM): A SIEM system will aggregate security events from across all platform layers like WAF alerts, authentication failures, privilege escalation attempts, unusual data access patterns and provide real-time security monitoring, automated alerting, and forensic audit capability. Vulnerability Management: A continuous vulnerability scanning programme will be maintained throughout the O&M period, ensuring that newly disclosed vulnerabilities in the platform's technology stack are identified and remediated within defined SLA timelines, in accordance with CERT-In advisories and the Department's security governance requirements.
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Layer 6: AI and Intelligence Layer	The AI and Intelligence Layer represent the most strategically differentiated component of the CIMS platform. It will comprise four distinct AI/ML capabilities: MSME Knowledge Graph: A persistent and continuously updated knowledge graph derived from MSME Coordinator field interactions, scheme documents, policy guidelines, and programme outcome data. The knowledge graph will structure relationships between MSMEs, schemes, geographies, sectors, and outcomes enabling the platform to generate context-aware scheme recommendations, identify underperforming geographies, and measure Centre-State programme alignment across state, district, block, and panchayat hierarchies. AI-Driven Document Ingestion and Training Content Generation: A Natural Language Processing (NLP) pipeline that automatically ingests scheme documents and policy notifications from GoI and State sources, parses and structures their content, and generates role-specific training modules and contextual advisories for delivery through the TMIS (without manual intervention). Predictive Analytics and Demand Forecasting: Machine Learning models trained on historical MSME production, employment, export, and compliance data, delivering forward-looking demand forecasts, inventory optimizations insights, and early-warning signals for MSMEs exhibiting financial stress indicators. AI-Powered Citizen Services: Including the bilingual voice assistance chatbot for citizen guidance on scheme access and application processes, and the AI-based Resume Builder within the Job Portal both of which leverage NLP capabilities developed and hosted within the AI layer.
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Technical Reference Architecture

Bihar CIMS & BI Platform — .NET Backend · PostgreSQL · Open-Source Stack · BSDC On-Premises

PRESENTATION LAYER

Web Portal React / Angular Responsive · Bilingual (Hindi/English)	Native Android App Kotlin Material Design · Bilingual	Native iOS App Swift Human Interface · Bilingual	AI Voice Chatbot Open-Source NLP Voice + Text · Bilingual
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APPLICATION LAYER — .NET Microservices

API Gateway — Authentication · Routing · Rate Limiting · Audit Logging Single controlled entry point for all internal & external API traffic						
MSME MIS .NET Microservice	TMIS .NET Microservice	HRMS .NET Microservice	Grievance .NET Microservice	Job Portal .NET Microservice	AI Knowledge .NET Microservice	BI & Analytics .NET Microservice
Workflow Engine Configurable · No-code	Notification Engine Email · SMS · In-App	Document Management Versioned · DigiLocker · eSign	Reporting Service PDF · Excel · CSV · Word			

INTEGRATION LAYER

Integration Middleware REST · Async · Batch	Message Broker Pub-Sub · Event Streaming	External Adapters Gov · Udyam · ONDC · TreDS	ETL Pipeline Scheduled · Real-time
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DATA & AI/ML LAYER

Operational DB PostgreSQL	Analytical DB PostgreSQL Columnar	Cache In-Memory Store	Knowledge Graph Open-Source Graph DB	Vector Store Open-Source Vector DB	NLP / ML Models Open-Source Frameworks
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INFRASTRUCTURE & DEPLOYMENT — BSDC On-Premises

BSDC Patna (Primary) On-Prem · Sovereign	Containers Docker · Kubernetes	CI/CD Pipeline Open-Source DevOps	Backup & DR RPO < 1h · RTO < 4h
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CROSS-CUTTING — Security · Identity · DevOps · Compliance

Security WAF · SIEM · TLS 1.3 · AES-256	Identity & Access SSO · RBAC · RLS · OLS · MFA	DevSecOps & Observability SAST · DAST · Logs · Metrics · Traces	Compliance CERT-In · ISO 27001 · GIGW · DPDP
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Technical Reference Architecture

Bihar CIMS & BI Platform — .NET Backend · PostgreSQL · Open-Source Stack · BSDC On-Premises

PRESENTATION LAYER

Web Portal

React / Angular

Responsive · Bilingual (Hindi/English)

Native Android App

Kotlin

Material Design · Bilingual

Native iOS App

Swift

Human Interface · Bilingual

AI Voice Chatbot

Open-Source NLP

Voice + Text · Bilingual

APPLICATION LAYER — .NET Microservices

API Gateway — Authentication · Routing · Rate Limiting · Audit Logging

Single controlled entry point for all internal & external API traffic

MSME MIS

.NET Microservice

TMIS

.NET Microservice

HRMS

.NET Microservice

Grievance

.NET Microservice

Job Portal

.NET Microservice

AI Knowledge

.NET Microservice

BI & Analytics

.NET Microservice

Workflow Engine

Configurable · No-code

Notification Engine

Email · SMS · In-App

Document Management

Versioned · DigiLocker · eSign

Reporting Service

PDF · Excel · CSV · Word

INTEGRATION LAYER

Integration Middleware

REST · Async · Batch

Message Broker

Pub-Sub · Event Streaming

External Adapters

GoI · Udyam · ONDC · TReDS

ETL Pipeline

Scheduled · Real-time

DATA & AI/ML LAYER

Operational DB

PostgreSQL

Analytical DB

PostgreSQL Columnar

Cache

In-Memory Store

Knowledge Graph

Open-Source Graph DB

Vector Store

Open-Source Vector DB

NLP / ML Models

Open-Source Frameworks

INFRASTRUCTURE & DEPLOYMENT — BSDC On-Premises

BSDC Patna (Primary)

On-Prem · Sovereign

Containers

Docker · Kubernetes

CI/CD Pipeline

Open-Source DevOps

Backup & DR

RPO < 1h · RTO < 4h

CROSS-CUTTING — Security · Identity · DevOps · Compliance

Security

WAF · SIEM · TLS 1.3 · AES-256

Identity & Access

SSO · RBAC · RLS · OLS · MFA

DevSecOps & Observability

SAST · DAST · Logs · Metrics · Traces

Compliance

CERT-In · ISO 27001 · GIGW · DPDP

5.6 DEPLOYMENT ARCHITECTURE

The CIMS platform will be deployed on infrastructure provided by the Department of Industries at the **Bihar State Data Centre (BSDC), Patna**, in strict compliance with the on-premises deployment mandate of the RFP. The deployment topology will be organized as follows: -

Primary Environment- Bihar State Data Centre: All production workloads such as application services, databases, integration middleware, AI/ML services, and the BI/Analytics engine will run on the primary environment at BSDC. The primary environment will be configured in a High Availability (HA) topology with redundant application servers, database clustering, and load-balanced web tier, ensuring that no single point of failure exists within the production environment.

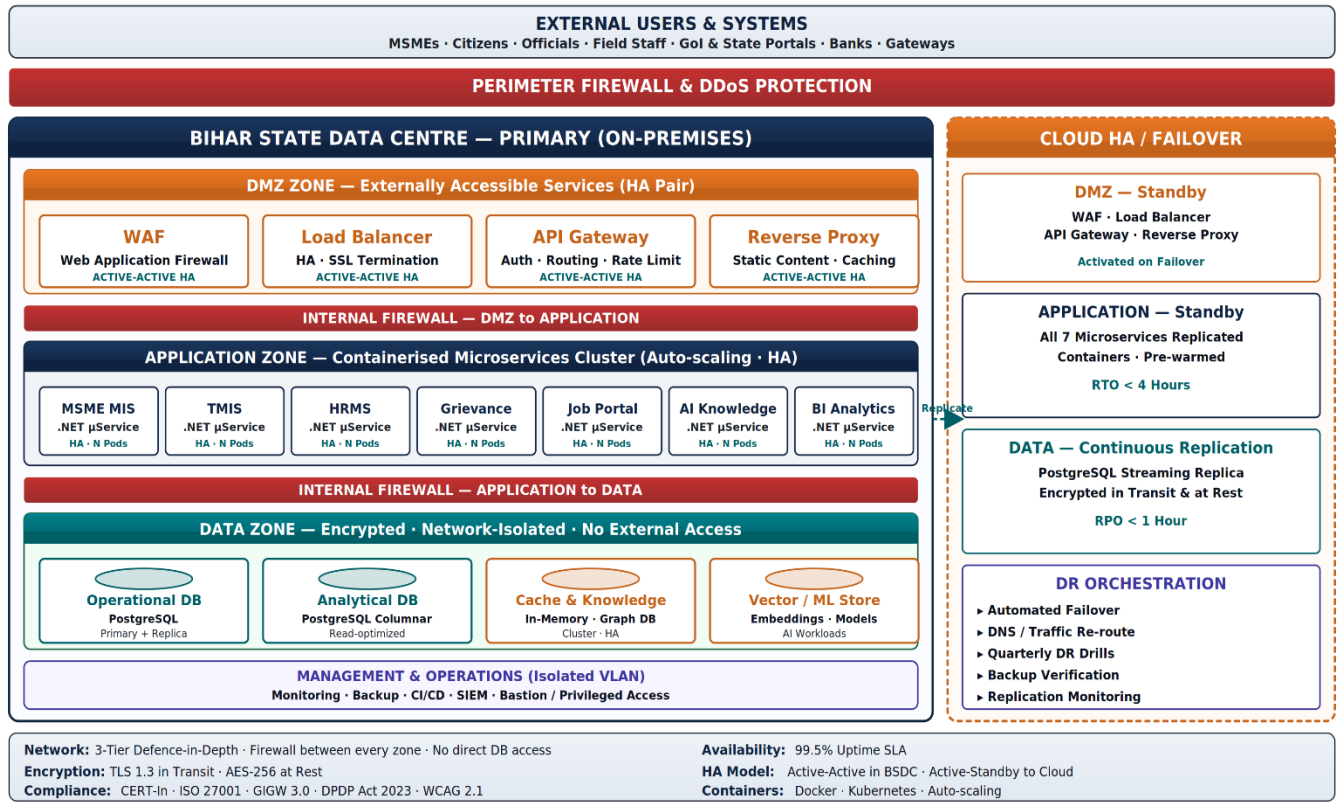
High Availability Failover: A cloud-based or secondary on-premises failover configuration will be maintained as a warm standby, capable of assuming production workloads within defined Recovery Time Objectives (RTO) in the event of a primary environment failure. The failover configuration will be subject to periodic failover drills to validate its operational readiness.

Development and Testing Environment: Dedicated development, testing, and staging environments will be maintained throughout the development phase, ensuring that production infrastructure is never exposed to development or testing activities.

Network Security Zones: The deployment will be organized into discrete network security zones - a DMZ for externally accessible services (web portal, API gateway, mobile backend), an application zone for internal services, and a data zone for databases and AI/ML stores with controlled, firewall-governed traffic flows between zones.

Deployment Architecture

Bihar CIMS & BI Platform — Zoned Deployment at BSDC with Cloud Failover



Deployment Architecture

Bihar CIMS & BI Platform — Zoned Deployment at BSDC with Cloud Failover

EXTERNAL USERS & SYSTEMS

MSMEs · Citizens · Officials · Field Staff · Gol & State Portals · Banks · Gateways

PERIMETER FIREWALL & DDoS PROTECTION

BIHAR STATE DATA CENTRE — PRIMARY (ON-PREMISES)

DMZ ZONE — Externally Accessible Services (HA Pair)

WAF

Web Application Firewall
ACTIVE-ACTIVE HA

Load Balancer

HA · SSL Termination
ACTIVE-ACTIVE HA

API Gateway

Auth · Routing · Rate Limit
ACTIVE-ACTIVE HA

Reverse Proxy

Static Content · Caching
ACTIVE-ACTIVE HA

INTERNAL FIREWALL — DMZ to APPLICATION

APPLICATION ZONE — Containerised Microservices Cluster (Auto-scaling · HA)

MSME MIS

.NET µService
HA · N Pods

TMIS

.NET µService
HA · N Pods

HRMS

.NET µService
HA · N Pods

Grievance

.NET µService
HA · N Pods

Job Portal

.NET µService
HA · N Pods

AI Knowledge

.NET µService
HA · N Pods

BI Analytics

.NET µService
HA · N Pods

INTERNAL FIREWALL — APPLICATION to DATA

DATA ZONE — Encrypted · Network-Isolated · No External Access

Operational DB

PostgreSQL
Primary + Replica

Analytical DB

PostgreSQL Columnar
Read-optimized

Cache & Knowledge

In-Memory · Graph DB
Cluster · HA

Vector / ML Store

Embeddings · Models
AI Workloads

MANAGEMENT & OPERATIONS (Isolated VLAN)

Monitoring · Backup · CI/CD · SIEM · Bastion / Privileged Access

CLOUD HA / FAILOVER

DMZ — Standby

WAF · Load Balancer
API Gateway · Reverse Proxy

Activated on Failover

APPLICATION — Standby

All 7 Microservices Replicated
Containers · Pre-warmed

RTO < 4 Hours

DATA — Continuous Replication

PostgreSQL Streaming Replica
Encrypted in Transit & at Rest

RPO < 1 Hour

DR ORCHESTRATION

- ▶ Automated Failover
- ▶ DNS / Traffic Re-route
- ▶ Quarterly DR Drills
- ▶ Backup Verification
- ▶ Replication Monitoring

Replicate

Network: 3-Tier Defence-in-Depth · Firewall between every zone · No direct DB access

Encryption: TLS 1.3 in Transit · AES-256 at Rest

Compliance: CERT-In · ISO 27001 · GIGW 3.0 · DPDP Act 2023 · WCAG 2.1

Availability: 99.5% Uptime SLA

HA Model: Active-Active in BSDC · Active-Standby to Cloud

Containers: Docker · Kubernetes · Auto-scaling

5.7 TECHNOLOGY STACK SUMMARY

Architecture Layer	Proposed Technology
Presentation	• Web Portal (React/Angular) • Native Android (Kotlin) • Native iOS (Swift) • AI Voice Chatbot (Open-Source NLP)
Application	• API Gateway → 7 .NET Microservices + 4 Platform Services (Workflow, Notification, DMS, Reporting)
Integration	• Integration Middleware • Message Broker • External Adapters • ETL Pipeline
Data & AI/ML	• PostgreSQL (OLTP) • PostgreSQL Columnar (OLAP) • In-Memory Cache • Open-Source Graph DB • Vector DB • NLP/ML Frameworks
Infrastructure	• BSDC Patna • Containers (Docker · K8s) • Open-Source CI/CD • Backup & DR
Cross-Cutting	• Security • IAM • DevSecOps & Observability • Compliance

5.8 NON-FUNCTIONAL ARCHITECTURE COMMITMENTS

The following non-functional characteristics are designed into the CIMS platform's architecture as first-class requirements:

Availability	The platform will be designed and deployed to deliver high availability for all citizen-facing and departmental operational services, with the deployment topology, redundancy configuration, and monitoring framework aligned to the uptime commitments defined in the O&M Service Level Agreement.
Performance	All user-facing transactions will be designed to meet defined response thresholds under normal and peak load conditions. The BI and Analytics layer will be optimized for sub-second dashboard rendering for standard views and acceptable response times for complex ad-hoc queries across large datasets.
Scalability	The microservices architecture and the horizontally scalable database and analytics tiers will ensure that the platform can accommodate growth in MSME registrations, user concurrency, data volumes, and module additions without requiring architectural redesign.
Interoperability	All external integration interfaces will adhere to open standards like REST, JSON, XML, ensuring that the platform can interoperate with current and future GoI digital infrastructure without proprietary lock-in.
Maintainability	The codebase will adhere to defined coding standards, documentation requirements, and modular design principles thereby ensuring that the platform remains maintainable and extensible throughout the 60-month engagement and beyond, by both Uneecops's O&M team and, over time, by the Department's own technical personnel.
Accessibility & Inclusivity	The web portal and mobile application will comply with accessibility standards, ensuring usability for individuals with varying levels of digital literacy and, to the extent feasible, for users with accessibility requirements.

6 ENGAGEMENT TEAM & WORK PLAN

6.1 TEAM COMPOSITION PHILOSOPHY

Unecops Technologies Limited will constitute a dedicated, full-lifecycle project team for the CIMS engagement structured to reflect the multi-module complexity of the platform, the parallel-track Agile delivery methodology, and the governance rigour demanded by a public sector engagement of this institutional significance.

The team will be organised across three functional tiers: Programme Governance, Solution Delivery, and Quality, Security, and Operations each staffed by professionals with domain expertise directly relevant to the functional and technical scope of the CIMS platform.

The team will be resident and operationally dedicated to this engagement throughout the development phase, with a structured transition to the O&M team composition following go-live at Milestone 4. All key personnel will be formally introduced to Udyog Mitra at the project inception stage, with their CVs submitted in the prescribed format as part of the technical bid documentation.

6.1.1 TIER-1: PROGRAMME GOVERNANCE

Project Manager	The Project Manager will serve as the single point of accountability for the end-to-end delivery of the CIMS engagement holding overall responsibility for schedule adherence, quality governance, stakeholder relationship management, risk and issue resolution, and formal communication with Udyog Mitra and the Department of Industries. The Project Manager will chair all formal project governance forums, produce and maintain the Master Project Plan, manage the change control process, and ensure that all contractual milestones are achieved within the RFP-mandated timelines. Minimum Qualification: BE / B. Tech / MCA / MBA with a minimum of five years of relevant project management experience in IT/ITeS or e-Governance engagements, preferably in the public sector. PMP or equivalent certification preferred.
Programme Management Office (PMO) Support	A dedicated PMO support function will assist the Project Manager in maintaining project governance artefacts including the risk register, issue log, stakeholder communication plan, meeting minutes, and milestone tracking dashboards ensuring that project governance documentation is current, accessible, and audit-ready throughout the engagement.

6.1.2 TIER-2: SOLUTION DELIVERY

Solution Architect	✓ The Solution Architect will hold technical authority over the end-to-end architecture of the CIMS platform ensuring architectural coherence, integration consistency, and technical quality across all seven modules and the shared platform layers. ✓ The Solution Architect will produce and govern all architectural artefacts, review and approve all module-level design decisions, provide technical guidance to module leads, and serve as the primary technical interface with Udyog Mitra's designated technical counterparts.
Module Lead- MSME MIS	✓ The MSME MIS Module Lead will be responsible for the design, development, integration, and testing of the MSME Management Information System including the public portal, registration and onboarding module, DIC profiling, scheme management, GIS mapping, e-commerce linkage, compliance monitoring, and AI-driven MSME analytics. ✓ The Module Lead will coordinate directly with the departmental subject matter experts from Udyog Mitra during requirements validation and UAT phases.
Module Lead- HRMS	✓ The HRMS Module Lead will govern the design, development, and delivery of the Human Resource Management System covering the complete employee lifecycle from recruitment through separation, payroll, attendance with biometric integration, performance management, and all associated reporting and analytics functions. ✓ The HRMS Module Lead will maintain close coordination with the Department's HR administration stakeholders throughout the requirements refinement and UAT process.
Module Lead- TMIS & Grievance Redressal	• This Module Lead will govern the concurrent delivery of the Training Management Information System and the Grievance Redressal System; two functionally distinct but architecturally related modules sharing common notification, communication, and analytics infrastructure. • The Lead will ensure that both the modules are delivered to specification, integrated with the AI Knowledge Platform (in the case of TMIS) and with the ODR and TReDS external platforms (in the case of Grievance Redressal), and operationally validated through structured UAT.
Module Lead- Job Portal & AI Resume Builder	✓ The Job Portal Module Lead will govern the design and delivery of the Integrated Job Portal and its AI-Powered Resume Builder including job seeker registration, employer onboarding, vacancy listing management, application management, and the NLP-driven resume generation capability. ✓ The Lead will coordinate with the AI/ML Engineer for the integration of the Resume Builder's NLP capability with the platform's AI layer.
BI and Analytics Lead	✓ The BI and Analytics Lead will govern the design, development, and delivery of the Business Intelligence and Analytics Platform including the semantic layer, dashboard library, ad-hoc query environment, KPI management framework, geospatial visualisation, predictive analytics, and all reporting capabilities. ✓ The BI and Analytics Lead will work in close collaboration with the Solution Architect on the data architecture and ETL pipeline design and will

	coordinate with all Module Leads to ensure that transactional module data is made available to the analytics layer in the required structure and quality.
AI and Machine Learning Engineer	✓ The AI/ML Engineer will be responsible for the design, development, and operationalization of all AI and Machine Learning capabilities within the CIMS platform including the MSME Knowledge Graph, the NLP-based document ingestion and training content generation pipeline, the AI-powered bilingual voice assistance chatbot, the AI Resume Builder engine, and the predictive analytics and demand forecasting models. ✓ The AI/ML Engineer will work within the platform's data architecture framework and will ensure that all AI outputs are traceable, governed, and linked to the appropriate platform modules.
Integration and API Specialist	✓ The Integration and API Specialist will govern the design, development, testing, and documentation of all integration interfaces between the CIMS platform and external systems — including Udyam Registration Portal, MSME Ministry portals, TReDS, ODR, ONDC, biometric devices, Payment Gateway, and all other GoI platforms identified in the integration architecture. ✓ The Specialist will maintain the Interface Control Document register and will coordinate with external system counterparts on API availability, versioning, and testing schedules.
UI/UX Designer	✓ The UI/UX Designer will be responsible for the design of the user interface and user experience across all platform modules — for both the web portal and the mobile applications (iOS and Android). ✓ Design work will be grounded in user research conducted with representative departmental users and MSME stakeholders and will adhere to accessibility standards and bilingual interface requirements. ✓ All UI/UX designs will be formally reviewed and approved by Udyog Mitra before development commences.
Mobile Application Developer(s)	✓ Dedicated mobile application development capability will be maintained for the native iOS and Android applications ensuring that the mobile interface delivers the full scope of citizen-facing and field-level departmental functionality with the performance, usability, and offline capability required for the Bihar field environment.

6.1.3 TIER-3: QUALITY, SECURITY & OPERATIONS

Quality Assurance Lead	✓ The QA Lead will govern the platform's end-to-end testing programme encompassing test strategy definition, test plan development, test case design and execution across all modules, defect management, regression testing, performance and load testing, and UAT facilitation. ✓ The QA Lead will operate the automated testing suite established from Sprint 1 and will maintain the defect tracking and resolution discipline throughout the development lifecycle. ✓ The QA Lead will produce formal test completion reports at each milestone, documenting test coverage, defect resolution status, and residual risk for review by both internal governance and Udyog Mitra.
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Security Specialist & DevOps Engineer	✓ The Security Specialist and DevOps Engineer will jointly govern the platform's CI/CD pipeline, infrastructure provisioning and configuration at the Bihar State Data Centre, application security posture, and operational monitoring framework. ✓ The Security Specialist will maintain the platform's security controls like WAF configuration, vulnerability scanning, patch management, and compliance with CERT-In guidelines and ISO 27001:2022. ✓ The DevOps Engineer will maintain the automated build, test, and deployment pipeline, the staging and production environment configurations, and the performance monitoring and alerting infrastructure.
Database Administrator/Expert (DBA)	✓ The Database Administrator will govern the design, implementation, and ongoing administration of the platform's multi-tier data architecture including the relational transactional database, the columnar analytical database, and the AI/ML data store. ✓ The responsibilities of the DBA will encompass database performance tuning, index management, backup and recovery operations, data archival, and database security configuration throughout both the development phase and the 52-month O&M period.
Change Management & Training Lead	• The Change Management and Training Lead will govern the organizational change management and capacity building programme for the CIMS engagement like developing and delivering role-based training programmes for all user categories, producing all training and documentation artefacts, facilitating user acceptance testing with departmental end users, and supporting the Department's ongoing institutional capability development throughout the O&M period.

6.1.4 TIER-4: OPERATIONS & MANAGEMENT TEAM (O&M TEAM)

- Following the formal go-live of the CIMS platform at Milestone 4, the engagement team will transition to its O&M configuration which will house a dedicated, experienced support and operations team structured to deliver the O&M Service Framework commitments across the 52-month post-go-live period. This will be a combination of onsite and offsite manpower
- The O&M team will maintain continuity of key technical knowledge from the development phase, ensuring that the team responsible for platform support is intimately familiar with its architecture, codebase, and operational characteristics.

6.2 WORK PLAN

The CIMS project work plan is structured in strict conformance with the milestone framework prescribed in the RFP. The four milestones that govern the engagement timeline are as follows: -

Milestone	Description	Timelines	Phase Narrative
Milestone 1	Contract Signing and Project Commencement	Week 0: Contract Effective Date	-
Milestone 2	Inception, Requirements Validation, Architecture Sign-off, Environment Setup	Weeks 1 to 8	- This phase will establish the complete foundational framework for the CIMS engagement ensuring that all requirements are validated, all architectural decisions are confirmed, and all delivery infrastructure is in place before development sprint work commences. - The Key activities during this phase will include: ✓ Formal project kick-off with Udyog Mitra and the Department of Industries including introduction of the full Uneecops project team, confirmation of project governance structures, communication protocols, and escalation pathways. ✓ Detailed requirements elicitation workshops with subject matter experts from all relevant departments and directorates validating the requirements documented in the RFP, capturing any nuances, exceptions, or refinements required for accurate platform design. ✓ Finalisation and formal sign-off of the Solution Architecture Document, for all GoI integration points, UI/UX Prototype designs, and the Test Strategy. Establishment of the development, testing, and staging environments at the Bihar State Data Centre (if required) including server

			provisioning, network configuration, development toolchain setup, and CI/CD pipeline initialisation. ✓ Finalisation and baseline of the Master Project Plan including the sprint calendar, milestone schedule, resource assignment plan, risk register, and communication plan for formal review and approval by Udyog Mitra.
Milestone 3	Core Development, Integration, System Testing — Substantial Completion	Week 9-24	<u>Iterative Development and Integration:</u> The core development phase will be executed across four parallel delivery tracks through a structured series of two-week Agile sprints with each sprint producing tested, demonstrable increments of working software across all seven modules simultaneously. The four parallel tracks are: - ✓ <i>Track 1- MSME MIS and External Integrations:</i> Progressive development and integration of the MSME MIS module from registration and onboarding through scheme management, GIS mapping, compliance monitoring, e-commerce linkage, and GoI portal integrations culminating in a fully functional, integration-tested MSME MIS by Week 24. ✓ <i>Track 2: HRMS and TMIS:</i> Concurrent development of the Human Resource Management System and the Training Management Information System including biometric device integration, payroll processing, performance management, course management, and AI-driven training content generation linkage. ✓ <i>Track 3: Grievance Redressal, Job Portal, and AI Capabilities:</i> Development of the Grievance Redressal System including multi-channel intake, escalation matrix, and ODR/TReDS integration alongside the Job Portal and AI Resume Builder, and the AI/ML layer encompassing the Knowledge Graph, NLP pipeline, chatbot, and predictive analytics models. ✓ <i>Track 4: BI/Analytics Platform and DevOps:</i> Development of the Business Intelligence and

			Analytics Platform including the semantic layer, ETL pipelines, dashboard library, ad-hoc query environment, KPI framework, and geospatial visualizations alongside the ongoing DevOps infrastructure management, security monitoring, and automated testing pipeline maintenance. Sprint reviews will be conducted at the conclusion of every two-week sprint with working software demonstrated to Udyog Mitra's designated review team, sprint outcomes documented, and feedback formally incorporated into the subsequent sprint backlog. At the conclusion of Week 24, a formal Milestone 3 review will be conducted presenting the substantially complete platform across all seven modules for departmental assessment prior to the commencement of UAT.
Milestone 4	UAT, Security Audit, Load Testing, Training, Go-Live	Week 25-35	This phase will transition the platform from development completion to full production readiness through a structured sequence of user acceptance testing, security validation, load testing, training delivery, and formal go live. The key activities will include: ✓ Structured User Acceptance Testing with representative end users from across the Department's directorates, at State, District, Block, and Sub-Division levels for all seven platform modules. ✓ UAT will be conducted in the staging environment, with a formally documented UAT plan, UAT test cases, and a defect log maintained throughout. ✓ All UAT observations will be triaged, prioritised, and resolved through the defect management process, with re-testing confirming resolution before UAT sign-off. ✓ The Application Security Audit conducted by the Department's appointed Certified Auditor covering all security vulnerability categories enumerated in the RFP, including SQL injection, XSS, penetration testing, configuration testing, network security, and all others specified.

			✓ All identified vulnerabilities will be remediated by Uneecops within 15 days of notification, with a formal remediation report submitted for the Auditor's re-assessment. ✓ Performance and Load Testing simulating a minimum of 100 concurrent users at peak load across all platform modules with results documented and submitted to Udyog Mitra as evidence of production readiness. ✓ Role-based training delivery for all categories of platform users administrators, departmental operational users, HR personnel, grievance officials, and data entry operators in both Hindi and English, through a combination of classroom and digital training formats. Training completion will be documented and submitted as a Milestone 4 deliverable. ✓ Data migration and MSME onboarding support assisting the Department in the structured transfer and validation of any existing MSME registration or scheme data into the CIMS platform prior to go-live. ✓ Formal Go-Live at Week 35 (Month 8) following successful completion of UAT sign-off, security audit clearance, load test certification, and training completion confirmation. ✓ The Go-Live event will be formally documented in a Go-Live Acceptance Report, signed by authorised representatives of both Uneecops Technologies and Udyog Mitra.
O&M Period	Operations, Maintenance, Support, Enhancements	Month 8 through Month 60 (52 Months)	✓ The 52-month O&M phase commences immediately upon the formal Go-Live confirmation at Milestone 4 and concludes at Month 60 of the total engagement. ✓ During this phase, Uneecops will provide the full O&M Service Framework as committed encompassing incident management and resolution within defined SLA timelines, service request fulfilment for configuration changes and user management requirements, proactive performance monitoring and optimization, security patch management and periodic re-audits, enhancement releases

			incorporating new schemes, policy changes, and departmental requirements as directed by Udyog Mitra, and continual service improvement through structured periodic reviews with the Department. ✓ Monthly operational reports will be submitted to Udyog Mitra throughout the O&M period, documenting platform availability, incident statistics, enhancement activities, and service improvement initiatives.
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Project Work Plan — Master Gantt Chart

60-Month Integrated Schedule | 57 Activities across 4 Phases | Weeks 1–35 Development + Months 8–60 Operations & Maintenance

Project Work Plan — Master Gantt Chart

60-Month Integrated Schedule | 57 Activities across 4 Phases | Weeks 1–35 Development + Months 8–60 Operations & Maintenance

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6.3 ENGAGEMENT PLAN

CIMS & BI PLATFORM — ENGAGEMENT PLAN & TEAM STRUCTURE

#	Tier / Function	Role	Key Responsibilities	Engagement Phase	Reporting To	Qualification Profile
T1	Programme Governance	Project Manager	Overall delivery accountability; milestone management; stakeholder governance; risk & issue management; formal communication with Udyog Mitra	All Phases (W1-M60)	Udyog Mitra CEO / PM	B. Tech and MBA; 11+ yrs IT/e-Gov
T2	Programme Governance	PMO Support	Project governance artefacts; risk register; issue log; meeting minutes; milestone dashboards; communication plan maintenance	All Phases (W1-M60)	Project Manager	Graduate; 2+ yrs PMO / project coordination experience
T3	Solution Delivery	Solution Architect	End-to-end architectural authority; design governance across all 7 modules; technical interface with Dept. counterparts; architecture documentation	Phase 1-3 (W1-W35)	Project Manager	BE/B. Tech/MCA; 8+ yrs solution architecture; e-Gov /BI experience
T4	Solution Delivery	Module Lead: MSME MIS & Developers	Design, development, integration and testing of MSME MIS module; GoI portal integrations; MSME stakeholder liaison during UAT	Phase 1-3 (W1-W35)	Solution Architect	BE/B.Tech/MCA; 5+ yrs full-stack development; e-Gov or MSME domain
T5	Solution Delivery	Module Lead: HRMS & Developers	Design, development and delivery of HRMS module; payroll, biometric integration, performance	Phase 1-3 (W1-W35)	Solution Architect	BE/B. Tech/MCA; 5+ yrs HRMS / ERP development; Govt. payroll rules knowledge

			management; HR stakeholder coordination			
T6	Solution Delivery	Module Lead: TMIS & Grievance & Developers	Concurrent delivery of TMIS and Grievance Redressal modules; AI content integration (TMIS); ODR/TReDS integration (Grievance)	Phase 1-3 (W1-W35)	Solution Architect	BE/B. Tech/MCA; 5+ yrs application development; training or grievance systems experience
T7	Solution Delivery	Module Lead: Job Portal & AI Resume & Developers	Design and delivery of Integrated Job Portal and AI Resume Builder; employer/job seeker onboarding; NLP integration coordination	Phase 1-3 (W1-W35)	Solution Architect	BE/B. Tech/MCA; 4+ yrs application development; AI/NLP integration experience
T8	Solution Delivery	BI & Analytics Lead & Developers	Design and delivery of BI/Analytics platform; semantic layer; ETL pipelines; dashboard library; KPI framework; predictive analytics	Phase 1-3 (W1-W35)	Solution Architect	BE/B. Tech/MCA; 6+ yrs BI/Analytics development; data modelling and on- prem BI deployment experience
T9	Solution Delivery	AI / ML Engineer & Developers	Knowledge Graph; NLP pipeline; AI chatbot; AI Resume Builder engine; predictive models; AI content generation for TMIS	Phase 1-3 (W1-W35)	Solution Architect	BE/B.Tech/MCA/ M.Tech; 4+ yrs AI/ML engineering; NLP and knowledge graph experience
T10	Solution Delivery	Integration & API Specialist & Developers	All GoI integration interfaces; API design and documentation; Interface Control Documents; external system coordination	Phase 1-3 (W1-W35)	Solution Architect	BE/B.Tech; 5+ yrs API / integration development; REST, JSON, middleware experience
T11	Solution Delivery	UI/UX Designer	Web portal and mobile application UI/UX design; bilingual interface; accessibility compliance;	Phase 1-2 (W1-W24)	Module Leads / Solution Architect	Design qualification; 4+ yrs UI/UX for web & mobile; Govt. design system knowledge

			prototype development for stakeholder review			
T12	Solution Delivery	Mobile Application Developers	Native iOS (Swift) and Android (Kotlin) development for all citizen-facing and field modules; offline capability implementation	Phase 2-3 (W10-W35)	Module Leads	BE/B. Tech/MCA; 3+ yrs native iOS and Android development
T13	Quality, Security & Ops	QA Lead	Test strategy; test plan and case design; automated testing suite; defect management; UAT facilitation; performance & load testing; test completion reports	All Phases (W4-M60)	Project Manager	BE/B.Tech/MCA; 5+ yrs QA/testing; automation and performance testing experience
T14	Quality, Security & Ops	Security Specialist	Application security posture; WAF; SIEM; vulnerability assessments; CERT-In compliance; ISO 27001; security audit coordination; patch management	All Phases (W1-M60)	Project Manager	BE/B.Tech/MCA; 5+ yrs application security; CISSP/CEH preferred; Govt. security compliance experience
T15	Quality, Security & Ops	DevOps Engineer	CI/CD pipeline; Bihar SDC environment management; infrastructure provisioning; server health monitoring; capacity management; BCDR drills	All Phases (W1-M60)	Project Manager	BE/B.Tech; 4+ yrs DevOps; Linux, containerisation, monitoring tools; on-prem deployment experience
T16	Quality, Security & Ops	Database Administrator	Multi-tier data architecture management; OLTP/OLAP/AI data stores; performance tuning; backup and recovery; data archival; database security	All Phases (W3-M60)	Solution Architect	BE/B.Tech/MCA; 5+ yrs DBA; relational and columnar DB experience; large-scale Govt. data environments preferred

T17	Quality, Security & Ops	Change Management & Training Lead	Organisational change management; role-based training (Hindi & English); training documentation; UAT support; administrator knowledge transfer	Phase 3 & O&M (W25–M60)	Project Manager	Graduate/post-graduate; 5+ yrs training delivery; Govt. e-Gov training programmes experience preferred
T18	Quality, Security & Ops	O&M Support Team	Incident management; service requests; application support; helpdesk; monthly operational reporting; continual service improvement	Phase 4 — O&M (M8–M60)	O&M Lead / Project Manager	Graduate; relevant technical certifications; ITIL preferred; prior Govt. application support experience

Note: The above team structure represents the roles proposed for this engagement. No commercial or financial information is included in this document.

7 DEMONSTRATION OF PROJECT FUNCTIONALITIES

Unecops Technologies Limited will deliver the Centralised Integrated Management System (CIMS) and Business Intelligence Platform as a comprehensively functional, natively integrated digital governance solution encompassing seven solution modules. The following section presents a module-wise description of the functionalities that will be delivered, articulating the features, operational workflows, and capabilities of each module from the perspective of its intended users within the Department of Industries, Government of Bihar.

All functionalities described herein will be delivered as integral, production-ready components of the CIMS platform; accessible through the unified web portal and mobile application (iOS and Android), governed by the platform's centralised identity, access management, and audit framework, and fully operational within the Bihar State Data Centre environment.

7.1 MSME MIS (MANAGEMENT INFORMATION SYSTEM)

7.1.1 OVERVIEW

- The MSME MIS will serve as the central enterprise data platform for Bihar's MSME ecosystem, the definitive digital interface between the State Government and its MSME constituency.
- It will provide public-facing self-service access for MSME enterprises and entrepreneurs, operational tools for departmental officials at all administrative levels, and a data-rich foundation for the platform's Business Intelligence and Analytics layer.

- The MSME MIS will be designed to serve the complete MSME lifecycle; from initial discovery and registration through scheme participation, compliance monitoring, market access, and outcome measurement.

7.1.2 PUBLIC ACCESS PORTAL

- The MSME MIS will provide unrestricted public access to a dedicated information portal through which any citizen, enterprise, or researcher can access information about MSME-relevant schemes, policies, guidelines, Government notifications, and departmental resources without requiring registration or authentication.
- The public portal will present information in a structured, navigable format in both Hindi and English, with step-by-step bilingual video guides explaining how to apply for key schemes, accompanied by process flow diagram illustrations for visual clarity.
- The portal will additionally host a resource library; a categorised, searchable directory of MSME policy documents, scheme guidelines, regulatory circulars, and Government notifications, enabling entrepreneurs to independently access the information they need to navigate the MSME support ecosystem.

7.1.3 MSME REGISTRATION & ONBOARDING

- The platform will provide a structured, end-to-end digital registration facility through which Micro, Small, and Medium Enterprises across Bihar will be onboarded onto the CIMS with a unique identification number.
- The registration process will be convergent with the Udyam Registration Portal of the Government of India; ensuring that Bihar's MSME database is aligned with the national MSME identification framework and eliminating the burden of duplicate registration on enterprises.
- The registration module will support the complete data lifecycle for each MSME entity from initial entry of enterprise details, subsequent updates to reflect changes in enterprise profile, scale, ownership, or sector classification, and deletion or deactivation of records as appropriate.
- User accounts will be created for each registered MSME, enabling authenticated access to scheme applications, grievance filing, training enrolment, and job portal services.
- User management will extend to District, Subdivision, Block, and Panchayat-level administrative accounts, enabling the Department to manage data entry, verification, and oversight responsibilities at every tier of the administrative hierarchy.

7.1.4 DIC PROFILING

- The platform will deliver a dedicated District Industries Centre (DIC) Profiling module that enables the systematic capture, maintenance, and analysis of enterprise-level intelligence at the DIC level.
- DIC profiles will aggregate MSME registration data, scheme participation history, compliance records, and field assessment observations — providing DIC officials with a comprehensive, up-to-date intelligence view of the MSME landscape within their jurisdiction.
- The profiling module will support the creation of sector-wise, cluster-wise, and size-wise enterprise profiles, enabling targeted scheme interventions and resource allocation decisions.

7.1.5 INTEGRATED DASHBOARD & SCHEME MANAGEMENT

- The MSME MIS will deliver an integrated, real-time dashboard environment that provides departmental officials with a unified view of all MSME-relevant schemes of both the State Government and the Central Government, drawing data from a separate, department-wise database maintained within the platform's data architecture.
- The dashboard will display scheme-wise beneficiary counts, application pipeline status, disbursement progress against allocation, and geographic distribution of beneficiaries, at State, District, Block, and Panchayat hierarchy levels.
- Scheme data will be maintained in real time, with updates to State and Central Government scheme parameters, eligibility criteria, and beneficiary lists reflected in the platform's database as and when they are published.
- The platform will additionally integrate with the scheme portals of the Ministry of MSME, Ministry of Textiles, Ministry of Food Processing Industries, and other line departments, enabling real-time data retrieval based on categorical filters without manual data entry.

7.1.6 AWARD AND RECOGNITION MANAGEMENT

- The platform will incorporate an Award and Recognition management facility through which the Department can design, administer, and communicate its MSME recognition programmes sector-wise best enterprise awards, export excellence recognition, innovation awards, and other institutionalized recognition frameworks.
- The module will support nomination intake, evaluation workflow, selection recording, and award notification with the results published on the public portal for stakeholder visibility.

7.1.7 GIS MAPPING & GEO-SPATIAL INTELLIGENCE

- A dedicated GIS Mapping and Geospatial Intelligence capability will be embedded within the MSME MIS, enabling the systematic geographic tagging and map-based visualization of all MSME units, industrial clusters, Industrial Areas, Special Economic Zones, land parcels, and Government-owned infrastructure assets across Bihar.
- Geo-tagged MSME data will be rendered on interactive maps with drill-down capability from State to District to Block to Panchayat level, supporting field monitoring, cluster development planning, infrastructure investment decisions, and geographic gap analysis.

7.1.8 DEMAND AND SUPPLY MODULE

- The platform will deliver a Demand and Supply Management module enabling systematic tracking and analysis of raw material demand, manufacturing output, product supply, export and import flows, and market pricing trends across Bihar's MSME sector.
- The module will support the capture and visualization of sector-wise production data, enabling the Department and individual enterprises to monitor market dynamics, identify supply-demand imbalances, and make data-informed production and inventory decisions.
- An integrated directory of exporters and importers with linkages to GoI trade entities will provide Bihar's MSME enterprises with structured access to national and international market connections.

7.1.9 E-COMMERCE & MARKET LINKAGE

- The CIMS platform will facilitate the creation of a departmental e-commerce interface linked to the ONDC network and other national e-commerce platforms through which registered MSME enterprises can showcase and sell their products and services to a national and international audience.
- The e-commerce module will support product catalogue creation, pricing management, order processing, and customer communication for MSME sellers, while enabling manufacturers and wholesalers to view MSME product catalogues, access pricing information, and initiate procurement orders directly through the portal.
- Integration with Trade Fair and Exhibitions management functionality will additionally enable MSME enterprises to access information on national and international trade fair opportunities and register their participation through the platform.

7.1.10 CREDIT LINKAGE & FINANCIAL MONITORING

- The platform will incorporate a Credit Linkage facility that connects Bihar's MSME enterprises with formal financial institutions and credit facilitation schemes providing an interface through which enterprises can explore credit products, assess eligibility, and initiate credit linkage applications.
- A complementary Financial Monitoring module will enable the tracking of key financial health indicators for registered MSMEs including revenue trends, profit margins, liquidity positions, and debt levels drawing on data submitted by enterprises and collected through scheme interactions.
- An AI-driven financial stress identification capability will automatically flag MSMEs exhibiting early warning indicators of financial distress, enabling the Department to initiate targeted support interventions proactively.

7.1.11 COMPLIANCE MONITORING

- A dedicated Compliance Monitoring module will enable the Department to track the regulatory compliance posture of registered MSME enterprises across multiple dimensions such as tax payment compliance, environmental regulatory compliance, labor law adherence, and business licensing currency.
- The module will integrate with relevant GoI compliance portals where APIs are available, supplemented by self-declaration and document upload facilities for compliance categories not yet covered by digital GoI infrastructure.
- Automated alerts will notify enterprises of approaching compliance deadlines and notify departmental officials of enterprises with overdue compliance obligations.

7.1.12 IPR, TRADEMARK, FSSAI AND COPYRIGHT LINKAGE

- The platform will provide an integrated information and linkage facility for MSME enterprises seeking to protect and leverage their intellectual property covering registration and status tracking for patents, trademarks, geographical indications, and copyrights, as well as FSSAI licensing for food-sector enterprises.

- The module will link to relevant GoI registries and portals, enabling Bihar's MSME enterprises to access IP protection pathways and FSSAI compliance information through the unified CIMS interface without navigating multiple external government portals independently.

7.1.13 DPR AND INCENTIVE CALCULATOR

- An integrated Detailed Project Report (DPR) preparation aid and Incentive Calculator will be provided within the MSME MIS, enabling entrepreneurs to estimate their eligibility for State and Central Government financial incentives like capital subsidies, interest subventions, technology upgradation grants, and export promotion support.
- This will be based on their enterprise profile and proposed investment parameters. The DPR facility will guide entrepreneurs through a structured project documentation process, improving the quality of scheme applications and reducing processing delays.

7.1.14 DATA COLLECTION, ANALYTICS AND AI-DRIVEN INTELLIGENCE

- The MSME MIS will incorporate comprehensive data collection tools enabling the systematic capture of MSME performance data such as 'financial indicators', 'employment levels', 'production capacities', 'export values', and 'compliance status at defined intervals.
- This data will feed the platform's Business Intelligence and Analytics layer for scheme effectiveness analysis, sector performance monitoring, and geographic benchmarking.
- The AI-driven analytical capability within the MSME MIS will automatically generate performance assessments for each registered MSME, identify enterprises that are under stress or in need of targeted support, surface forecasting insights on sector demand trends, and generate structured reports for policy and programme management use; all accessible through the integrated MIS dashboard with export in PDF, Excel, and CSV formats.

7.1.15 COMMUNICATION & NOTIFICATION

- The MSME MIS will incorporate a multi-channel communication facility delivering scheme notifications, compliance reminders, application status updates, award announcements, and policy alerts to MSME enterprises and departmental users through Email and SMS.
- A Bulk Messaging capability will enable authorized departmental officials to broadcast targeted communications to defined segments of the MSME database; by sector, district, enterprise size, or scheme participation status for policy dissemination, event announcements, and programme mobilization purposes.

7.1.16 AI POWERED BI-LINGUAL VOICE ASSISTANT CHATBOT

- An AI-powered, bilingual voice assistance chatbot; operable in both Hindi and English will be embedded within the MSME MIS public portal, providing citizens and MSME entrepreneurs with intelligent, conversational guidance on scheme eligibility, application processes, registration procedures, and grievance filing.

- The chatbot will be powered by the platform's Natural Language Processing layer and will be continuously updated through the AI Knowledge Platform as new schemes and policy changes are published ensuring that the guidance provided is current, accurate, and contextually relevant.

7.2 HUMAN RESOURCE MANAGEMENT SYSTEM (HRMS)

7.2.1 OVERVIEW

- The HRMS module will deliver a comprehensive, end-to-end human resource governance platform for the employees of the Department of Industries across all its directorates covering the complete employee lifecycle from recruitment and onboarding through day-to-day attendance and payroll management, performance assessment, asset management, and eventual separation and retirement processing.
- The HRMS will be designed for the operational realities of a multi-directorate government department supporting hierarchical approval workflows, role-based data access, biometric integration, and regulatory compliance with applicable Government of Bihar HR policies and service rules.

7.2.2 RECRUITMENT & ONBOARDING

- The HRMS will provide a structured digital recruitment management facility covering the complete recruitment lifecycle; from vacancy notification and application intake through screening, selection, offer, and formal onboarding.
- The Recruitment workflows will be configurable to reflect the Department's applicable recruitment rules and approval structures.
- Upon selection and acceptance of offer, the onboarding process will be fully digitized capturing all mandatory joining documents, creating the employee's digital service record, establishing their HRMS account with appropriate role-based access, and initiating their inclusion in the payroll and attendance management framework.

7.2.3 EMPLOYEE PROFILE MANAGEMENT

- A comprehensive, structured employee profile will be maintained for every employee of the Department within the HRMS; capturing personal details, educational qualifications, service history, current posting, grade and pay particulars, asset assignments, training participation, performance assessment records, and leave balances.
- Employee profiles will be fully updatable by authorized HR administrators, with all modifications captured in a complete audit trail.
- Profile data will serve as the foundational master data for all other HRMS modules, ensuring consistency and eliminating redundant data entry across payroll, attendance, performance, and reporting functions.

7.2.4 ATTENDANCE MANAGEMENT & BIO-METRIC INTEGRATION

- The HRMS will deliver a real-time attendance management facility integrated with the Department's biometric devices enabling automated, tamper-resistant recording of employee attendance at all departmental offices across Bihar. Biometric attendance data will flow directly into the HRMS through a secure integration layer, eliminating manual attendance marking and the associated risks of proxy attendance.
- The system will generate daily, weekly, and monthly attendance reports for each employee and aggregate department-wide attendance analytics for supervisory and HR management review.
- Exception management for late arrivals, early departures, and unexplained absences will be automated, with configurable notification and escalation rules.

7.2.5 LEAVE MANAGEMENT

- A structured, workflow-enabled Leave Management module will govern the application, approval, tracking, and reporting of all categories of leave applicable to departmental employees under Government of Bihar service rules.
- Employees will be able to submit leave applications through both the web portal and the mobile application, with automatic routing to the designated approving authority based on the employee's posting and reporting structure.
- Leave balances will be automatically updated upon approval and will be viewable in real time by both the employee and their approving authority.
- The system will enforce leave policy rules like encashment limits, carry-forward caps, and mandatory leave requirements automatically, reducing administrative burden and ensuring policy compliance.

7.2.6 TOUR MANAGEMENT

- The HRMS will incorporate a Tour Management module enabling the structured recording and approval of official travel by departmental employees capturing tour purpose, destination, duration, and travel details.
- Tour records will be linked to the employee's attendance record for the tour period and will feed the payroll module for travel allowance and daily allowance computation.
- A reporting facility will enable supervisors and HR administrators to monitor tour frequency, duration, and geographic distribution across the department.

7.2.7 PAYROLL MANAGEMENT

- The HRMS will deliver a comprehensive, rule-based Payroll Management facility covering the complete payroll cycle for all departmental employees including basic pay computation based on pay matrix grade, allowance calculation (Dearness Allowance, House Rent Allowance, Travel Allowance, and other applicable allowances), statutory deduction management (Income Tax, GPF, NPS, professional tax, and loan recoveries), and net salary computation.
- The payroll processing will be automated drawing on attendance data, leave records, and tour records maintained within the HRMS and will produce pay slips, bank transfer files, and statutory compliance reports (Form 16, GPF statements, NPS statements) in the required formats.

- The payroll module will be configurable to reflect applicable Government of Bihar pay commission recommendations and policy revisions.

7.2.8 PERFORMANCE MANAGEMENT SYSTEM

- A structured Performance Management System will be embedded within the HRMS, enabling the Department to conduct periodic performance appraisal cycles for all eligible employees in accordance with applicable service rules.
- The module will support the definition of performance parameters and key result areas, self-appraisal submission by employees, assessment by the reporting officer, review by the reviewing officer, and final rating recording all through a digital, workflow-governed process.
- Quarterly Progress Reports for individual employees will be generated by the system as a structured output of the performance management cycle.
- Aggregate performance analytics including departmental performance score distributions, appraisal completion rates, and high-performer identification will be available through the integrated dashboards.

7.2.9 ASSET MANAGEMENT

- The HRMS will incorporate an Asset Management module enabling the systematic recording, tracking, and periodic reconciliation of Government assets assigned to departmental employees including computing equipment, vehicles, communication devices, and office furniture.
- The Asset assignment records will be maintained against each employee's profile, with a formal handover and takeover process documented within the system.
- Asset condition assessments and disposal records will be captured through the module, providing the Department with a complete, auditable asset register.

7.2.10 HOLIDAY CALENDAR MANAGEMENT

- A centralized, administratively maintainable Holiday Calendar will be maintained within the HRMS capturing all Gazetted Holidays, Restricted Holidays, and Departmental Closure days applicable to different categories of employees and different locations across Bihar.
- The holiday calendar will automatically integrate with the attendance management, leave management, and payroll modules, ensuring consistent, rule-compliant computation of working days, leave entitlements, and pay across the entire employee base.

7.2.11 REPORTING & ANALYTICS

- The HRMS will generate a comprehensive suite of standard operational reports like attendance reports, leave balance statements, payroll summaries, tour reports, asset utilization reports, and Quarterly Progress Reports across multiple dimensions and time periods, with export capability in PDF and Excel formats.
- In addition to this, the HRMS data will feed the platform's Business Intelligence and Analytics layer, enabling advanced HR analytics like 'workforce distribution analysis, leave utilization patterns, payroll trend analysis', 'performance distribution reporting', and 'vacancy management dashboards'

accessible to senior HR officials and departmental leadership through the integrated BI dashboard environment.

7.2.12 ROLE MANAGEMENT & MULTI-LEVEL ACCESS CONTROL

- The HRMS will implement a rigorous Role Management framework defining discrete access profiles for different categories of HRMS users including Employees (self-service access), Reporting Officers (team-level visibility and approval authority), HR Administrators (department-wide HR data management), Payroll Administrators (payroll processing and financial reporting), and Super Administrators (system configuration and audit access).
- The access will be strictly role-defined and location-scoped, ensuring that no user can access HR data beyond their authorized jurisdiction.

7.3 TRAINING MANAGEMENT INFORMATION SYSTEM (TMIS)

7.3.1 OVERVIEW

- The TMIS will provide an integrated, end-to-end training governance platform enabling Udyog Mitra and its associated training entities to manage the complete training lifecycle, right from trainer empanelment and course design through learner registration, delivery, attendance, assessment, certification, and outcome analytics.
- The TMIS will be tightly integrated with the AI Knowledge Platform, which will automatically populate the TMIS with structured, policy-derived training content ensuring that the training curriculum remains current and aligned with evolving scheme and policy frameworks without requiring manual content development effort.

7.3.2 TRAINER MANAGEMENT

- The TMIS will maintain a structured Trainer Registry capturing comprehensive profiles for all empaneled trainers including personal details, qualifications, domain expertise, subject certifications, training delivery experience, geographic availability, and performance history.
- The trainer empanelment will be workflow-governed, with defined eligibility criteria and an approval process.
- The Trainer performance assessed through learner feedback, assessment outcome data, and delivery quality metrics will be tracked within the system and will inform future trainer assignment decisions.

7.3.3 COURSE & CURRICULUM MANAGEMENT

- The authorized administrators will be able to create, maintain, and publish training courses and curricula through the TMIS by defining course titles, subject areas, learning objectives, delivery modes (online or offline), duration, pre-requisite requirements, and associated assessment frameworks.
- The subject areas will be manageable supporting the addition, modification, and removal of subject categories as the training mandate evolves.

- The AI Knowledge Platform will supplement administrator-created content by automatically generating structured training modules from ingested scheme documents and policy guidelines with AI-generated content available for administrator review, approval, and publication through the TMIS content management interface.

7.3.4 LEARNER REGISTRATION & ENROLMENT

- The MSME entrepreneurs, departmental employees, and other eligible learners will be able to register for available training programmes through the TMIS self-service interface accessible through both the web portal and the mobile application.
- The enrolment process will capture learner details, validate eligibility against defined course prerequisites, confirm enrolment, and communicate confirmation along with programme details through Email and SMS notification.
- The enrolment data will be linked to the learner's MSME registration profile or HRMS employee profile as applicable, creating a longitudinal training participation record for everyone.

7.3.5 TRAINING DELIVERY MANAGEMENT

- The TMIS will manage all aspects of training programme logistics and delivery including venue identification and scheduling for offline programmes, virtual platform linkage for online programmes, batch management, trainer assignment, and communication of programme details to enrolled learners and assigned trainers.
- A Training Calendar will be maintained within the system providing an enterprise-wide view of all scheduled programmes, their dates, venues, trainers, and enrolment status accessible to administrators, trainers, and learners through role-appropriate interfaces.
- Secluded Training management for residential or immersive training programmes with specific accommodation and logistics requirements will be supported through dedicated fields within the programme configuration module.

7.3.6 ATTENDANCE TRACKING WITH BIO-METRIC INTEGRATION

- Learner attendance for offline training programmes will be recorded through integration with biometric devices at training venues, ensuring accurate, tamper-resistant attendance capture.
- Online programme attendance will be tracked through the digital delivery platform integration.
- Attendance data will be available in real time to programme administrators and will automatically determine learner eligibility for assessment and certification based on the minimum attendance threshold defined for each programme.

7.3.7 ASSESSMENT & CERTIFICATION

- The TMIS will incorporate a structured assessment management facility enabling the design and delivery of post-training assessments including 'objective-type, short-answer types', and 'practical evaluation frameworks' through both online and offline modes.
- Assessment results will be automatically scored where objective assessment formats are used, with manual scoring facilitation for other formats.

- Upon successful completion of a programme and its associated assessment, the TMIS will automatically generate a digital Certificate of Completion which will be branded with Udyog Mitra's institutional identity, personalized with the learner's name, programme details, and completion date, and digitally signed by the authorized issuing authority.
- All training documents like course materials, assessment papers, and certificates will be uploadable and retrievable through the TMIS document management facility.

7.3.8 REPORTING, ANALYTICS AND TRAINING PAYMENTS

- The TMIS will generate comprehensive reports on all dimensions of training activity like programme-wise enrolment and completion statistics, trainer performance summaries, district-wise and sector-wise training participation, assessment score distributions, and learner feedback analysis.
- These reports will be available in standard formats for operational management use and will additionally feed the BI and Analytics platform for strategic training effectiveness analysis.
- Where training programmes involve fee-based participation, the TMIS will incorporate a training-related payment integration connecting to the platform's Payment Gateway facility to manage fee collection, payment confirmation, and refund processing in a structured, accountable manner.

7.4 GRIEVANCE REDRESSAL SYSTEM

7.4.1 OVERVIEW

- The Grievance Redressal System will provide a comprehensive, multi-channel, escalation-governed platform through which MSME enterprises, citizens, and other stakeholders can lodge grievances and complaints with Udyog Mitra and the Department of Industries, Bihar and through which the Department can manage, resolve, and report grievances in a structured, time-bound, and accountable manner.
- The system will be integrated with national dispute resolution infrastructure including ODR platforms and TReDS ensuring that Bihar's MSME grievance management is embedded within the broader GoI grievance governance ecosystem.

7.4.2 MULTI-CHANNEL GRIEVANCE INTAKE

- The Grievance Redressal System will support the intake of grievances through multiple channels like the web portal, the mobile application, Email, SMS, and a dedicated call center interface ensuring that all MSME enterprises and citizens can access the grievance mechanism through their preferred and most accessible channel.
- All grievances, regardless of intake channel, will be captured within the unified grievance management platform with a unique reference number, enabling consistent tracking and reporting across all intake modes.

7.4.3 GRIEVANCE CLASSIFICATION & ROUTING

- Upon intake, every grievance will be automatically classified by category like ‘delayed payment’, non-compliance by buyers’, regulatory harassment’, scheme denial’, ‘infrastructure complaint’, or other defined categories and routed to the designated responsible authority within the Department based on the classification and the geographic jurisdiction of the complainant.
- The routing logic will be configurable by authorized administrators, enabling the Department to update routing rules as its organizational structure and responsibility assignments evolve.

7.4.4 ESCALATION MATRIX MANAGEMENT

- A formally configured, multi-level Escalation Matrix will govern the lifecycle of every grievance within the system.
- Wherever, a grievance is not acknowledged, actioned, or resolved within defined turnaround time thresholds at the assigned level, the system will automatically escalate the grievance to the next level of the hierarchy — alerting both the escalating authority and the escalated authority through Email and SMS notification.
- The escalation matrix will be configurable for different grievance categories and jurisdictional levels, ensuring that the escalation structure reflects the Department's actual accountability framework.

7.4.5 GRIEVANCE TRACKING & RESOLUTION

- Both the grievant and the assigned departmental official will have real-time visibility into the status of each grievance through the platform interface.
- Grievants will receive automated status update notifications at each stage of the resolution process like acknowledgement, assignment, investigation, and resolution through their registered email and mobile number.
- The resolution recording will be structured requiring the responsible official to document the resolution action taken, supporting evidence, and the outcome before the grievance is formally closed.
- Grievances resolved within the defined turnaround time will be flagged as on-time closures while those resolved beyond the threshold will be flagged as overdue, with the delay recorded in the system's analytics.

7.4.6 FEEDBACK MECHANISM

- Upon resolution closure, the grievant will be invited to provide structured feedback on their satisfaction with the resolution process and outcome through a brief, structured feedback instrument accessible through the web portal, mobile application, or SMS response.
- The feedback data will be aggregated and analyzed within the Grievance Redressal System's reporting module, providing the Department with an ongoing, data-driven measure of grievance resolution quality and citizen satisfaction.

7.4.7 REPORTING, BULK COMMUNICATION AND EXTERNAL INTEGRATION

- The Grievance Redressal System will generate comprehensive reports on all dimensions of grievance activity like intake volumes by channel and category, pending grievance registers, turnaround time performance, escalation frequency, resolution rates, and satisfaction scores, in PDF and Excel formats, with real-time data feeding the BI and Analytics dashboard for senior management review.
- A Bulk Messaging facility will enable authorised officials to communicate with multiple grievant simultaneously for programme updates, resolution status communications, or policy announcements relevant to pending grievance categories.
- The system will be integrated with the Online Dispute Resolution (ODR) platform, TReDS, and other GoI grievance redressal mechanisms enabling grievances involving delayed payments, trade receivables disputes, or cross-jurisdictional issues to be routed or escalated to the appropriate national platform where necessary, ensuring that Bihar's MSME grievance management is seamlessly connected to the broader national dispute resolution infrastructure.

7.5 INTEGRATED JOB PORTAL WITH AI-POWERED RESUME BUILDER

7.5.1 OVERVIEW

- The Integrated Job Portal will provide a structured, AI-enabled employment facilitation platform connecting Bihar's job seekers with employment opportunities within the State's MSME sector serving simultaneously as a talent sourcing channel for MSME employers and as a skills-to-opportunity matching engine for job seekers.
- The portal will be embedded within the broader CIMS platform linking job seeker profiles to MSME employer registrations and enabling the Department to monitor and report on MSME sector employment generation as a scheme outcome metric.

7.5.2 JOB SEEKER REGISTRATION & PROFILE MANAGEMENT

- The Job seekers across Bihar will be able to register on the Job Portal through a guided, structured registration process capturing educational qualifications, skills, prior work experience, sector preferences, geographic availability, and language proficiency.
- The registered job seekers will maintain updatable digital profiles within the platform, accessible to MSME employer entities that have posted vacancy listings for which the job seeker's profile is relevant.

7.5.3 AI-POWERED RESUME BUILDER

- Upon completion of profile registration, the Job Portal's AI-Powered Resume Builder will automatically generate a professionally formatted, contextually appropriate resume for the job seeker drawing on the structured profile data entered during registration and supplementing it with AI-generated language enhancements, skill categorization, and presentation optimization.
- The Job seekers will be able to review, edit, and customize the AI-generated resume through an intuitive interface and download it in a standard digital format for use in their job search beyond the CIMS platform.

- The Resume Builder will leverage the platform's Natural Language Processing capability to continuously refine its output quality based on employer feedback signals captured within the portal.

7.5.4 MSME EMPLOYER ONBOARDING & JOB LISTING

- The Registered MSME enterprises, authenticated through their CIMS MSME registration profile will be able to onboard onto the Job Portal as employers and publish vacancy listings for positions available within their enterprises.
- The Job listings will be structured capturing job title, role description, required qualifications, experience requirements, salary range, location, and application deadline.
- The listings will be published on the portal's publicly accessible job board and will be visible to all registered job seekers, with AI-assisted matching surfacing the most relevant listings to job seekers whose profiles align with vacancy requirements.

7.5.5 APPLICATION MANAGEMENT AND DEPARTMENTAL VISIBILITY

- The Job seekers will be able to apply for listed positions directly through the portal with their registered profile and AI-generated resume submitted as the basis of the application.
- The MSME employers will receive and manage applications through the employer portal interface, with communication tools to acknowledge receipt, invite for interview, and provide application outcome notifications.
- The Department of Industries will maintain visibility of all job listings and application volumes through the departmental interface, enabling the portal's data to feed MSME sector employment generation metrics within the BI and Analytics platform.

7.6 BUSINESS INTELLIGENCE AND ANALYTICS PLATFORM

7.6.1 OVERVIEW

- The Business Intelligence and Analytics Platform will serve as the intelligence engine of the entire CIMS ecosystem aggregating, structuring, and presenting data from all transactional modules and external GoI data sources through a unified, on-premises analytical environment.
- It will deliver interactive dashboards, self-service ad-hoc analytics, KPI management, geospatial visualization, AI-driven insight generation, and comprehensive reporting capabilities to all authorized users across the Department's administrative hierarchy, from district-level officials to the most senior policy makers.
- The Detailed functional specifications for the BI and Analytics platform are provided in Annexure-4 of this proposal.

7.6.2 UNIFIED MSME AND SCHEME ANALYTICS DASHBOARD

- A purpose-built executive analytics dashboard will aggregate key MSME sector performance metrics with total registered enterprises, sector-wise distribution, scheme application pipeline, disbursement status, employment generation, geographic coverage, and financial health indicators into a visually compelling, interactive single-screen overview.

- The dashboard will support drill-down from State-level aggregates to District, Block, and Panchayat hierarchies with a single interaction, enabling both macro-level policy assessment and granular operational monitoring from the same interface.

7.6.3 DEPARTMENTAL HR ANALYTICS DASHBOARD

- A dedicated HR Analytics dashboard will provide departmental leadership and HR administrators with an integrated view of workforce metrics headcount by directorate and grade, attendance trends, leave utilisation, payroll expenditure analysis, performance score distributions, and vacancy management status.
- The HR analytics environment will enable data-driven workforce planning decisions and support the Department's human resource governance responsibilities.

7.6.4 TRAINING & CAPACITY BUILDING ANALYTICS

- The TMIS data will be presented through dedicated analytics views within the BI platform covering training programme reach (participants by district, sector, and enterprise size), completion and dropout rates, assessment performance distributions, trainer effectiveness scores, and training expenditure versus capacity building outcome metrics.
- These analytics will enable the Department to optimize its training programme portfolio and allocate training resources to the highest impact areas.

7.6.5 GRIEVANCE RESOLUTION ANALYTICS

- Grievance data from the Grievance Redressal System will be synthesized into structured analytics views displaying grievance intake trends by category and channel, average resolution turnaround times by category and jurisdiction, escalation frequency, overdue grievance registers, and citizen satisfaction scores.
- These analytics will enable departmental leadership to identify systemic grievance drivers, assess the performance of grievance resolution across different administrative units, and implement targeted service quality improvements.

7.6.6 AI GENERATED GAP ANALYSIS AND PERFORMANCE REPORTS

- The platform's AI and Machine Learning layer will automatically generate structured gap analysis and performance assessment reports identifying underperforming districts, sectors, or scheme categories based on real-time field data, and surfacing actionable recommendations for programme managers and policy officials.
- These AI-generated reports will be accessible through the integrated MIS dashboard and will be exportable in PDF, Excel, and CSV formats for distribution and archiving.

7.6.7 PREDICTIVE ANALYTICS & DEMAND FORECASTING

- The BI platform will incorporate predictive analytics capabilities leveraging Machine Learning models trained on historical MSME production, employment, export, and market data to generate forward-

looking demand forecasts for key MSME product categories, early warning signals for enterprises exhibiting financial stress indicators, and projections of scheme utilization and budget requirements.

- These predictive outputs will be presented as structured dashboard views and scheduled reports, enabling the Department to shift from reactive scheme management to proactive, intelligence-led programme governance.

7.7 ANALYTICAL REPORTS/CUSTOMIZED DASHBOARDS FOR STAKEHOLDERS

We will deploy a three-tier MIS dashboard architecture comprising data ingestion layer, analytical processing engine, and visualization layer. The core capabilities of the same shall include: -

- **Real-Time Data Integration:** Automated data pipelines from Udyam portal (API integration), scheme application databases, Health Clinic query management system, training attendance records, and implementing agency uploads with 15-minute refresh cycles.
- **Role-Based Dashboards:** Customized views for State Admin, SPIU, Department Users, and MIS Experts with granular permission controls.
- **Live Analytics Modules** shall include: -
 - Scheme-Wise Application Tracking (District/Category/Status Segregation)
 - Expert query resolution metrics (pending vs. resolved, domain-wise breakdown, response time analytics)
 - ONDC sales visualization (category-wise, district-wise, MSME-level performance). Aggregated insights displayed on the Patna MSME One Dashboard:
 - District-wise ONDC participation rates
 - Category-wise sales performance
 - Individual MSME sales statistics
 - Market Trends Analysis
 - Training participation statistics (attendance rates, completion metrics, district coverage)
 - DLI (Disbursement Linked Indicator) tracking with automated alerts for milestone achievement
- The reports shall be downloadable in Excels/CSVs etc. and shall have export functionality with configurable filters on “date ranges”, “district”, “scheme”, “category”

7.8 BACK-END FUNCTIONALITIES

7.8.1 MASTERS' MANAGEMENT

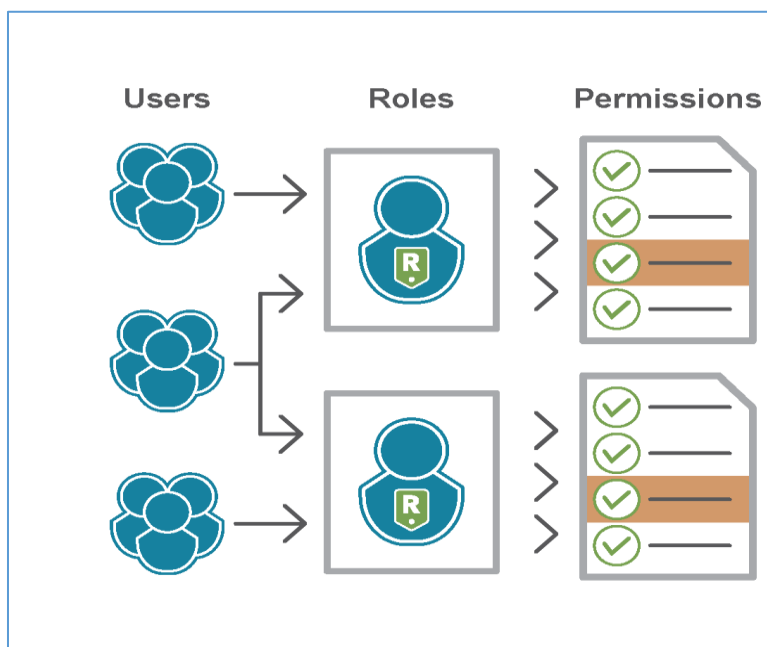
- This will ensure that all the master entries can be maintained at database level. This will help the department in using the information in a more effective way wherever required.
- Provision to maintain “Schemes Masters”; the schemes information with eligibility criteria and benefits provided shall be updated into the system

- Provision to add more schemes dynamically into the system like:-
 - MSME Enterprise Master (linked to Udyam Reg. No.)
 - Employee Master (HRMS)
 - Administrative Hierarchy Master (State → Division → District → Sub-Division → Block → Panchayat)
 - Scheme & Programme Master (State + Central)
 - Reference Data Master (sector codes, enterprise size categories, NIC codes, qualification codes)
 - Trainer & Course Master
 - Venue Master
 - Asset Master
 - Document Type Master
 - Holiday Calendar Master

7.8.2 ROLE BASED USER ACCESS MANAGEMENT

- This module will help the Administrator in managing the authorized users to manage & monitor the Complete Solution. This will have following features: -
 - Administrator will be able to Add/edit/block/unblock the users. For adding user, admin need to fill a form with basic details e.g. Name, email, mobile no, password etc.
 - On successful addition of user, user details will be sent to respective users by email.
 - Admin will be able to assign the Specific role to user.
 - The user list would be as follows: -
 - Replace user list with the following indicative role groups (final RBAC matrix to be finalised in SRS)
 - Platform-Level: Super Administrator, Platform Data Administrator, Security Administrator
 - Departmental – State Level: Principal Secretary / Director, MIS Expert, SPIU Users, Scheme Administrators
 - Departmental – Field Level: District Industries Centre (DIC) GM, Sub-Division Officer, Block-Level Officer, Panchayat-Level Coordinator
 - Users: HR Administrator, Reporting Officer, Employee, Payroll Officer
 - TMIS Users: Training Coordinator, Trainer, Learner / Trainee, Assessment Officer
 - Grievance Module: Grievance Officer (Tier-1, Tier-2), Escalation Authority, Nodal Officer (GRV-04)
 - Job Portal Users: Job Seeker, MSME Employer, Department Reviewer
 - Citizen-Facing: MSME Entrepreneur (Registered), MSME Applicant, Public Visitor (Read-only)
- Users created by admin will be able to login into system and do the specified role assigned to them. **The exact set of departmental users and their roles (rights and access privileges) shall be decided after conducting a detailed stakeholder consultation with department at regular intervals.**
- **By Incorporating their insights and addressing their specific needs, the framework will effectively safeguard sensitive information and functionalities, supporting the successful**

operation of the digital portal. This consultation will also ensure that the framework is comprehensive, secure, user-friendly and aligned with the strategic goals and regulatory requirements of the department.



7.8.3 PAYMENTS SECTION

- Provision for Integration with a MeitY-empanelled / RBI-authorized Payment Gateway with support for Net Banking, UPI, Debit / Credit Cards, NEFT/RTGS, and BBPS.
- Integration with Training-related payments including learner fee collection, trainer payouts, and venue payments.
- Provision to view the payment details for each record
- Provision to check payment failures/multiple payments
- Provision to view payment transaction audit-log.
- Provision to maintain an SMS and Email transaction audit log
- Provision for reconciliation of payments through CSV files received from the bank
- Provision for Refund / chargeback management workflow with maker-checker controls.
- Provision for multi-bank reconciliation support (auto-match + exception queue).
- Provision for GST invoice generation for fee-bearing services where applicable.

7.8.4 INFORMATION & FEEDBACK CENTRE (OPEN ACCESS FOR PUBLIC)

This module shall function as a platform for the MSMEs in the state to access information about various MSME schemes, incentives, state and central government initiatives and updates on all upcoming events and exhibitions.

The content of this section may need to be changed frequently. To manage the content of those pages, a Content Management Solution shall be implemented. The CMS shall be used for managing the content of pages which get updated frequently where the users can contribute and publish the content easily with very little knowledge about webpage designing.

7.8.4.1 DEVELOPMENT OF RESPONSIVE UI DESIGN

A Responsive UI Design will be developed to engage Desktop, Tablet and Mobile users using a single layout changing design, Responsive Design is a web development approach that suggests web pages should respond to the context in which they're loaded (primarily screen size) and change their user interface layout accordingly.

7.8.4.2 DEVELOPMENT OF USER INTERFACE DESIGN

This will ensure professional look and feel for the Web Application. This will be done by working together with Department in ensuring that all its expectations related to the overall aesthetics, content presentation structure and content organization structure are achieved. While preparing designs we will also make sure that they are developed as per the relevant guidelines. The design will be made responsive to providing optimal viewing experience easy reading and navigation with a minimum of resizing, panning, and scrolling across a wide range screen size of devices (from smart phones, tabs to desktop computer monitors).

7.8.4.3 DEVELOPMENT OF THREE (3) COLOR DESIGN THEMES BASED ON APPROVED UI DESIGN

This will ensure that the website has fresh look & feel on continuous basis. Three design themes based on the approved UI design will be developed to facilitate this. Website administrator will be able to adapt new look through the website management system control panel.

7.8.4.4 DEVELOPMENT OF GUIDELINES COMPLIANT TEMPLATE BASED ON IA AND UI DESIGN

This will ensure that the website is developed as per GIGW guidelines and is universally accessible. Following will be done to ensure this:

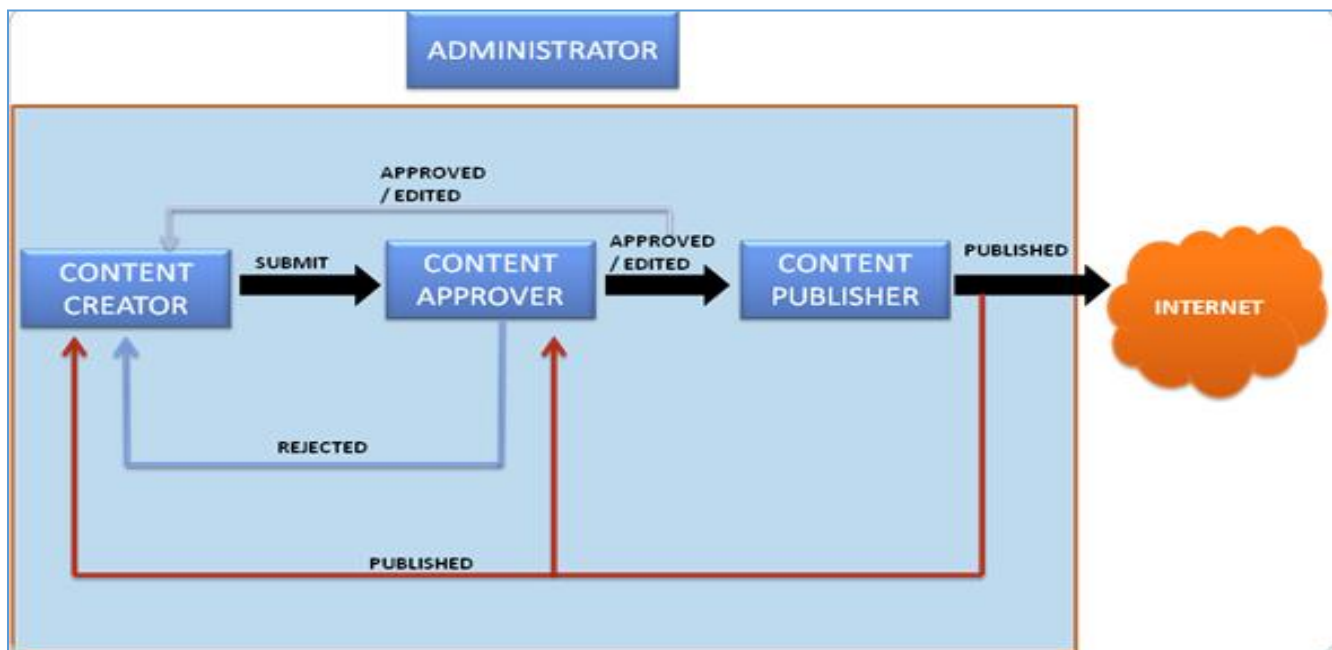
- Development of screen reader friendly website.
- Restructuring overall content with proper heading structure.
- Ensuring compatibility with popular browsers like Internet Explorer, Mozilla Firefox and smart mobile phones.

7.8.4.5 DEVELOPMENT OF A CONTENT MANAGEMENT SYSTEM (CMS)

This will help Department- in making changes in the static website content. This will have following features:-

- **Dynamic menus:** Menus and submenus can be created based on the page-tree as pages are added and subtracted.
- **Bi-lingual support:** Content of both languages English & Hindi can be managed from the same control panel. Administrator will be able to add content for both the languages from the same page.
- **Metadata Insertion:** Can be inserted for each page or globally, and for different languages.

- **Search Engine Friendly URL's:** A rewrite engine is software that modifies a web URL's appearance (URL rewriting). Rewritten URLs (sometimes known as short, fancy URLs, or search engine friendly - SEF) are used to provide shorter and more relevant-looking links to web pages.
- **Rich Text Editor:** Multiple RTEs with Microsoft word like icons support for Mac/safari and Mozilla Firefox on all platforms; WYSIWYG editors are highly configurable-buttons and CSS styles can be added and removed.
- **CSS Styled Content:** All aspects of the core functionality can be styled on the website. Most extensions can be styled through CSS.
- **Online Administration:** All administration is handled online via a current web browser.
- **Audit Trail:** Administrators have access to one log in the backend or individual logs on each page where they can view changes that have been made to the database.
- **Minimal Training Required:** Editing content is as easy as editing in Microsoft word. Menu links are automatically generated as an editor adds pages and most content elements features wizards for further ease of use.
- **User Management:** Administrator can add/edit or can manage users of CMS as content creator or contributor and content reviewer/publisher.
- **Site Map:** A hierarchical visual model of the pages of a Web site. Site maps help users navigate through a Web site that has more than one page by showing the user a diagram of the entire site's contents.
- **Role based CMS:**



- **Two or more stage workflow for content uploading (creator > publisher or creator> approver> publisher)**
- **CMS Features:**
 - Dynamic Menus
 - Meta-Data Insertion

- Short Search Engine Friendly URLs
 - Rich Text Editor
 - CSS Styled Content
 - Online Administration
 - Audit Trail
 - User Management
 - Sitemap
- Automatic archival of content (for time sensitive content like tender notices, recruitment notices, announcements, events etc.)
- Bilingual content management: The entire portal excluding application part will be provided in Hindi also. The visitors would be able to view the content with ease without any requirement of font download. The portal will allow admin to enter content for all modules & CMS pages separately in English/ Hindi.
- Dashboard for administrator
- Responsive design
- Audit Trail: This will be an automated system. At the time of any updation/addition into the portal through Administrative Section the System will create a Log; this would help the Super Administrator to generate the Report for a specified time period pertaining to any Login access into the system, any updating/addition into the portal, any unsuccessful attempts of Login into the system.
- Printer Friendly
- Secure, vulnerability free
- Site Search (including searchable documents): There would be a comprehensive search function for the proposed portal. The portal will be optimized for Search Engines. Full text search will be provided in the portal for all the content entered through the CMS
- Based on the latest standards and best practices for web sites like layout, navigation, accessibility, usability etc.

7.8.4.6 NAVIGATION & EASE OF USE

Users and visitors will usually leave a website if there are problems with navigation, or if it is not simple to use. To tackle ease of use, Uneecops information architects will analyze the information flow on the Website and how the user will interact with the various areas. To reach the below:

- Ease of use
- Smooth usage flow
- Minimum steps for reaching needed information
- General positive user experience

Use the latest techniques and technologies to make it easy for the non- sophisticated Internet users to quickly get familiar with using your service instead of spending a long time to find out how it works.

Through the user-friendly graphical user interface, familiar icons and simple business-related self-explanatory words; the users will know where to click to move forward.

7.8.4.7 ACTS/POLICIES/REGULATIONS REGARDING MSMEs/CIRCULARS/NEWSLETTERS

- This will allow the Administrator to add Acts/Policies/Regulations regarding MSMEs
- Administrator will be able to View/Add/Edit/Delete Messages through WYSIWYG editor.

7.8.4.8 WITHIN SITE SEARCH MODULE

- This module will allow website visitors to search for relevant information like documents, policies and other relevant information within the website.
- Site search will work based on keywords and will search the entire database driven content.
- Search results will be displayed in the form of a listing and will also have link to relevant pages within the website.

7.8.4.9 CONTACT US

This module will help the Administrator to manage contact information on the website. This module will have the following features:

- Contact Information will be displayed on Google Map. On clicking on that, visitors will be redirected to the Google map page.
- This page will display a map of Patna with locations of DICs, technical institutions, and incubation centers and the list of all the offices of the department along with a list of key personnel as well as their contact information.
- For contacting the personnel, there will be a form with fields like “Name”, “Email”, “Verify OTP”, “Phone”, “Message”, “Enter Captcha for added authentication”

7.8.4.10 POPULATION OF CONTENT

- This will ensure that the newly developed website has all the relevant content available on the new website. We will help the department in restructuring its content as per the GIGW guidelines. We will do one-time integration of department’s content into the newly developed website
- **The department will need to supply the final version of content within 1-2 weeks after approval of design concept to make sure that it is integrated in the website. The department will also need to cross-check to ensure correctness of the content before making new website live.**

7.8.4.11 OTHER CMS WEB PAGES

- Other Customized Modules containing various information modules will be developed as per the existing portal.

- The Administrator will be able to edit the contents of these sections for viewing by public at large.

7.8.5 DYNAMIC APPLICATION WORKFLOW, REVIEW AND FILTERING

- Provision to configure workflows across various internal processes of the Department of Industries, Government of Bihar for State and Central scheme operations including Bihar Industrial Investment Promotion Policy (BIIPP) incentive applications, MSME Cluster Development Programme, Mukhyamantri Udyami Yojana, PMEGP, and other notified schemes managed by Udyog Mitra and its directorates (BIADA, BICICO, BSFC, etc.)
- Provision for Configurable BPMN-aligned workflow engine supporting maker-checker, parallel approval, and conditional routing.
- Provision to screen the applications, flag incomplete or inconsistent entries, and prioritize applications based on a pre-defined criterion
- Provision to de-duplicate applications (by analyzing patterns and anomalies) and generate timely alerts in case of suspicious cases indicative of fraud or inconsistencies. the system can prompt for additional verification like OTP etc. or else assign it for manual intervention.
- Provision to tag each application with a unique identification number and use it to track the progress of the application.
- Provision for workflow versioning and audit wherein every workflow change is subject to a formal change-control mechanism

7.8.6 SLA MANAGEMENT

- Provision for defining the timelines for every step in the application process
- Auto-escalation mechanism to be incorporated in case the application is not accepted/rejected in the stipulated time-period.
- Provision for generating alerts based on time-out of assigned time to take action on a particular request
- Provision for generating reports on the pending actions, completed actions, long pending actions etc.
- Provision for linking the cross-cutting SLA module to grievance resolution, TAT tracking, escalation matrix and HRMS leave/tour SLAs.

7.8.7 ALERTS & NOTIFICATIONS

- SMS/E-mails Gateway can be integrated to send SMS & event-based reminders/notifications/alerts to various stakeholders at various events like e.g. verification, dependencies, escalations etc.
- Provision for configuring the alerts and notifications at designated events/junctures for timely updation to relevant stakeholders.
- Provision will be made for setting automated alert(s)/notification(s) etc. at identified events in the workflow for data consolidation in the form of App based as well as E-mail & SMSs

7.8.8 AUDIT TRAIL MANAGEMENT

- Availability of complete audit trail of all transactions (for e.g. add, update and delete) using transaction log reports, so that errors in data, intentional or otherwise, can be traced and reversed.
- The Audit Trail will be maintained to reflect modifications made by respective stakeholders. The Audit Trail will capture the login ID, date and time stamp of the operations conducted on the portal.
- A system level report containing a detailed audit log of all user activities will be available for downloading in excel and pdf format.
- The audit log would comprise of Module Name, Activity Type, Username, E-mail ID, Role details, last access date & time, last login date & time, Activity date & time, Logout Time & IP Address.

7.8.9 DOCUMENT MANAGEMENT

- Provision for all modules across the system to archive documents that meet specified criterion like age of documents or document type(s).
- Provision for easily archived documents, index documents, search/retrieve documents etc.
- Provision for retrieving documents through search or browsing the documents in directory
- Provision for version control with full revision history
- Provision for AES-256 encryption at rest; TLS 1.3 in transit
- Provision for configurable retention policies with secure cryptographic deletion on expiry.
- Provision to classify documents as “Public / Internal / Confidential / Restricted” with access control aligned to RBAC
- Provision for virus/malware scanning on every upload
- Provision for Digi locker integration for citizen document fetch & e-sign workflows

7.8.10 SINGLE SIGN-ON

- The platform's Identity and Access Management framework will implement SSO using industry-standard federated identity protocols (SAML 2.0 and OAuth 2.0 / OpenID Connect) enabling integration with whichever Government of Bihar or Government of India SSO infrastructure is nominated by Udyog Mitra during the SRS phase.
- The Identity Provider integration will be configurable rather than hard-coded, ensuring that the platform can interoperate with NIC-led SSO infrastructures (MeriPehchaan / Janparichay), Bihar State Service Delivery Gateway, the Single Window System SSO, or any future State / Central SSO infrastructure designated by the Department, without architectural rework.
- The SSO framework will be implemented in conjunction with RBAC, Row-Level Security (RLS), and Object-Level Security (OLS) as mandated under RFP

7.9 DEVOPS, INFRASTRUCTURE AND SECURITY

7.9.1 OVERVIEW

- This module represents the cross-cutting technical discipline that will govern the platform's development pipeline, production deployment, ongoing operational health, and security posture throughout the full 60-month engagement.

- While not a citizen-facing functional module, it is the operational foundation upon which the reliability, security, performance, and continuity of all other six modules depend.

7.9.2 DEVELOPMENT & DEPLOYMENT PIPELINE (DEVOPS)

- Uneecops will establish and maintain a fully automated, Continuous Integration and Continuous Deployment (CI/CD) pipeline for the CIMS platform enabling rapid, reliable delivery of software updates, bug fixes, and enhancement releases throughout the development phase and the 52-month O&M period.
- The pipeline will incorporate automated code quality checks, static code analysis, automated unit and regression test execution, and staged environment promotion ensuring that only tested, approved code reaches the production environment at the Bihar State Data Centre.

7.9.3 INFRASTRUCTURE MANAGEMENT AT BIHAR STATE DATA CENTRE

- All production infrastructure like application servers, database servers, analytical data stores, AI/ML compute, integration middleware, and network components will be provisioned, configured, and maintained within the Bihar State Data Centre environment provided by the Department.
- Uneecops's infrastructure management responsibilities will encompass server health monitoring, capacity management, operating system and middleware patch management, database administration, backup and recovery operations, and periodic infrastructure review and optimization to ensure that the platform's performance and availability commitments are sustained as data volumes and user concurrency grow over the O&M period.

7.9.4 PERFORMANCE MONITORING & SLA MANAGEMENT

- Uneecops will maintain a real-time performance monitoring environment, tracking application response times, database query performance, API latency, ETL pipeline throughput, BI dashboard rendering times, and system resource utilisation across all platform layers continuously.
- Performance metrics will be reported to the Department through a structured monthly operational report, with proactive alerts triggered whenever metrics approach defined SLA threshold boundaries.
- Load testing will be conducted periodically during the O&M period to validate that the platform continues to meet its performance commitments as user concurrency and data volumes evolve.

7.9.5 BUSINESS CONTINUITY & DISASTER RECOVERY

- Uneecops will implement and maintain a formally documented Business Continuity and Disaster Recovery (BCDR) framework for the CIMS platform defining Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) for all platform components and establishing tested procedures for rapid service restoration in the event of infrastructure failure, data corruption, or other business continuity events.

- The High Availability failover configuration will be subject to periodic failover drills to validate its operational readiness, with drill results documented and reviewed with the Department.

7.9.6 O&M SERVICE FRAMEWORK

- The 52-month Operations and Maintenance period commencing from go-live and concluding Month 60 of the engagement will be governed by a formally documented O&M Service Framework covering incident management, service request fulfilment, change management, problem management, and continual service improvement.
- The framework will define tiered incident severity classifications with associated response and resolution SLAs, a structured change management process for all platform modifications, a user support mechanism for departmental users and system administrators, and a formal continual service improvement programme that systematically reviews operational performance and implements targeted improvements across the O&M period.

7.10 OTHER NON-FUNCTIONAL REQUIREMENTS

7.10.1 RESPONSIVENESS/ADAPTABILITY

- The portal will be designed to be device independent and work seamlessly on different devices and their updated version such as Desktop, Laptop, Mobiles, tablets, iPad, etc.
- The re-developed portal shall support all major & latest browsers on both web & mobile devices/smartphones/tablets/phablets etc.
- The UI/UX level changes will involve addition/removal of buttons on the existing implemented solution as per the business requirements

7.10.2 SCALABILITY

- The system's deployment architecture would be designed as such to handle an increasing number of users and documents over time. In order that the system seamlessly handles increasing load over a period of time, the architecture of the same shall be designed as such that it is amenable to both horizontal and vertical scaling e.g. due to rapidly growing quantum of data, the solution proposed can scale upwardly to accommodate the increase in data being parsed and churned out.

7.10.3 OPEN STANDARDS & INTER-OPERABILITY

- The platform shall be open, modular and scalable (Standards, Open API, Micro-Services), to allow changes in sub-system level without affecting other parts, architected to work completely within a heterogeneous compute, storage and multi-level environment.
- The envisaged portal will support commonly used open source or proprietary platforms and support all standard databases and avoid any technology provider lock-in.

7.10.4 REPORTING

- Application will have reporting flexibility, scheduling of reports and processes, and the ability of the system to design/create specific reports quickly.

7.10.5 SECURITY FEATURES

7.10.5.1 OVERALL SOLUTION SECURITY

Overview of security parameters which would help to achieve the above-mentioned features as per the RFP:

- **Identification:** Establishing the claimed identity of an entity.
- **Data Encryption:** Ensuring data content is indecipherable during its travel and storage. This includes encryption of stored data and any communication between modules. All facial data shall be encrypted during storage.
- **Data Confidentiality:** Ensuring the protection and non-disclosure of data because of the nature of information.
- **Data Availability:** Ensuring that data are present, accessible, and obtainable, in a timely fashion and that data are fit and ready for use.

7.10.5.2 DATABASE SECURITY

- Data will be stored in encrypted format
- Appropriate user permission will be given on DB level
- Protection from malicious attempts to steal (view) or modify data
- 100% Web service-based Database connectivity.

7.10.5.3 APPLICATION SERVER SECURITY

- Ensure against SQL injection attacks will be taken
- Ensure against script injections and cross site scripting (XSS)
- Access to Host Server through VPN only
- Secure from latest OWASP top-10 vulnerabilities through CERT-In Certification.

7.10.5.4 PROTECTION FROM VIRUS/WORM ATTACKS

- Application Security
- Authentication and authorization mechanisms to allow specific users access only to required data

7.10.5.5 DATA SECURITY

- Multi-Layered Authentication for content Updating and publication by authorized administrators: System will have Module-wise / department wise Role based access and menu / operation restriction for increasing the security aspects
- Use of Strong Password Policy

8 TRAINING & HAND-HOLDING SUPPORT

- Comprehensive training/handholding must be conducted at department headquarters. The same will be Classroom based (CBT) and instructor led (ILT).
- The training may be conducted via online collaboration platforms like Zoom, Google Meet, MS-Teams etc.
- All infrastructure related to the offline training like seating space, air conditioning, IT infrastructure etc. will be provided by the department along with a list & profile of audiences.
- The training will likely be conducted in batches on train-the-trainer model wherein master trainers shall be created for imparting future, refresher/renewal trainings or fresh half-yearly/quarterly trainings during the support period
- The training shall be conducted in a hybrid mode i.e. both virtual and physical and all training venues shall be restricted to the department office in Patna, Patna only in case of physical sessions.

9 SUPPORT & MAINTENANCE ROADMAP

The support and maintenance shall be conducted for a period of 52 months post go-live

9.1 AMC/TECHNICAL SUPPORT FOR A PERIOD OF 52 MONTHS (POST GO-LIVE)

We will provide primarily offsite support (& onsite mode in case of emergent requirement on hop-on & hop-off modes) from the date of Go-Live as per severity of case to ensure SLA & performance compliance. The support period shall cover the following: -

- **Address, analyze and fix any technical glitches** within the existing features.
- **Resolution of errors/bugs** (if any), software updates, changes in the software that may be necessary due to legal/statutory changes etc.
- **Rectification of bugs** like content formatting issues, image placement, link mismatch etc. immediately after user request.
- **Maintenance and upkeep of any upgrades** of the solution will be a part of the ongoing maintenance period.
- **Technical Assistance** through email and phone.
- **Fixing Security Vulnerabilities** whenever any issue is reported or in the case of cyber-attacks.
- **Training & Handholding** of user representatives on Content Management System workflow on demand basis.

- **Addressing Server related issues** to ensure uptime by coordinating with data center.
- **Addressing Performance related issues** to fix issues like no response, slow response, app crashing etc.
- **Addressing Browser Compatibility related issues** to ensure smooth functioning of website on updated versions of all popular web browsers.
- **Addressing Device Compatibility Related Issues** to ensure smooth functioning of website on all Personal Computers, Smart Phones, Tablet PC's and disabled specific devices.
- **Implement software updates and new version releases** during the currency of the contract provided that the new release does not adversely affect systems operation or performance or require extensive reworking of the systems.
- **Quality Assurance/Testing** of any changes being implemented
- **Hand holding the stakeholders** – Regarding data entry, API integration, Sandboxing, onboarding etc.
- **Conducting Minor Code-Level Changes like** changing static images, text, updating your privacy policy, site terms of use, disclaimers etc. and other small enhancement requests that continuously improvise and optimize the existing solution
- **Continuous Production Support** that includes bug fixing, managing, monitoring, and maintenance of Production Applications/Workflows/E-mail Notifications and taking regular back-ups throughout the entire duration of contract.

9.2 ENHANCEMENTS/CUSTOMIZATIONS UNDER CHANGE REQUESTS

- In case any new enhancements or customizations/major code level changes are to be conducted in the website during the maintenance phase, the same shall be analyzed and if the outcome of the analysis mandates utilization of resources beyond the reasonable thresholds, a fresh proposal with a change request shall be submitted to the department consisting mainly of Business Analyst, Developer, Designer & Tester.
- Such changes can be of the class like: -
 - Customization of various Forms Workflows, and Views, including data migration across business verticals basis changes being implemented in the existing forms/fields for onboarding new teams or any other requirements.
 - Additions/ modifications in any of the modules not specified elsewhere in SoW mentioned above
- Change request will not contain bug fix and update release. Bug fix and update release should be reported and handled by Uneecops at no additional cost and should form a part of the regular operations & maintenance.
- Uneecops shall raise the change request as per the agreed terms and the same shall be finalized in mutual discussion with department.

10 CONDUCTION OF SECURITY AUDIT

- The new Web & mobile application will be audited by Cert-in empaneled auditor, and all the vulnerabilities will be fixed. Upon completion, security clearance certificate of cert-in empaneled auditor will be submitted.
- Uneecops' shall conduct one security audit before going live with this project and shall subsequently be conducted once every year (each for web and mobile application) until Uneecops and the department are contracted.

11 PROJECT PLAN WITH WORK BREAKDOWN STRUCTURE (WBS) & APPROACH FOR DATA GOVERNANCE

11.1 PROJECT PLAN & WORK BREAK-DOWN STRUCTURE

11.1.1 OVERVIEW

The Work Breakdown Structure presented herein constitutes the formal decomposition of the total scope of the CIMS and Business Intelligence Platform engagement into a hierarchically organized set of deliverable-oriented work packages structured across four phases, seven functional modules, and all cross-cutting technical disciplines that together comprise the full 60-month engagement.

The WBS has been developed in strict conformance with the milestone framework mandated by the tender and reflects the parallel-track Agile delivery methodology that Uneecops Technologies will employ throughout the development phase. This WBS is organized across four levels of decomposition as follows: -

- **Level 1-Phase:** The highest level of decomposition, aligning to the four engagement phases Inception and Architecture, Iterative Development and Integration, UAT and Go-Live, and Operations and Maintenance.
- **Level 2-Work Stream:** Each phase is decomposed into discrete work streams like functional modules, cross-cutting technical disciplines, and governance work streams that map directly to the delivery tracks and organisational responsibilities of the project team.

The WBS does not contain any commercial, financial, or effort-related information. It is a scope decomposition instrument only providing the evaluation committee and departmental project stakeholders with a complete, transparent view of the work that Uneecops Technologies will execute across the 60-month engagement.

The detailed, fully formatted WBS is presented in the accompanying Excel file. The following section provides a narrative overview of the major work streams and their constituent activity groups.

11.1.2 WBS NARRATIVE OVERVIEW

Phase-1: Inception, Requirements Validation, and Architecture (Weeks 1-8)	This phase establishes the complete foundational framework for the engagement. Its work streams encompass: - • Project Governance Establishment covering kick-off, governance structure, communication planning, and risk framework initialization • Requirements Validation covering structured stakeholder workshops across all seven functional modules, requirements documentation, and formal sign-off with Udyog Mitra • Solution Architecture covering the production of all architectural artefacts including the Solution Architecture Document, Data Architecture Document, Security Architecture Document, and Deployment Architecture Document • UI/UX Design covering user research, wireframing, prototype development, and design system establishment. • Integration Architecture covering Interface Control Document production for all GoI integration points • Infrastructure Setup covering environment provisioning at the Bihar State Data Centre, CI/CD pipeline initialization, and automated testing framework establishment and • Project Planning covering Master Project Plan baseline, sprint calendar, and risk register establishment.
Phase-2: Iterative Development and Integration (Weeks 9-24, Milestone 3)	This phase constitutes the core development activity of the engagement, executed across four parallel delivery tracks through a series of two-week Agile sprints. Its work streams encompass • Module Development for each of the seven modules namely MSME MIS, HRMS, TMIS, Grievance Redressal, Job Portal and AI Resume Builder, BI and Analytics Platform, and the AI and Machine Learning layer; • GoI and External Integration Development covering the implementation and testing of all integration interfaces with Udyam, MSME Ministries, TReDS, ODR, ONDC, biometric devices, and payment gateway • Mobile Application Development covering native iOS and Android application development across all citizen-facing and field-level modules • System Integration Testing covering cross-module integration validation and end-to-end scenario testing and

	• DevOps and Security covering continuous pipeline management, security monitoring, and infrastructure optimizations throughout the development phase.
Phase-3: UAT, Security Audit, Training, and Go-Live (Weeks 25-35, Milestone 4)	This phase transitions the platform from development completion to full production readiness. Its work streams encompass • User Acceptance Testing covering UAT planning, structured UAT execution with departmental end users across all modules and administrative levels, defect management and re-testing, and formal UAT sign-off • Application Security Audit covering the formal audit by the Department's appointed Certified Auditor, vulnerability remediation, re-audit, and security compliance certification; • Performance and Load Testing covering test planning, execution at defined concurrency levels, and performance certification; • Training and Change Management covering training plan development, role-based training delivery in Hindi and English, and training documentation production • Data Migration and Onboarding covering data migration planning, execution, validation, and MSME onboarding support; • Documentation covering the production of all system and user documentation deliverables; and Go-Live — covering production readiness verification, go-live execution, and Go-Live Acceptance Report sign-off.
Phase 4: Operations and Maintenance (Month 8 through Month 60)	This phase governs the 52-month post-go-live O&M period. Its work streams encompass • Service Management covering incident management, service request fulfilment, and helpdesk operations within defined SLA tiers; Infrastructure and Platform Operations covering continuous performance monitoring, database administration, backup and recovery, and capacity management • Security Operations covering security patch management, periodic vulnerability assessments, CERT-In compliance, and periodic re-audits • Enhancement and Change Management covering the intake, assessment, development, testing, and deployment of platform enhancements incorporating new schemes, policy changes, and departmental requirements • AI and Analytics Operations covering AI model retraining, knowledge graph updates, and analytical content governance;

	• Governance and Reporting covering monthly operational reporting to Udyog Mitra, quarterly service review meetings, and continual service improvement and • Engagement Closure covering final knowledge transfer, documentation handover, and formal engagement closure at Month 60.
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11.1.3 WORK BREAKDOWN STRUCTURE (WBS)

CENTRALISED INTEGRATED MANAGEMENT SYSTEM (CIMS) — WORK BREAKDOWN STRUCTURE (KEY DELIVERABLES VIEW)						
WBS ID	Level	Phase	Work Package / Activity	Key Deliverables	Milestone Alignment	Quality Gate
1.0	Level 1 — Phase	Phase 1	INCEPTION, REQUIREMENTS VALIDATION & ARCHITECTURE (Weeks 1–8)	Approved architecture; validated requirements; provisioned environments	Milestone 2 — Week 8	Architecture & Design Sign-off
1.1	Level 2 — Work Stream	Phase 1	Project Governance Establishment	Charter; Governance Framework; Communication Plan	Weeks 1–2	PM + Client sign-off
1.2	Level 2 — Work Stream	Phase 1	Requirements Validation	Validated Requirements Document; Functional Spec baseline	Weeks 1–5	Udyog Mitra sign-off
1.3	Level 2 — Work Stream	Phase 1	Solution Architecture	Technical Design Document containing various types of architectures	Weeks 3–7	Architecture Review Board sign-off
1.4	Level 2 — Work Stream	Phase 1	UI/UX Design	Design System; Web Portal Prototype; Mobile App Prototype	Weeks 2–7	Udyog Mitra UX approval
1.5	Level 2 — Work Stream	Phase 1	Integration Architecture	Updated Technical Design Document containing cascaded	Weeks 3–7	Integration Spec. + client sign-off

				changes on various types of architectures		
1.6	Level 2 — Work Stream	Phase 1	Infrastructure Setup	Bihar SDC Dev/Test/Staging environments; CI/CD pipeline; Test Strategy	Weeks 1–8	Environment Readiness Certificate
1.7	Level 2 — Work Stream	Phase 1	Project Planning	Master Project Plan with Data Governance Framework	Week 8	PM + Udyog Mitra approval
2.0	Level 1 — Phase	Phase 2	ITERATIVE DEVELOPMENT & INTEGRATION (Weeks 9–24)	Substantially-complete, integration-tested CIMS platform across all 7 modules	Milestone 3 — Week 24	Milestone 3 Review Sign-off
2.1	Level 2 — Work Stream	Phase 2	Track 1 — MSME MIS Development	MSME MIS module — all features + Gol integrations	Weeks 9–24	Module Lead + QA per sprint
2.2	Level 2 — Work Stream	Phase 2	Track 1 — Gol and External Integrations	11 Gol integration interfaces — tested and operational	Weeks 11–23	Integration Spec. + QA sign-off
2.3	Level 2 — Work Stream	Phase 2	Track 2 — HRMS Development	HRMS — lifecycle, payroll, biometric, performance	Weeks 9–24	Module Lead + QA per sprint
2.4	Level 2 — Work Stream	Phase 2	Track 2 — TMIS Development	TMIS — trainer, learner, AI content, certification, payments	Weeks 9–24	Module Lead + QA per sprint
2.5	Level 2 — Work Stream	Phase 2	Track 3 — Grievance Redressal System	Grievance system — multi-channel intake, ODR/TReDS integration	Weeks 9–22	Module Lead + QA per sprint
2.6	Level 2 — Work Stream	Phase 2	Track 3 — Job Portal & AI Resume Builder	Job Portal + AI Resume Builder	Weeks 9–24	Module Lead + QA per sprint
2.7	Level 2 — Work Stream	Phase 2	Track 4 — BI & Analytics Platform	On-prem BI/Analytics — dashboards,	Weeks 9–24	BI Lead + QA per sprint

				KPIs, ad-hoc, geospatial, predictive		
2.8	Level 2 — Work Stream	Phase 2	AI / ML Platform Development	AI/ML Layer — Knowledge Graph, NLP pipeline, predictive models	Weeks 9–22	AI/ML + Solution Architect sign-off
2.9	Level 2 — Work Stream	Phase 2	Mobile Application Development	Native iOS + Android applications (citizen and field modules)	Weeks 10–24	Module Leads + QA sign-off
2.10	Level 2 — Work Stream	Phase 2	System Integration Testing	Integration Test Reports — all modules and 11 GoI ICDs	Weeks 20–24	QA Lead sign-off
3.0	Level 1 — Phase	Phase 3	UAT, SECURITY AUDIT, TRAINING & GO-LIVE (Weeks 25–35)	Go-Live Acceptance; Security Compliance; Trained user base	Milestone 4 — Week 35 (Month 8)	Go-Live Acceptance Sign-off
3.1	Level 2 — Work Stream	Phase 3	User Acceptance Testing	UAT Sign-off (7 modules); Defect Closure Certificates	Weeks 25–33	Udyog Mitra UAT sign-off
3.2	Level 2 — Work Stream	Phase 3	Application Security Audit	Security Compliance Certificate; Remediation Report	Weeks 25–33	Certified Auditor sign-off
3.3	Level 2 — Work Stream	Phase 3	Performance and Load Testing	Load Test Certification — ≥100 concurrent users at peak	Weeks 28–32	QA Lead + DevOps sign-off
3.4	Level 2 — Work Stream	Phase 3	Training, Change Management & Documentation	Training Completion Records; Full Documentation Set v1.0	Weeks 29–34	Training Lead + PM sign-off
3.5	Level 2 — Work Stream	Phase 3	Data Migration and Go-Live	Go-Live Acceptance Report; Signed Go-Live Certificate	Week 35 — Milestone 4	Udyog Mitra authorised sign-off

4.0	Level 1 — Phase	Phase 4	OPERATIONS & MAINTENANCE (Month 8 – Month 60 52 Months)	Monthly Operational Reports; Enhancement releases; Annual reviews; Final Handover	Engagement Closure — Month 60	Final Handover Certificate
4.1	Level 2 — Work Stream	Phase 4	Service Management	Monthly incident logs; service request records; helpdesk reports	M8–M60 (Continuous)	Monthly SLA compliance
4.2	Level 2 — Work Stream	Phase 4	Infrastructure & Platform Operations	Infrastructure health reports; Backup logs; BCDR drill reports to be facilitated with co-operation from BSEDC team	M8–M60 (Continuous)	Monthly operational report
4.3	Level 2 — Work Stream	Phase 4	Security Operations	Security patch records; VA reports; CERT-In compliance status	M8–M60 (Continuous)	Security compliance status
4.4	Level 2 — Work Stream	Phase 4	Enhancement & Change Management	Enhancement releases; deployed platform updates; updated docs	M8–M60 (As required)	Change Advisory Board approval
4.5	Level 2 — Work Stream	Phase 4	AI, Analytics & Data Governance Operations	AI retraining records; Quarterly DQ reports; Annual classification & access reviews	M8–M60 (Periodic)	Data Governance Council review
4.6	Level 2 — Work Stream	Phase 4	Governance, Reporting & Engagement Closure	Monthly Operational Reports; Quarterly Review Minutes; Final Handover Certificate (M60)	M8–M60 / Closure at M60	Final Handover Certificate

11.1.4 DATA GOVERNANCE APPROACH

11.1.4.1 INTRODUCTION & GOVERNANCE IMPERATIVE

The CIMS platform will, at full operational maturity, serve as the primary data repository and intelligence engine for one of India's most significant State-level MSME development programmes. It will hold and process data of extraordinary institutional sensitivity and operational criticality like MSME enterprise registration and financial records, individual employee HR and payroll data, citizen grievance records, training participation histories, and analytical datasets informing State policy and programme decisions. The governance of this data is therefore not a peripheral technical consideration; it is a core institutional responsibility that Uneecops Technologies treats with the same seriousness as functional delivery.

The Data Governance Framework proposed herein establishes the principles, structures, policies, and operational procedures through which the CIMS platform's data assets will be managed, protected, and sustained in a condition of integrity, accuracy, accessibility, and regulatory compliance throughout the full 60-month engagement and beyond.

The framework is designed in alignment with the Government of India's National Data Governance Framework Policy, the Information Technology Act 2000 and its 2008 Amendment, CERT-In guidelines on Information Security, ISO 27001:2022 (Information Security Management), and ISO 20000-1:2018 (IT Service Management) ensuring that Bihar's CIMS data governance posture meets both national and international standards applicable to public sector digital infrastructure.

11.1.4.2 DATA GOVERNANCE STRUCTURE

Data Governance Council

Uneecops Technologies recommends the establishment of a Data Governance Council as a joint governance body between Udyog Mitra and the Uneecops project team responsible for oversight of all data governance policies, resolution of data ownership disputes, approval of data classification changes, and periodic review of data governance effectiveness. The Council should meet quarterly, chaired by a senior representative of Udyog Mitra, and should include designated data owners from each of the Department's directorates and the Uneecops Project Manager and Solution Architect.

Data Owners

A Data Owner will be designated for each major data domain within the CIMS platform corresponding to the organizational unit of the Department that holds primary institutional accountability for that data. Data Owners are responsible for approving access requests to their data domain, defining data quality standards and acceptable value ranges, reviewing and approving data classification designations, and authorizing retention and archival decisions for their data assets. The following data ownership structure is proposed: -

Data Domain	Proposed Data Owner
MSME Enterprise Data	Director, Directorate of MSME
Scheme and Programme Data	Director, Directorate of Industries
HR and Payroll Data	Head, Human Resources — Department of Industries
Training and Capacity Building Data	Designated Training Authority — Udyog Mitra
Grievance and Dispute Resolution Data	Chief Grievance Officer — Udyog Mitra
Job Portal and Employment Data	Designated Employment Authority
BI and Analytics Data	Head, Analytic Cell — Department of Industries

Data Stewards

Each Data Owner will designate one or more Data Stewards typically senior departmental officials within their directorate who exercise day-to-day operational responsibility for data quality monitoring, data entry standard enforcement, access review management, and first-level data issue resolution within their domain.

Platform Data Administrator

Unecops's designated Database Administrator and nominated data management personnel will serve as the Platform Data Administrator responsible for the technical implementation of data governance policies within the CIMS platform, including access control configuration, audit log management, data quality rule implementation, and retention policy enforcement.

11.1.4.3 DATA QUALITY STANDARDS

The integrity and utility of the CIMS platform's analytical and operational outputs are entirely dependent on the quality of the data it holds. Unecops will implement a structured Data Quality Management programme governing the accuracy, completeness, consistency, timeliness, and validity of all data within the platform across five quality dimensions: -

Accuracy	All data entered the CIMS platform must accurately represent the real-world entity or event it describes. Validation rules will be configured at the point of data entry for all mandatory and critical data fields enforcing format standards, value range constraints, reference data consistency, and cross-field logical dependencies. For example, MSME registration data will be validated against the Udyam Registration Portal at the point of onboarding, ensuring that the enterprise's CIMS profile is consistent with its GoI-registered identity.
Completeness	Mandatory data fields will be enforced at the point of entry preventing the creation of incomplete records that would impair the platform's reporting, analytics, and scheme eligibility assessment functions. Data completeness dashboards will be available to Data Stewards and departmental administrators, enabling proactive identification and resolution of records with missing mandatory information.
Consistency	Data values that are shared across multiple modules such as the MSME unique identifier, employee identifiers, district and block codes, and scheme reference codes will be maintained as governed master data entities within the platform's Master Data Management layer, ensuring that the same entity is represented consistently and without duplication across all modules and analytical views.
Timeliness	Data refresh cycles will be defined for all data categories distinguishing between real-time updates (MSME registration changes, grievance status updates, attendance records), daily batch updates (payroll calculations, scheme application status), and periodic updates (financial performance data, compliance monitoring data). ETL pipelines will be monitored for successful execution within defined windows, with alerts triggered to the Platform Data Administrator upon pipeline failure or data freshness threshold breach.
Validity	All coded and categorical data fields scheme codes, sector classifications, district and block codes, employee grade codes, and other reference data will be constrained to approved reference data lists maintained within the platform's reference data management layer. Entries outside approved reference value sets will be rejected at the point of entry, preventing the accumulation of invalid coded data that would compromise analytical accuracy.

11.1.4.4 MASTER DATA MANAGEMENT

The CIMS platform will govern five categories of Master Data entities that serve as the authoritative, shared reference point for multiple modules and analytical contexts:

MSME Enterprise Master	The definitive registry of all MSME enterprises registered on the CIMS platform, each uniquely identified by a CIMS-assigned identifier that is cross-referenced to the Udyam Registration Number from the GoI Udyam portal. The MSME Enterprise Master will serve as the authoritative source of enterprise identity across the MSME MIS, Job Portal, Grievance System, and BI/Analytics modules. All module-level MSME records will reference the Enterprise Master rather than maintaining independent enterprise identification — ensuring consistency and eliminating duplication.
Employee Master	The definitive registry of all departmental employees across the Department of Industries and its directorates maintained within the HRMS module and serving as

	the authoritative source of employee identity for attendance, payroll, performance, asset management, and HR analytics functions. The Employee Master will be aligned with the Department's service record system, with new employee records created only upon formal onboarding approval within the recruitment workflow.
Administrative Hierarchy Master	The authoritative reference for Bihar's administrative hierarchy State, Division, District, Sub-Division, Block, and Panchayat used consistently across all modules for geographic scoping, data entry location assignment, and hierarchical dashboard drill-down. Updates to the administrative hierarchy (boundary changes, new administrative units) will be managed through a formal change process, ensuring consistent reflection across all modules.
Scheme & Programme Master	The authoritative reference for all State and Central Government schemes managed through the CIMS platform capturing scheme identifiers, names, sponsoring ministry or department, eligibility criteria, and operational status. The Scheme Master will be maintained by the designated data stewards for the Scheme and Programme data domain, with additions and modifications subject to a formal approval workflow.
Reference Data Master	The centralized repository of all coded reference data used across the platform sector classifications, enterprise size categories, educational qualification codes, geographic codes, and all other reference value sets. The Reference Data Master will be governed by the Platform Data Administrator, with all additions and changes requiring Data Owner approval and a formal release process to ensure consistent propagation across all modules.

11.1.4.5 DATA CLASSIFICATION FRAMEWORK

All data within the CIMS platform will be classified into one of four sensitivity tiers determining the access controls, encryption requirements, transmission restrictions, audit intensity, and retention policies applicable to that data:

- **Tier 1: Public Data:** Information that is intentionally made available to the public through the CIMS portal like scheme descriptions, policy documents, MSME resource library content, job listings, and trade fair information. No access restriction beyond standard web access controls. Minimal audit requirements.
- **Tier 2: Internal Operational Data:** Data that is accessible to authorised departmental users within their role and jurisdictional scope but is not intended for public disclosure like MSME registration details, scheme application records, training enrolment data, and grievance records. Subject to RBAC access controls, standard encryption at rest and in transit, and standard audit logging.
- **Tier 3: Sensitive Departmental Data:** Data of heightened sensitivity that requires stricter access control and enhanced audit logging like employee HR records, payroll data, individual financial performance data, and grievance resolution details involving specific parties. Subject to stricter RBAC controls with mandatory Data Owner approval for access beyond standard roles, enhanced encryption, and comprehensive audit logging of all access and modification events.
- **Tier 4: Restricted Data:** The most sensitive data within the platform individual employee salary and deduction details, personally identifiable information, security audit findings, and system configuration documentation. Accessible only to specifically named individuals with documented justification. Subject to the most stringent access controls, mandatory encryption, complete audit trails, and periodic Data Owner review of access authorizations.

11.1.4.6 DATA RETENTION & ARCHIVAL POLICY

A formally documented Data Retention and Archival Policy will govern the lifecycle of all data within the CIMS platform from creation through active use, archival, and eventual secure deletion. The policy will be established in alignment with applicable Government of Bihar record retention rules, the Information Technology Act 2000 and 2008, and the RFP's mandate for a 60-month engagement horizon. The following retention principles will apply: -

- **Transactional Operational Data:** All transactional records like MSME registrations, scheme applications, grievance records, training enrolment records, and attendance records will be retained in the active database for the full duration of the engagement and for a defined post-engagement period, as specified by the Data Owner and consistent with applicable Government record retention requirements.
- **HR and Payroll Data:** Employee HR and payroll records will be retained in accordance with applicable Government of Bihar service record retention rules with completed payroll cycles archived to the archival tier after defined periods while remaining accessible to authorised HR and audit personnel through the platform's archival retrieval function.
- **Analytical and BI Data:** Historical analytical datasets and report snapshots will be retained in accordance with the defined data freshness and historical comparability requirements of the Department's BI and policy functions with older historical data migrated to the archival tier as active storage capacity thresholds are approached.
- **Audit Logs:** All audit log data like access logs, modification logs, approval workflow logs, and security event logs will be retained for a minimum period defined by CERT-In guidelines and the Information Technology Act, to support forensic investigation, compliance audit, and accountability review requirements.
- **Secure Deletion:** Data scheduled for deletion upon expiry of defined retention periods and with formal Data Owner approval will be subject to secure, cryptographic deletion procedures that ensure data cannot be reconstructed or recovered following its authorized deletion.

11.1.4.7 AUDIT TRAIL FRAMEWORK

A comprehensive, tamper-evident audit trail will be maintained across all layers of the CIMS platform capturing a complete, chronological record of every significant event within the system that bears on data integrity, access control, and operational governance.

- **What is Audited:** The audit trail will capture all of the following event categories: user authentication events (successful and failed logins, password changes, session timeouts); data access events (records viewed, reports executed, dashboard sessions); data modification events (records created, updated, or deleted with before and after values for all changed fields); approval workflow events (approvals granted, rejections issued, escalations triggered); system configuration changes (access control policy

changes, ETL configuration changes, notification rule changes); and security events (WAF alerts, privilege escalation attempts, unusual access pattern detections).

- **Audit Log Integrity:** All audit log entries will be written to a dedicated, separately access-controlled audit log store to which no application-level process has delete or modify permissions. Log entries will be cryptographically signed to ensure their integrity cannot be compromised without detection. Audit log backups will be taken independently of application data backups, ensuring that audit records are preserved even in scenarios where application data requires restoration from backup.
- **Audit Log Access and Review:** Access to audit log data will be restricted to the Platform Data Administrator, designated Security Administrator, and formally authorized audit personnel from Udyog Mitra. Regular audit log reviews will be conducted with automated anomaly detection alerts supplementing periodic manual review by the security team to identify unusual access patterns, potential data integrity violations, or security incidents requiring investigation.

11.1.4.8 REGULATORY COMPLIANCE FRAMEWORK

The CIMS platform's data governance posture will be maintained in continuous compliance with the following regulatory and standards frameworks throughout the 60-month engagement:

- **Information Technology Act 2000 and IT Amendment Act 2008:** All data processing activities within the CIMS platform will comply with the provisions of the IT Act governing data protection, cybersecurity obligations, and the responsibilities of intermediaries handling sensitive personal information.
- **CERT-In Guidelines on Information Security:** The platform's security controls, incident response procedures, vulnerability management programme, and audit practices will be aligned to the most current CERT-In guidelines with any new advisories published during the O&M period incorporated into the platform's security management programme within the timelines specified by Uneecops's security management SLA.
- **ISO 27001:2022 — Information Security Management:** As a CMMi Level 5 certified organisation with ISO 27001:2022 certification, Uneecops will apply the ISMS framework of ISO 27001 to the CIMS engagement ensuring that information security risks are systematically identified, assessed, treated, and monitored in accordance with the standard's requirements.
- **ISO 20000-1:2018 — IT Service Management:** The O&M service management practices incident management, change management, service request fulfilment, and continual service improvement will be governed in accordance with the ISO 20000-1 framework, ensuring a structured, measurable, and continuously improving service delivery posture throughout the post-go-live period.
- **National Data Governance Framework Policy:** Where applicable provisions of the Government of India's National Data Governance Framework Policy are relevant to the CIMS platform's data management practices, Uneecops will ensure that its data governance approach is aligned to the policy's requirements supporting the Department in meeting its obligations as a government data custodian.

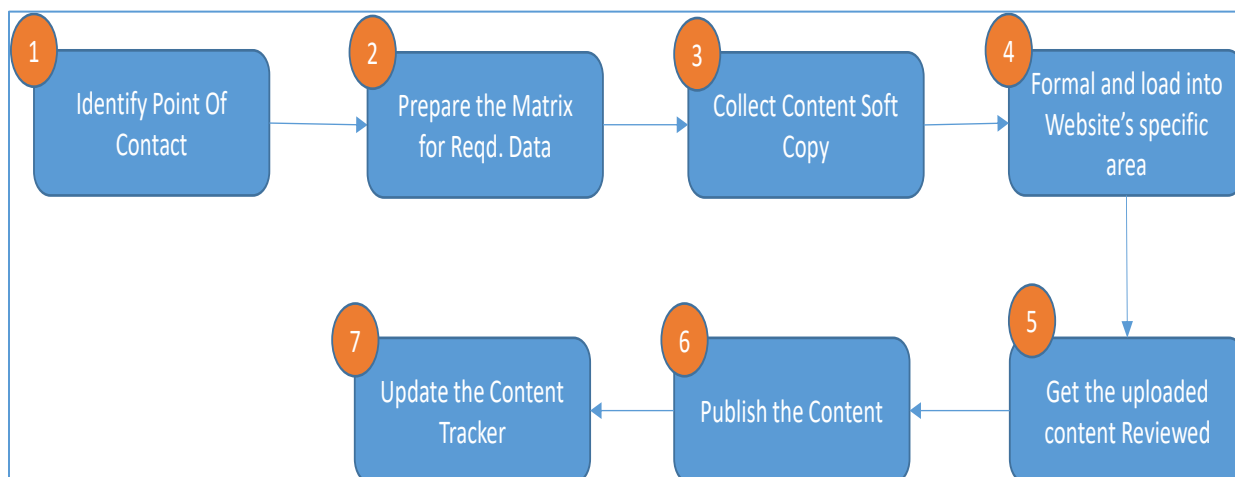
11.1.4.9 DATA GOVERNANCE DURING O&M

Data governance is not a one-time implementation exercise it is a continuously exercised institutional discipline. Throughout the 52-month O&M period, Uneecops will maintain the following ongoing data governance activities:

- Quarterly Data Quality Reports will be produced and presented to the Data Governance Council assessing data completeness, accuracy, and consistency across all modules, identifying quality degradation trends, and recommending corrective actions.
- Annual Data Classification Reviews will be conducted with Data Owners reviewing and reconfirming the classification designations of all major data entities, to ensure that classification remains appropriate as the platform's data landscape and the Department's risk environment evolve.
- Retention Policy Reviews will be conducted bi-annually ensuring that retention periods remain aligned with applicable regulations and the Department's evolving archival requirements.
- Access Authorization Reviews will be conducted semi-annually with Data Owners reviewing and reconfirming all user access authorizations within their data domain, ensuring that access rights remain appropriate and that no orphaned access persists beyond role change or separation event.

11.1.4.10 DATA MIGRATION PLAN

- Existing data (Masters through data dumps) related to the existing scheme applications (in-process) & approvals shall be ported to the envisaged portal & displayed in the format as finalized for the new portal
- Uneecops shall prepare and share the requisite migration plan and architecture and then move the data to the new target environment as per the defined schema and table structure in the SRS document.
- The data migration plan will have clearly defined timelines, potential risks, issues or challenges that might arise during the migration process. It will also bear a data contingency plan to address any unforeseen issues or challenges
- This will be done after analyzing the data formats, structures, and relationships in the existing system to ensure compatibility with the envisaged application's data model
- Uneecops shall create a mapping document that defines the correspondence between data elements in the legacy systems and the data structures of the envisaged application
- Uneecops shall ensure minimum business downtime at the time of data migration by identifying, configuring, and developing the data upload/download programs for the data migration
- Uneecops (if required) shall perform data cleansing activities to eliminate duplicates, resolve inconsistencies, and address any data quality Issues
- Uneecops shall prepare and share the requisite migration architecture and then move the cleansed and transformed data to the new target environment as per the defined schema and table structure in the SRS document.
- The indicative data migration strategy is as shown below. Uneecops would follow a structured process for content uploading into the database. The diagram below depicts the high-level steps: -



- The migrated data will be subject to reconciliation, verification, and validation to ensure accuracy & integrity of the data within the envisaged web application.
- The migrated data will be subject to rigorous testing including data integrity checks, validation rule testing and data consistency verification.
- Uneecops shall perform post migration clean-up activities like archiving, purging, or removing redundant data to add to the extra layer of accuracy.
- Uneecops shall continuously monitor the application to identify any anomalies arising due to data integrity and address them promptly.

12 ASSUMPTIONS & DEPENDENCIES

- The exact set of departmental users and their roles (rights and access privileges) shall be decided after conducting a detailed stakeholder consultation with the Department, covering State, District, Block, and Panchayat-level user categories. By incorporating their insights and addressing their specific operational needs, the Identity & Access Management framework will effectively safeguard sensitive information and functionalities, supporting the secure operation of the CIMS portal and aligning with the strategic goals and regulatory requirements of the project.
- It is assumed that the project will be primarily executed/undertaken offsite (Uneecops premises) and the whole team need not necessarily be placed onsite before Go-Live of the platform. Certain key resources can be deployed onsite at Patna and shall hop in and out of Patna as and if required during requirements elicitation, UAT, training delivery, and Go-Live cut-over.
- The APIs and associated technical documentation for all eleven (11) GoI/external portals with which integration must be conducted namely Udyam Registration Portal, MSME Ministry portal, Textiles portal, Food Processing portal, ONDC, TReDS, ODR, Single Window System, biometric device interfaces, Payment Gateway, and IPR/FSSAI portals shall be provided by the Department, latest by the Interface Control Document freeze in Phase 1.
- All data dumps for data migration (at Master and Transaction levels, including legacy MSME records, DIC profiling, scheme records, training records, and grievance archives, if any) shall be provided by the Department in clean state and in the format requested by Uneecops, in alignment with the Phase 3 data-migration window (Weeks 28–34).

- The iOS and Android based mobile applications shall be developed in accordance with the RFP requirement, with offline capability and a synchronisation framework. The mobile application will deliver, at minimum, all citizen-facing and field-officer-facing transaction modules (MSME registration and profile management, scheme application and tracking, grievance filing, training enrolment, learner attendance capture, and BI dashboard consumption with offline-capable reading). Module decompositions optimized for mobile form-factors such as administrative configuration screens better suited to web shall be deferred to web-only interfaces. The exact feature decomposition between web and mobile shall be jointly finalized at the SRS phase, governed by mobile UX best practices and stakeholder consultation
- For the e-commerce capability of the MSME MIS module, Uneecops shall integrate the platform with the ONDC network using ONDC protocol for fetching market activity, catalogue, and order information for the purpose of dashboarding on the CIMS application. MSME sellers shall continue to transact through the existing ONDC seller apps; Uneecops shall not develop a parallel seller-side application.
- The Knowledge Graph schema, NLP pipeline, AI chatbot, AI Resume Builder, and predictive ML models shall be built by Uneecops using established open-source frameworks. The initial seed corpus (policy documents, scheme catalogue, training material, departmental circulars) for fine-tuning and content ingestion shall be supplied by the Department in machine-readable format.
- AI-driven training content generation for the TMIS module should be predicated on Department-supplied authoritative source documents. Uneecops shall not be responsible for the factual content of the source material; quality assurance of generated content shall be a joint responsibility, with final approval resting with the Department's training authority.
- In case it is decided that DSCs shall be used for e-Signing of applications and approvals, the cost of procurement of DSC tokens, their utility software, and the KYC formalities for departmental signatories shall be the responsibility of the Department.
- Uneecops shall purchase the SSL certificate(s) in the name of the Department and integrate the same into the CIMS portal to ensure secure transactions. The payment for the SSL certificate(s) shall be made in actuals and reimbursed by the Department.
- The infrastructure (IT and physical) for carrying out training/handholding of departmental stakeholders will be provided by the Department. The exact quantum of trainings, batch size, and locations shall be discovered at the time of requirement gathering by Uneecops and finalized in consultation with the Department. Furthermore, physical training shall be held at the Department's Headquarters in Patna only, subject to logistics and travel arrangements for trainees by the Department.
- We hereby assume that training shall be conducted in a hybrid mode both virtual and physical and that all physical training venues shall be restricted to the Department's office in Patna. Training shall be delivered in Hindi and English, with role-based modules for State, District, Block, and Panchayat-level users.
- Static content of the platform like page labels, menu items, system messages, navigation elements, and instructional content shall be implemented bilingually in Hindi and English. Operational data entry shall be in English by default for departmental administrative modules including HRMS records, TMIS administrative entries, scheme master data definitions, and BI configuration. Hindi data entry shall be supported in citizen-facing intake channels grievance submission, the AI-powered bilingual voice chatbot's conversational interface, and free-text MSME profile fields where the underlying user constituency operates predominantly in Hindi. The platform's NLP layer and BI rendering engine shall transparently display Hindi-origin data to authorized users wherever the underlying data is provided

in Hindi. Hindi static content transcripts for the platform interface shall be supplied by the Department in machine-readable format during the SRS phase..

- In case any new integration requirement (beyond the 11 specified GoI/external integrations) emerges in the future, the same shall be treated through the change-request mechanism with associated cost and timeline implications.
- It is assumed that no data digitization or scanning work is required to be carried out within the scope of this engagement.
- It is assumed that for data migration, all source data shall be provided to Uneecops in clean state and in the format requested by Uneecops. Data cleansing of legacy records is not within the scope of work of Uneecops; remediation of malformed source data, if encountered, shall be handled through a change request.
- It may be noted that the proposed technology stack can be changed (if required) as per mutual discussion and written approval from the Department, provided that all functional, non-functional, security, and compliance requirements continue to be met.
- Uneecops shall only be responsible for providing support to the envisaged CIMS portal and not for the integrated GoI portals or external systems with which the platform interfaces. Issues arising from external systems shall be coordinated by Uneecops on a best-effort basis but shall not constitute a CIMS SLA breach.
- The production environment shall be hosted on the Bihar State Data Centre (SDC). Compute, storage, network bandwidth, and Disaster Recovery configuration shall be provisioned by the BSEDC operator per the sizing tabled at the requirement gathering stage. Uneecops shall configure, deploy, and operate the CIMS platform within the SDC-allocated capacity envelope. Capacity expansion, if required during O&M, shall be governed by a change request.
- The performance benchmark of supporting 100+ concurrent users at peak load shall be validated through a controlled performance/load test in Phase 3. The exact concurrency target, transaction mix, and response-time thresholds shall be finalized in consultation with the Department during Phase 1 (Test Strategy preparation).
- UAT participants shall be representative users from State, District, Block, and Panchayat levels, nominated by the Department. UAT shall be deemed accepted upon formal sign-off by the Department's authorized signatory; defects shall be classified per agreed severity rules, and only Severity-1 and Severity-2 defects shall be blockers for Go-Live.
- The application security audit shall be conducted by the Department's empaneled Certified Auditor in accordance with RFP §4.2 vulnerability categories. Uneecops shall remediate identified vulnerabilities within fifteen (15) days of formal notification per RFP timelines and submit the platform for re-audit prior to issuance of the Security Compliance Certificate.
- AI/ML model retraining and Knowledge Graph refresh shall be performed at a quarterly cadence during O&M. Major model retrains driven by significant policy or content shifts beyond the standard quarterly cadence shall be addressed through a change request.
- While the CIMS platform is logically multi-modular (7 modules across 4 tracks), the platform shall be deployed as a single-tenant solution for Udyog Mitra. Should multi-tenant capability be required for additional Government of Bihar departments in the future, the same shall be evaluated through a change request.
- It is assumed that internet bandwidth at the Department's offices and at Bihar SDC is adequate for the platform's web, mobile, and integration loads. Bandwidth augmentation, if found necessary post-deployment, shall be the responsibility of the Department.

- Any commercial component requiring per-user or per-instance licensing (if recommended during architecture finalization) shall be provided by the Department.
- It is assumed that internet bandwidth at the Department's offices and at Bihar SDC is adequate for the platform's web, mobile, and integration loads. Bandwidth augmentation, if found necessary post-deployment, shall be the responsibility of the Department
- Procurement and renewal of any commercial software licenses (database, BI front-end, OS, monitoring tools- if commercial variants are mandated by departmental policy) shall be the Department's responsibility unless explicitly included in the financial bid.
- The Department shall provide a single point of contact (Programme Director or Nodal Officer) for day-to-day governance, escalation routing, and inter-stakeholder coordination across the engagement life cycle.
- Recurring third-party operational expenditures associated with citizen, and departmental notifications and integrations shall be governed as follows. The platform's Notification Engine shall integrate with a DLT-registered, TRAI-compliant SMS gateway and a Department-nominated SMTP infrastructure; SMS unit charges and email infrastructure charges over the engagement period shall be borne by the Department directly to the respective service providers, against monthly volume reports generated by the platform.
- Similarly, for Geo-spatial mapping and GIS rendering; the associated API consumption and subscription charges shall be borne by the department.
- On similar lines, Aadhaar-based per-transaction e-Sign charges shall be borne by the Department.
- It is assumed that the hosting of this system will be on Bihar State Data Centre and the charges for consumption of cloud services shall be borne by the department directly. The cloud specifications mentioned in the deployment diagram are only indicative in nature and may be subject to changes as per the actual requirement discovery in the SRS phase of the project.

13 ROLES AND RESPONSIBILITIES

13.1 UDYOG MITRA, DEPARTMENT OF INDUSTRIES, GOVERNMENT OF BIHAR

- **Nominate dedicated officials** (the Programme Director / Nodal Officer and supporting departmental coordinators) to act as the primary points of contact for all project-related activities, decisions, and approvals throughout the 60-month engagement.
- **Provide timely access** to all institutional information, legacy data, departmental workflows, and existing system details required for design and development of the CIMS platform.
- **Review and approve project deliverables** including the SRS, Solution Architecture Document, Data Architecture Document, Security Architecture Document, Interface Control Documents, UI/UX prototypes, Test Strategy, training plans, and Go-Live readiness artefacts within mutually agreed timeframes to avoid schedule slippage.
- **Facilitate coordination with internal directorates**, the Bihar State Data Centre, line departments (DIT, GST/Commercial Tax, Labour, MSME Ministry, Textiles, Food Processing), ONDC, TReDS, ODR, and other GoI/external integration counterparties.
- **Provide content, guidelines, and domain-specific input** for the AI Knowledge Platform including scheme catalogue, policy documents, departmental circulars, training collateral, and Hindi-language transcripts for the multi-lingual rendering layer.

- **Control scope variations**, ensuring that changes are routed through a formal change management process to prevent uncontrolled scope expansion.
- **Ensure timely feedback and clarifications** on queries raised by the Agency, including those related to functional requirements, domain rules, and integration points.
- **Monitor and review the overall project progress**, conduct periodic evaluation meetings, and provide strategic direction as required.
- **Facilitate release of funds** to the Agency under various cost heads based on approved milestones and in accordance with the payment schedule.
- **Support scheduling and execution of UAT**, including nomination of end-users and timely validation of test cases, results, and acceptance signoffs.
- **Provide administrative support**, access, and permissions necessary for project development, deployment, and subsequent AMC operations.
- Provision of **DSC tokens and KYC** for e-Sign (if required)
- Provision for **Hindi content transcripts** for multi-lingual rendering
- Provision of **physical infrastructure for training at Patna**
- Provision for the **Bihar SDC access enablement** like firewall rules, VPN tunnels, IP whitelisting, capacity provisioning
- Provision for the nomination of **Data Governance Council, Data Owners and Data Stewards** by data domain
- Provision for approval of the **Change Advisory Board (CAB)** participation during O&M

13.2 UNEECOPS TECHNOLOGIES LIMITED

- **Design, develop, configure, test, and deploy** the Patna ICMS Portal as per the technical and functional requirements of the RFP, in conformance with Milestone 4 (Go-Live) at T + 8 months and the full 60-month engagement timeline as per RFP
- **Prepare and submit detailed project plans**, including work breakdown structure, timelines, risk plans, dependency charts, and resource allocation to support timely execution.
- **Submit all deliverables** like SRS, UI/UX, architecture, test plans, UAT documents, deployment manuals - within the agreed schedule.
- **Maintain regular communication with the Department**, providing progress updates through periodic status reports, review meetings, dashboards, and escalation matrices.
- **Designate a single point of contact (SPOC)** for all communication, coordination, and clarifications throughout the project and O&M period.
- **Provide technical support and promptly resolve issues**, bugs, or system errors reported by the Department or identified during development, testing, or O&M period.
- **Facilitate User Acceptance Testing (UAT)** by providing UAT environments, test scripts, supporting data, and incorporating feedback in a time-bound manner.
- **Coordinate with external agencies and systems**, including Udyam, MSME Ministry, Textiles, Food Processing, ONDC, TReDS, ODR, Single Window System, biometric devices, Payment Gateway, IPR/FSSAI, and any other API-based integrations approved by the Department.
- **Ensure compliance with cybersecurity standards**, including data protection, access control, encryption, logging, audit trails, security testing, and MeitY guidelines.
- **Provide training and handholding support** to department officials and nominated users, including training materials, instructional videos, and user manuals.
- **Deploy and manage required resources** during development and O&M as outlined in the contract.
- **Host and maintain the application** during O&M, including monitoring, performance tuning, backups, incident management, SLA compliance, versioning, and patch management.
- **Handle change requests** through a structured Change Control mechanism and implement approved enhancements per Department guidance.
- **Ensure proper documentation and knowledge transfer**, including source code, configuration files, architecture documentation, and O&M manuals at Go-Live.
- **Assist the Department in future scalability and interoperability planning**, ensuring architecture remains modular, secure, and easily extensible.
- **Deploy the CIMS platform on-premises at the Bihar State Data Centre** in a High Availability topology with a warm-standby Disaster Recovery configuration meeting the agreed RTO/RPO targets, in conformance with the RFP's deployment-model directive
- **Conduct the security audit via the Department-appointed CERT-In empaneled auditor**, remediate all vulnerabilities within fifteen (15) days of notification, secure the Compliance Certificate prior to Go-Live, and conduct annual re-audits thereafter for both web and mobile applications.
- **Ensure regulatory and standards compliance throughout the 60-month engagement** with GIGW 3.0, WCAG 2.1, DPDP Act 2023, CERT-In guidelines, IT Act 2000 (and Amendment 2008), ISO 27001:2022, and ISO 20000-1:2018.
- **Procure SSL certificate(s) in the Department's name** for secure platform transactions,

- **Deliver role-based training in hybrid mode and bilingual delivery (Hindi/English)** across State, District, Block, and Panchayat-level users on a train-the-trainer model; physical sessions restricted to the Department's office at Patna only.
- **Submit the complete handover package at Milestone 5 (Month 60)** including the full source code, configuration artefacts, architecture and O&M documentation, security and AI model artefacts, training records, and the Final Handover Certificate as per the RFP in satisfactory functional condition.

14 ANNEXURE-4: TECHNICAL SPECIFICATIONS COMPLIANCE MATRIX

Features	Supported (Y/N)	If not supported, partially supported (Y/N)	Remarks (Unecops Compliance Statements)
The platform should have the ability to publish on premises to ensure all data is retained within the State Data Centre.	Y — Fully Supported	Not Applicable	Unecops Technologies will design, develop, and deploy the entire CIMS and Business Intelligence Platform exclusively on the infrastructure provided by the Department of Industries, Bihar, at the Bihar State Data Centre (BSDC), Patna. The proposed solution architecture is on-premises-first by design all application services, transactional databases, analytical data stores, AI/ML workloads, the BI engine, and the integration middleware will be provisioned, configured, and operationalised within the BSDC environment. No component of the platform whether data storage, application processing, or analytics computation will be hosted, processed, or transmitted to any external cloud environment as part of the primary production configuration. A cloud-based High Availability failover layer may be provisioned as a secondary, warm-standby configuration for business continuity purposes; however, the Bihar State Data Centre will remain the primary, authoritative, and sovereign deployment environment always. All MSME data, departmental HR data, grievance records, training records, and analytics datasets will reside exclusively within the BSDC, ensuring complete data sovereignty for the Government of Bihar and full compliance with the on-premises mandate of this RFP.

The platform should provide integrated end-user access ranging from interactive dashboards, ad-hoc queries, self-service data visualisation capabilities, augmented analytics, geospatial mapping and visualizations, etc.	Y — Fully Supported	Not Applicable	The Business Intelligence and Analytics Platform proposed by Uneecops will deliver a Single Integrated Analytics Environment that provides all categories of authorised users from senior policy officials to district-level departmental officers with seamless, role-appropriate access to the complete spectrum of analytical capabilities through a unified web-based interface and mobile application, without requiring any thick-client software installation. The integrated end-user access environment will encompass the following capabilities, all accessible from within a single, unified platform interface: - • Interactive Dashboards: Purpose-built, configurable dashboards for each user role; MSME performance dashboards for scheme administrators, operational dashboards for district officials, executive dashboards for senior departmental leadership, and HR analytics dashboards for HR administrators. All dashboards will support real-time data refresh, drill-down navigation, and interactive filtering. • Ad-hoc Query and Analysis: Business users will be empowered to construct their own analytical queries against the platform's data model using a natural language interface and an intuitive drag-and-drop query builder without requiring SQL knowledge or technical assistance. Ad-hoc queries will execute against the in-memory columnar analytical data store, ensuring responsive query

			performance even across large datasets. • Self-Service Data Visualisation: End users will be able to create, customise, and publish their own data visualisations like charts, graphs, heat maps, and tables- using a drag-and-drop visualisation canvas. The platform will intelligently recommend the most appropriate visualisation type based on the nature and dimensionality of the selected data fields. • Augmented Analytics: The platform will incorporate AI-driven augmented analytics capabilities including automated insight generation, anomaly detection, and trend identification that surface relevant insights proactively to users without requiring them to formulate specific analytical questions. • Geospatial Mapping and Visualisation: Full geospatial capability will be provided, supporting the mapping and visualisation of MSME units, clusters, Industrial Areas, SEZs, land parcels, scheme beneficiary distribution, and grievance hotspots on interactive maps using longitude and latitude coordinates. Geospatial views will be integrated with all relevant dashboards and will support drill-down from state to district to block to panchayat level.
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The intelligence platform should have an ad-hoc query and analysis environment that works against a logical view of information from multiple data sources in a pure Web environment. It should provide the capability to combine data from multiple applications or databases in a single calculation. It should make insights accessible to anyone, anytime, and anywhere with even mobile Business Intelligence.	Y — Fully Supported	Not Applicable	The BI and Analytics Platform proposed by Uneecops will deliver a pure web-based, multi-source ad-hoc query and analysis environment that operates against a unified logical data model abstracting the underlying complexity of the platform's multi-tier data architecture and presenting authorised users with a consistent, governed, business-vocabulary view of all available information assets. Logical Data Model and Semantic Layer: A robust semantic layer will be maintained above the physical data stores, translating raw database structures into business-meaningful dimensions, measures, and hierarchies. This semantic layer will ensure that users interact with concepts such as "MSME Scheme Applications," "District-wise Beneficiary Count," and "Training Completion Rate" rather than database table names and technical field identifiers. The semantic layer will be centrally governed and will serve as the single source of analytical truth across all modules. Multi-Source Data Combination: The analytical environment will support the combination of data from multiple source systems including the MSME MIS, HRMS, TMIS, Grievance System, Job Portal, and external GoI data feeds within a single analytical calculation or visualisation. Cross-database joins, data blending, and derived metric computation across heterogeneous data sources will be supported natively, without requiring manual data extraction or offline reconciliation.
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			• Pure Web Delivery: The entire ad-hoc query and analysis environment will be delivered through a standards-compliant web browser interface requiring no client-side software installation beyond a modern web browser. The platform will be compatible with all major web browsers and will render consistently across desktop and laptop devices used in government offices. • Mobile Business Intelligence: The full suite of analytical and dashboard capabilities including interactive dashboards, ad-hoc query results, KPI scorecards, and geospatial views will be accessible through the CIMS mobile application on both iOS and Android devices. Mobile BI will support offline analysis for scenarios where field officials require access to analytical content in areas with limited network connectivity, with automatic data synchronisation upon restoration of connectivity.
The platform should provide capabilities to create KPIs to measure progress and performance over time and graphically communicate strategy and strategic dynamics using Strategy Maps and Custom Views. In addition, intuitive and dense visualizations must be available.	Y — Fully Supported	Not Applicable	The proposed BI and Analytics Platform will deliver a comprehensive KPI Management and Strategy Communication framework that enables authorised administrators and analysts to define, configure, monitor, and visualise key performance indicators across all dimensions of departmental operations from MSME scheme effectiveness to HR performance, training outcomes, and grievance resolution efficiency. • KPI Creation and Management: The platform will provide a governed KPI authoring environment in which authorised administrators can define KPIs with associated targets, thresholds, weightages, and measurement frequencies. KPIs will

			be version-controlled and approval-governed, ensuring that the Department's performance measurement framework evolves in a structured and auditable manner. • Threshold-Based Performance Monitoring: All KPIs will support the configuration of performance thresholds distinguishing between on-target, at-risk, and below-target performance states. Threshold breaches will trigger visual alerts within the dashboard environment and configurable notifications through the platform's multi-channel notification engine. • Strategy Maps: The platform will support the creation of Strategy Maps that graphically communicate the causal relationships between strategic objectives, enabling senior officials to visualise how operational performance at the district and block level contributes to the Department's overarching strategic goals. Strategy Maps will be linkable to the underlying KPI data, enabling drill-down from strategic objective to operational metric with a single interaction. • Custom Views: Each authorised user will be able to configure and save personalised dashboard views tailored to their specific role and analytical focus areas, while remaining within the access permissions defined by their RBAC profile. Custom views will be saved against the user's profile and will persist across sessions.
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			• Dense and Intuitive Visualisations: The platform will support a comprehensive library of visualisation types including bar charts, line charts, scatter plots, heat maps, tree maps, waterfall charts, bullet charts, geospatial maps, and custom infographic-style visualisations ensuring that the most information-dense and analytically appropriate visualisation is available for every data scenario.
The platform should provide the capability to create reports that capture a series of snapshots of the visualizations platform or reports, allowing the information to be viewed offline.	Y — Fully Supported	Not Applicable	The proposed platform will deliver a robust Report Snapshot and Offline Viewing capability that ensures authorised users retain access to critical analytical information even in the absence of live network connectivity to the BSDC-hosted platform. • Report Snapshots: The platform will support the scheduled and on-demand generation of report snapshots point-in-time captures of dashboard states, visualisation outputs, and tabular reports that preserve the analytical content of a given moment for historical reference, audit, and offline review. Snapshots will be stored in the platform's document management repository with appropriate access controls and retention policies. • Scheduled Report Distribution: Authorised administrators will be able to configure scheduled report runs that automatically generate and distribute report snapshots to defined recipient lists via email with the report attached in PDF, Excel, or CSV format as specified. This ensures that senior officials and policy makers receive timely analytical

			summaries without requiring active platform access. • Offline Viewing: Report snapshots and exported analytical content will be viewable offline in standard formats like PDF for formatted reports and dashboards, Excel and CSV for tabular data requiring further analysis. The mobile application will additionally support the caching of recently accessed dashboards for offline reference in field scenarios. • Export Formats Supported: All reports, dashboards, maps, graphs, and tabular outputs will be exportable in the following formats namely PDF, Microsoft Excel, CSV, and image formats (PNG/JPEG), ensuring compatibility with the widest range of downstream consumption and distribution workflows used within the Department.
The platform should provide an alerting engine that captures and distributes notifications via multiple channels in response to pre-defined business events and/or data exceptions to speed exception-based decision making. When a condition is triggered, the user should be presented with a list of possible actions to take.	Y — Fully Supported	Not Applicable	The proposed platform will incorporate a centralized, multi-channel, event-driven Alerting and Notification Engine that enables the Department to establish a proactive, intelligence-led operational posture moving from reactive data review to automated exception management across all modules of the CIMS platform. • Multi-Channel Alert Distribution: The alerting engine will distribute notifications through Email, SMS, and in-application push notifications simultaneously, ensuring that critical alerts reach designated recipients regardless of whether they are actively using the platform at the time of the event. Alert routing will be governed by the recipient's role,

			administrative jurisdiction, and configured notification preferences. Pre-Defined Business Event Triggers: Authorised administrators will be able to define alert conditions against any metric, KPI, or data field within the platform's analytical data model including threshold breaches (e.g., district-level scheme application processing delays exceeding defined timelines), data quality exceptions (e.g., MSME registration records with incomplete mandatory fields), and milestone events (e.g., scheme budget utilisation reaching 80% of allocation). Alert conditions will be version-controlled and subject to an approval workflow to ensure governance discipline. Exception-Based Decision Support: When an alert condition is triggered, the platform will present the receiving user within both the web and mobile interface with a structured exception notification that includes the nature of the exception, the current value versus the defined threshold, the relevant data context, and a configurable list of recommended actions from which the user can initiate a response directly. This action-oriented alert design ensures that exception management is efficient and auditable. Alert Audit Trail: All alert events trigger time, distribution list, delivery status, and user response will be captured in the platform's unified audit log, supporting governance, accountability, and performance review requirements.
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The tool should have the capability to search for existing content based on full indexing of Dashboards, Analyses, Views, Prompts, KPIs, Scorecards, etc. The metadata should be indexed and searchable.	Y — Fully Supported	Not Applicable	The proposed BI and Analytics Platform will maintain a comprehensive, full-text metadata index across all analytical content objects within the platform ensuring that authorised users can rapidly locate relevant dashboards, reports, analyses, KPIs, scorecards, and views through a unified, Google-like search experience without requiring navigation through complex folder hierarchies. • Full Content Indexing: The platform's search engine will maintain a continuously updated full-text index covering the names, descriptions, tags, and metadata attributes of all content objects including Dashboards, Analyses, Ad-hoc Views, Prompts and Filters, KPI Definitions, Scorecards, Strategy Maps, Report Templates, and Data Source Definitions. The index will be refreshed automatically upon the creation, modification, or publication of any content object. • Metadata Search and Discovery: Users will be able to search for analytical content using natural language terms, business vocabulary, module names, and metadata tags with the search engine returning ranked, relevance-ordered results across all content types. Search results will respect the user's RBAC access permissions, ensuring that users can only discover and access content for which they are authorised. • Content Governance Through Metadata: The metadata framework will support the tagging of content objects with structured attributes including owning department, subject domain, data currency, and approval status enabling the
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			Department to maintain a well-governed analytical content library as the platform's content corpus grows over the 60-month engagement period.
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The platform should have the capability for query reuse and caching both at the Web Server and within the Analytic Server layer.	Y — Fully Supported	Not Applicable	The proposed platform will implement a multi-tier query caching and reuse architecture that materially improves dashboard rendering performance, reduces redundant computational load on the analytical data store, and ensures a consistently responsive user experience, particularly during peak usage periods when multiple concurrent users access common dashboards and reports simultaneously. • Web Server Layer Caching: At the web server tier, rendered dashboard outputs, report results, and visualisation data payloads will be cached for defined configurable durations. When multiple users request the same dashboard or report within the cache validity window, the platform will serve the cached result directly eliminating redundant round trips to the Analytic Server layer and significantly reducing response times for high-frequency, commonly accessed content. • Analytic Server Layer Query Caching: At the Analytic Server tier, the results of frequently executed analytical queries will be maintained in an in-memory query result cache. When a subsequent query identical or semantically equivalent to a previously executed query is submitted, the platform will serve the cached result set rather than re-executing the query against the analytical data store. The query cache will be intelligently invalidated upon underlying data refresh, ensuring that cached results never present stale data to users. • Query Reuse and Intelligent Pre-Fetching: The platform will support
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			the definition of shared, reusable query objects, canonical analytical queries that form the basis of multiple dashboards and reports. By maintaining these shared queries as governed, centrally cached objects, the platform ensures consistency of results across all dependent visualisations and eliminates the performance overhead of duplicate query execution. The platform will additionally support intelligent pre-fetching of query results for high-priority dashboards during off-peak periods, ensuring near-instantaneous rendering during business hours. • Cache Governance: Cache validity periods, invalidation rules, and pre-fetching schedules will be configurable by authorised platform administrators, enabling the Department to balance analytical freshness requirements against performance optimisation objectives for different categories of analytical content.

The platform should provide a central management server that enables consistent diagnostics and troubleshooting.	Y — Fully Supported	Not Applicable	The proposed CIMS and BI Platform will incorporate a Centralized Platform Administration and Operations Console which is a dedicated management server environment that provides authorised system administrators with a unified, browser-based interface for platform-wide diagnostics, health monitoring, configuration management, and troubleshooting across all application services, data and integration components. • Centralised Diagnostics: The management console will aggregate diagnostic data from all platform layers, application servers, database services, the BI and analytics engine, the integration middleware, the AI/ML service tier, and the notification engine into a single, real-time operational view. System administrators will be able to assess the health status of every platform component from a single screen, without requiring separate access to individual server environments. • Troubleshooting and Root Cause Analysis: The console will provide structured access to application logs, error logs, query execution logs, API call logs, and ETL pipeline logs with filtering, search, and correlation capabilities that enable administrators to trace the root cause of operational issues across interconnected platform components. Log retention and archival will be governed by defined policies, ensuring that historical diagnostic data is available for post-incident analysis. • Configuration Management: All platform configuration parameters including data source connections,
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			notification settings, user access policies, cache validity periods, and alert thresholds will be managed through the centralised console, ensuring that configuration changes are applied consistently, documented, and subject to an approval workflow. No configuration changes will be permitted through direct server access outside the governed console environment. • Performance Monitoring and Alerting: The management console will continuously monitor system performance metrics like CPU utilisation, memory consumption, query response times, ETL pipeline throughput, and API latency and will trigger automated alerts to designated administrators when metrics breach defined thresholds, enabling proactive intervention before performance degradation impacts end users.
The platform should generate optimized, native queries for each data source.	Y — Fully Supported	Not Applicable	The proposed BI and Analytics Platform will incorporate an intelligent query generation engine that automatically translates user-initiated analytical requests whether originating from dashboard interactions, ad-hoc query builders, or scheduled report runs into optimized, data-source-native queries that exploit the specific performance capabilities of each connected data source. • Native Query Generation: Rather than generating generic, lowest-common-denominator SQL that is then submitted uniformly to all data

			sources, the platform's query engine will generate data-source-specific query syntax that leverages the native optimizations features of each connected system. For relational databases, this will include optimized JOIN strategies, index-aware query plans, and push-down computation. For OLAP sources such as SQL Server Analysis Services or SAP BW, native MDX queries will be generated to exploit cube-level aggregations and hierarchical pre-computation. For flat file and REST API sources, the platform will generate optimized extraction requests that minimize data transfer volumes. • Query Optimisation and Push-Down: Where the connected data source supports server-side computation, the platform will push analytical calculations, aggregations, filters, derived metrics down to the data source layer rather than retrieving raw data and computing in the application tier. This push-down optimisation strategy materially reduces data transfer volumes, improves query response times, and reduces computational load on the platform's application servers. • Adaptive Query Planning: The query engine will maintain query execution statistics and will adaptively refine query plans based on observed performance patterns ensuring that the platform continues to deliver optimal query performance as data volumes grow and query patterns evolve over the 60-month engagement period.
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Reporting and Ad-hoc Query should be available in a Web-based environment that is designed for users who want to create new analysis from scratch or modify and change existing analysis that appears on dashboard pages.	Y — Fully Supported	Not Applicable	The proposed platform will deliver a fully web-based, self-service reporting and ad-hoc analysis environment accessible through any modern web browser without client-side software installation specifically designed to empower non-technical departmental users to construct, modify, and publish their own analytical content independently. • New Analysis Creation from Scratch: The web-based analysis authoring environment will provide users with an intuitive, drag-and-drop canvas on which they can construct new analyses by selecting data fields from the governed semantic layer, applying filters and calculations, choosing appropriate visualisation types, and arranging content into dashboard layouts. The authoring experience will guide users through the analytical construction process with contextual suggestions and smart defaults enabling departmental officials with no formal data analysis background to produce meaningful, accurate analytical content. • Modification of Existing Dashboard Analysis: Users with appropriate permissions will be able to open any dashboard page in an edit mode, modify existing visualisations — changing chart types, adjusting filters, adding or removing data fields, restructuring layouts and save their modifications either as a personal view or, subject to an approval workflow, as a published update to the shared dashboard. This ensures that dashboards remain living, evolving analytical assets rather than static reports that require IT intervention to update.
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			• Governed Self-Service: While the environment is designed to empower self-service, it will operate within a governed framework ensuring that all self-service analytical content is created against the approved semantic layer, that sensitive data fields are accessible only to authorised users, and that published analytical content is subject to a review and approval process before being shared with broader audiences.
The analysis and reporting layer should provide a single user interface for ad-hoc query, reporting, and analysis against relational, OLAP, and flat file data sources.	Y — Fully Supported	Not Applicable	The proposed BI and Analytics Platform will present a single, unified user interface for all analytical interactions whether the underlying data resides in the relational OLTP database, the OLAP analytical store, flat file sources such as Excel spreadsheets and CSV files, or external REST API data feeds. The multi-source complexity of the data architecture will be entirely abstracted from the end user through the semantic layer, which presents a coherent, consistent business vocabulary regardless of the physical data source from which any given metric or dimension originates. • Unified Interface Across Data Source Types: Users interacting with the analysis and reporting environment will experience no difference in interface, navigation, or analytical capability whether they are querying MSME registration data from the relational transactional database, scheme performance aggregates from the OLAP store, training attendance records from an uploaded Excel file, or real-time scheme data retrieved from a GoI API feed. The interface provides a consistent analytical experience across all source types.

			• Cross-Source Analysis: The unified interface will support the creation of analyses that combine data from multiple source types within a single visualisation or report for example, correlating MSME financial performance data from the relational store with sector-level benchmark data from an imported flat file, or comparing departmental scheme disbursement data from the OLAP store with GoI portal data retrieved through a live API connection. Cross-source joins and data blending will be handled transparently by the platform's integration and query layer. • Single Sign-On Across All Data Contexts: The user's identity and access permissions established through the platform's SSO and RBAC framework will govern their access to data from all source types through the unified interface. No separate login or authentication is required to access different data sources within the analytical environment.
The platform should provide the capability for Dashboards to be personalized and automatically display data based on the user's identity or role. The platform should provide the capability wherein Measures, descriptive attributes, filters, sorting patterns, sub-totals, charts, and pivot table views can be added, deleted, or changed from within the dashboard, after a user makes all the	Y — Fully Supported	Not Applicable	The proposed platform will deliver a role-driven, personalization-enabled dashboard environment in which every user's analytical experience is automatically contextualized to their identity, administrative role, and jurisdictional scope while simultaneously empowering users to further personalize their analytical workspace within governed boundaries. • Identity and Role-Driven Personalization: Upon login, the platform will automatically present each user with a dashboard experience personalized to their role profile, a District Collector will see district-level MSME performance

changes, the new analysis can be saved and shared with a group of users.			dashboards scoped to their district; a scheme administrator will see scheme-specific operational dashboards; a senior departmental official will see an executive summary view aggregating performance across all directorates. This personalization is driven automatically by the user's identity and RBAC profile requiring no manual configuration by the user. Dashboard Customisation Capabilities: Within the boundaries of their access permissions, users will be able to personalise their dashboard views by adding or removing measures and descriptive attributes, applying and saving personal filters, modifying sort orders and sort directions, adding sub-totals and grand totals to tabular views, switching between chart types, and creating and saving pivot table views against available data dimensions. All such customisations will be applied without requiring IT support or technical intervention. Save and Share with User Groups: Following personalisation, users will be able to save their customised analysis as a named personal view retaining it for future sessions and optionally share it with defined groups of users within the platform. Shared analyses will be subject to the platform's content governance workflow, ensuring that shared analytical content is appropriate, accurate, and approved before wider distribution. Shared views will be accessible to all members of the designated group through the platform's content library and search interface.
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The platform should provide complete interactivity with dashboard content by selecting prompted values and filtering data; drilling on charts or tables to access detail; changing the sort order or sort direction of columns; maintaining context and moving to a different analysis by automatically passing constraints or selecting columns to display.	Y — Fully Supported	Not Applicable	The proposed platform will deliver a fully interactive dashboard environment in which every element of the dashboard like charts, tables, maps, KPI tiles, and filter controls is a live, interactive object that responds to user actions in real time, enabling fluid, context-preserving analytical navigation across the platform's complete data landscape. Prompted Value Selection and Dynamic Filtering: All dashboards will support configurable prompt controls date range pickers, district selectors, scheme selectors, sector filters, and custom parameter inputs through which users can dynamically constrain the data displayed across all dashboard components simultaneously. Filter selections will be applied instantly, with dashboard content refreshing in real time to reflect the applied constraints. Drill-Down and Drill-Through: Users will be able to drill down within charts and tables to access progressively greater levels of detail for example, clicking on a state-level MSME scheme disbursement bar will drill down to district-level data, and further to block-level data, and further to individual beneficiary records where authorised. Drill-through from a summary chart to a detailed tabular report will also be supported, enabling users to move seamlessly from high-level visualizations to granular record-level investigation within a single analytical session. Column Sort Order and Direction: All tabular views within dashboards and reports will support interactive column-level sorting users will be able to click any column header to
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			sort the table by that column, and click again to reverse the sort direction. Multi-column sort sequences will also be supported for complex ranking and ordering scenarios. Context-Preserving Navigation: When a user navigates from one analysis or dashboard to another by clicking a linked data element, selecting a menu option, or following a drill-through path the platform will automatically pass the active filter constraints from the originating dashboard to the destination analysis. This context-preservation ensures that the user's navigational journey through the platform's analytical content remains coherent and data-consistent, eliminating the need to re-apply filters at each navigational step. Column Selection and View Configuration: Users will be able to interactively select which columns to display within tabular views showing or hiding data fields as required for their specific analytical purpose without modifying the underlying report definition or requiring IT intervention.
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The Visualization platform should provide support for real-time collaboration.	Y — Fully Supported	Not Applicable	The proposed platform will incorporate a structured real-time collaboration framework within the visualisation and analytics environment, enabling multiple authorised users to engage with, annotate, discuss, and act upon analytical content simultaneously transforming data review from an individual activity into a collaborative, governance-driven institutional process. Collaborative Annotations and Comments: Authorised users will be able to add contextual annotations and comments directly to dashboard elements like charts, data points, KPI tiles, and map regions tagging colleagues for their attention and initiating structured analytical discussions in context. Annotations will be timestamped, attributed to the authoring user, and retained as a persistent record within the platform's audit framework. • Shared Analytical Sessions: The platform will support the sharing of live dashboard states including active filter configurations and drill-down positions enabling a user to share their current analytical view with a colleague who can then navigate the same data context independently. This capability is particularly valuable for structured analytical review sessions between district officials and state-level administrators. • Workflow-Integrated Collaboration: Collaborative interactions within the platform including annotation responses, analytical approvals, and exception acknowledgements will be integrated with the platform's notification engine, ensuring that all collaborative actions generate appropriate alerts
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			to relevant parties and are captured within the platform's audit trail.
The platform should provide a highly customized report format	Y — Fully Supported	Not Applicable	The proposed platform will deliver a Report Formatting and Customization framework that provides authorised report designers and administrators with comprehensive control over every aspect of report presentation ensuring that analytical outputs can be formatted to conform precisely to the Department's institutional

			standards, branding requirements, and stakeholder communication norms. • Pixel-Level Format Control: Report designers will have precise control over all formatting attributes page size and orientation (A4, Letter, custom), margin settings, header and footer content (including report title, generation timestamp, filter parameters applied, and authorised signatory fields), font family, font size, font weight, and colour for all text elements, background colours and shading for header rows, alternating row formatting, and border styles for tables and data grids. • Conditional Formatting: The platform will support the application of conditional formatting rules automatically highlighting data cells, rows, or visualisation elements based on their values relative to defined thresholds, targets, or peer comparisons. Conditional formatting will support colour-coded status indicators (green/amber/red), bold text emphasis, icon overlays, and data bar representations enabling at-a-glance performance assessment within highly formatted tabular reports. • Department Branding and Standardisation: Report templates will be configurable to incorporate the Government of Bihar and Udyog Mitra branding elements including official logos, colour schemes, and header/footer formats ensuring that all reports produced by the CIMS platform carry a consistent, professionally formatted institutional identity.
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			• Template Library: A governed library of approved report templates will be maintained within the platform, from which users can select appropriate formats for different reporting contexts like executive summaries, detailed operational reports, scheme-specific disbursement statements, and HR analytics reports ensuring consistency across the Department's analytical output while allowing appropriate variation for different use cases.
The Report Builder should generate multiple layouts.	Y — Fully Supported	Not Applicable	The proposed platform's Report Builder will support the generation and management of multiple, distinctly configured report layouts enabling the Department to produce analytical and operational reports in diverse formats tailored to the specific communication, presentation, and distribution requirements of different audiences and contexts. • Tabular Layout: A structured, row-and-column tabular layout for detailed operational reports like scheme beneficiary registers, payroll statements, attendance records, grievance logs, and training completion rosters with full support for grouping, sub-totals, grand totals, sorting, and conditional formatting. • Cross-Tab/Matrix Layout: A crosstab or pivot-table layout for multi-dimensional analytical reports presenting data across two or more-dimensional axes simultaneously, such as scheme disbursement by district and by quarter, or training completion by sector and by trainer. Cross-tab layouts will support hierarchical row and column

			groupings with sub-total and grand total aggregation at every level. • Dashboard Layout: A freeform, multi-panel layout combining charts, tables, KPI tiles, geospatial maps, and text elements within a single, visually composed page designed for executive briefing documents, scheme performance summaries, and monthly departmental review reports intended for printing or PDF distribution. • Narrative / Composite Layout: A layout that combines structured data tables and visualisations with narrative text sections enabling the production of analytical reports that present data-supported written commentary alongside visual evidence, suitable for formal policy submissions, scheme evaluation reports, and ministerial briefing notes.
Breach of KPI benchmarks should have the ability to trigger actions e.g. Email alerts.	Y - Fully Supported	Not Applicable	The proposed platform will deliver a KPI Benchmark Monitoring and Action Triggering framework that ensures no KPI breach goes unnoticed or unaddressed, embedding a proactive, intelligence-driven accountability mechanism at the heart of the Department's performance management system. • Automated KPI Breach Detection: The platform's alerting engine will continuously monitor all defined KPIs against their configured benchmark thresholds evaluating KPI values at defined frequencies (real-time, hourly, daily, or weekly, as configured) and detecting breaches at the moment they occur. KPI

			benchmarks will support multiple threshold levels for example, a warning threshold at 80% of target and a critical threshold at 60% of target enabling graduated alerting responses proportionate to the severity of the performance deviation. • Email Alert Triggering: Upon detection of a KPI benchmark breach, the platform will automatically trigger an Email alert to the configured recipient list for that KPI which may include the responsible departmental official, their immediate supervisor, and any designated oversight authority. Email alerts will contain the KPI name, the current value, the benchmark threshold, the magnitude of the breach, the relevant time period, and a direct link to the dashboard page displaying the full KPI context. • Multi-Channel Action Triggering: In addition to email alerts, KPI benchmark breaches will also trigger SMS notifications to designated mobile numbers, in-application push notifications within the CIMS web and mobile interfaces, and where configured automated workflow actions such as the creation of a pending task for the responsible official within the platform's workflow engine. This multi-channel, multi-action response framework ensures that KPI breaches translate directly into accountable, time-bound response obligations. • Escalation on Non-Response: Where a KPI breach alert has been issued and no acknowledgement or remedial action has been recorded
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			within a defined escalation window, the platform will automatically escalate the alert to the next level of the organisational hierarchy ensuring that persistent performance deficits are brought to senior leadership attention without requiring manual escalation. • Audit Trail of KPI Breach Events: Every KPI benchmark breach including the breach detection timestamp, the alert distribution list, delivery confirmation, user acknowledgement, and any remedial action recorded will be captured in the platform's unified audit log, supporting the Department's performance governance and accountability requirements.
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